

Content Relatability and Standardized Testing: Evidence from Texas*

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Abstract

One goal of standardized tests is to measure aptitude across heterogeneous students with minimal bias. However, how different students relate to topics or characters that appear in exam content could lead to differential test scores even when aptitude is the same. We study how differential content relatability can impact test scores using item-level data from reading comprehension exams in Texas. Using time-use data and natural language processing techniques, we first build a novel measure of race- and gender-based relatability to topics in the exam’s text passages. A one standard deviation increase in exam-level topic relatability across race predicts a 0.05 standard deviation change in student exam performance, with null effects for gender. We find student test scores improve on passages with a higher share of characters matching either the student’s race or gender. Our estimates suggest that equalizing the relatability of passages in these standardized tests could reduce the Black-white and Hispanic-white test score gaps by up to 9 percent and 10 percent, respectively. We then counterfactually estimate close to 11,000 Black students and 37,000 Hispanic students during our sample period were designated to be at a lower reading comprehension level due to relatability.

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1 Introduction

Inequality along demographic dimensions is well-documented and widespread in educational contexts. In the United States, achievement disparities are observed among children as early as primary school and are especially notable in standardized testing (Fryer & Levitt, 2004; Fryer & Levitt, 2013; Bond & Lang 2013). For example, there is a 14 percentage point difference between White and Black students and a 3 percentage point difference between female and male students on 3rd and 4th grade reading comprehension exams in the state of Texas. In response, some observers and policymakers have called for a deeper understanding of this testing gap and the mechanisms that produce these differences. While some have suggested abandoning standardized testing entirely, claiming that it systematically disadvantages some groups, others insist that standardized measurement is essential to accountability and progress, especially regarding racial and economic equality in education.¹

If standardized tests fail to measure achievement consistently across student backgrounds, it becomes essential to identify the test attributes that contribute to such mismeasurement and quantify their impact. One possible factor is the degree to which educational content is *relatable* to certain groups of students, whereby students learn or perform better when encountering topics that are familiar and interesting or characters with whom they identify. Studies in educational psychology show that interest in a topic can impact performance on reading comprehension tests and that these interests diverge by race and gender (Bray & Barron, 2004; Asher, 1979). Other existing research documents race and gender gaps in identity representation in educational materials, such as children’s literature (Adukia et al., 2023).²

Empirically analyzing the relationship between test performance and the relatability of content poses several challenges. First, student familiarity or interest with topics is inherently qualitative and typically unobserved in existing datasets. Even when quantitative representations of students’ topic familiarity or identity representation are available, they are likely correlated with other test constructs that exam designers intentionally vary, such as vocabulary difficulty or text length. Second, students’ topic familiarity and interest may correlate with ability, which poses a threat to identification. Finally, topics and character

¹See, for example, discussions by the National Education Association (“The Racist Beginnings of Standardized Testing”, <https://www.nea.org/nea-today/all-news-articles/racist-beginnings-standardized-testing>) and in The Atlantic (“Are Standardized Tests Racist, or Are They Anti-Racist”, <https://www.theatlantic.com/science/archive/2023/01/should-college-admissions-use-standardized-test-scores/672816/>). There has also been an academic discussion of such topics, including by psychometricians (Boykin, 2023) and economists (Card & Giuliano, 2016).

²This work, along with ours, contributes to a growing literature which uses text analysis in causal social science research (Gentzkow & Shapiro, 2010; Loughran & McDonald, 2011; Hassan et al., 2017).

identities may co-vary, so examining either attribute in isolation risks omitted variable bias.

In this paper, we study how differential relatability to exam content across students impacts exam performance and estimation of demographic test score gaps. Our two measures of content relatability reflect a student’s connection to topics and character identities in the text, which may vary across race and gender. We build a novel measure of topic relatability using time-use data and natural language processing techniques. We complement this with identity relatability, constructed with race and gender predictions of characters in the text. Applying these measures to text passages in reading comprehension exams in Texas, we obtain causal estimates of topic and identity relatability on test scores. Using these estimates, we calculate the share of demographic test score gaps explained by relatability.

The core motivating concern is that two students of equal ability but different demographic backgrounds may receive different test scores. This arises from an interaction between a student’s characteristics and passage text attributes, driven by content the exam designer is not intending to screen for. We focus on multiple factors composing this systematic student-passage “match,” such as personal interest in passage topics, familiarity with those topics, or shared identity with passage characters. Topic relatability corresponds to both familiarity and interest factors, whereas identity relatability captures demographic representativeness of a passage.

The estimation data cover the universe of 3rd to 8th grade public schools students from 2013 to 2019 in the U.S. state of Texas. The administrative data contain reading comprehension exam responses for every student and test question. Questions are directly linked to reading passage text, providing passage-level test performance for each student. We also observe the race and gender of every student, which we use to build our relatability measures. We supplement this testing data with detailed time-use information from the American Time Use Survey (ATUS) and name-demographic group databases from the U.S. Social Security Administration (SSA) and state voter registration files.

Our topic relatability measure is constructed by combining demographic-level exposure to topics with the salience of topics in a passage. We start by selecting a set of leisure activities from the ATUS and group them into topic categories (e.g. basketball, soccer). Topic exposure for a race or gender group is the share of ATUS respondents in that group who spend any time on activities related to that topic. The salience of a topic in a passage is the share of the passage’s words directly related to the topic. We join topic exposure and topic salience together to create the demographic-passage level topic relatability measure. To obtain the casual effect of topic relatability on test scores, we isolate variation in relatability that is as-good-as-random. This leads directly to our shift-share empirical strategy, which argues that while topic exposure is endogenous (the “shares”), only topic salience (the

“shifts”) needs to be quasi-random subject to appropriate controls (see Borusyak, Hull, and Jaravel 2022). For example, we acknowledge that topical exposure to soccer across races is not random, but it is random whether Black fourth graders see a basketball passage in one year and a soccer passage in another.

We find that topic relatability of a reading comprehension passage for race causally raises student performance on questions connected to the passage. A standard deviation increase in race topic relatability in a passage leads to a 1.9 percentage point increase in the share of correct answers on that passage. This is equivalent to a 0.05 standard deviation increase in test scores from a standard deviation increase in exam-level relatability. While these effect sizes are small in absolute magnitude, they are moderately sized in comparison to other factors on test outcomes.³ Race topic relatability effects are strongest for lower-achievement students. We also find suggestive evidence that race topic relatability effects are strongest for students in more racially homogenous schools—those who are less likely to be exposed to a diverse set of topics. Contrary to the race topic relatability results, we find that gender topic relatability is not predictive of test scores. Given that cross-gender exposure is relatively higher than cross-racial exposure, the school and gender results highlight the potential importance of peer effects on mediating the impact of topic relatability. Our results are robust to alternative topic relatability measures, including using time-use data for children to construct the measure.

We supplement the topic relatability results by assessing the impact of identity relatability. We define identity relatability to be the share of characters matching the student’s race or gender. To construct this measure, we first use a large language model (LLM) to identify all characters in a text. We predict race and gender for each character: for race, this is done by linking first and last names to SSA data; for gender, the LLM imputes gender based on pronouns or titles. The results for both race and gender are quite similar. For race, we show moving from zero own-race characters to all own-race characters results in a 1.0pp change for a student’s test performance. For gender, we show moving from all opposite to all same gender characters yields a 0.7pp change in test performance. These results are robust to using different LLMs, different imputation methods for race and gender, and weighting characters by number of mentions when calculating the race or gender share.

We consider the total combined effect of relatability across our two measures. To fill the gaps left by using our structured measures of relatability, we invite respondents on an online survey platform to provide an unstructured assessment of passage relatability. This flexible survey-based measure is positively associated with identity relatability and is predictive of

³For instance, Chetty, Friedman, and Rockoff (2014) find that a standard deviation increase in teacher value-added raises English test scores by 0.1 standard deviations.

test scores. Estimating the test score impacts of topic relatability, identity relatability, and survey-based measures in one regression, we find the topic and identity relatability measures maintain their effect sizes, suggesting that these effects are separate from one another.

We investigate the extent to which content relatability contributes to disparities in test outcomes across demographic groups, focusing particularly on differences across race. We find that simply setting topic and identity relatability to be the same for all racial groups would lead to a 9% smaller Black-White test gap and a 10% smaller Hispanic-White test gap. Around one-third of the estimated topic relatability effect on test gaps is due to the selection of more White-relatable topics in our sample of passages. Results on average test score differences mask the role of test score thresholds in exacerbating racial disparities. We counterfactually suppose how each student’s state-determined reading performance category would change if they had received a more relatable exam from our sample. We find almost 11,000 Black students and almost 37,000 Hispanic students could have achieved a higher reading comprehension standard if relatability had been more equalized across groups.

We contribute to a literature documenting and explaining demographic gaps in educational outcomes, a core question in the economics of education. One strand of this literature examines race, asking what covariates might explain the gap that begins in very early childhood (Fryer & Leavitt 2004; Fryer & Leavitt 2013) and examines potential sensitivity of group differences in test scores due to scaling decisions (Bond & Lang, 2013; Bond & Lang, 2018; Nielsen, 2023). The relationship between segregation and the SAT racial test gap is examined in Card & Rothstein (2007), opening questions about peer effects which we address in our paper. Another strand of this literature examines gender, documenting trends of female students performing better than male students on reading exams—this is consistent across geography (Pope & Sydnor 2010), present within socio-economic status (Cobb-Clark & Moschion 2017), and robust to many other detailed controls (Lundberg 2020). Recent work by Brown et al. (2022) also considers the role of cognitive endurance in explaining the socioeconomic test score gap. As these papers have documented many patterns that explain variation in the demographic test gaps in question, we propose another variable that may improve accurate measurement of racial or gendered testing gaps. In particular, we consider whether test content itself may affect estimated gaps on exams administered with real-world stakes.⁴

The literature has also explored the representation of race, culture, gender in educational materials, and how this might shape student success. Recent papers have documented es-

⁴There is evidence suggesting the importance of test setting when estimating performance gaps: Ofek-Shanny (2024) demonstrates with a field experiment with Israeli students that estimation of gaps is more credible on higher-stakes exams. This motivates our decision to use real-world exams with consequences in this paper.

sential facts about race and gender learning materials, such as the under-representation of certain identities using text analysis and computer vision tools (Adukia et al., 2023; Lucy et al., 2020) or developing novel survey methods for detecting stereotypes (Baldazzi et al., 2025). Furthermore, researchers find that racial or ethnic coded content in educational text can have an impact on students’ outcomes and beliefs (Dee & Penner, 2017; Cantoni et al., 2017). Other papers study exams directly. Dobrescu et al. (2021) uses an experiment varying the cultural context in a standardized test in Australia, while Dee and Domingue (2021) test the theory of stereotype threat from Steele and Aronson (1995) in practice by looking at the impact of a culturally insensitive test question in the state of Massachusetts. For gender, Good et al. (2020) finds symmetric gender test effects when educational materials feature a gender identity-match, but Cohen et al. (2023) finds gender-neutral language only improves women’s performance. Finally, in a study closely related to our own, Duquenois (2022) finds that students of low socioeconomic status do worse on monetarily themed math practice and test questions.

We add to this literature in several ways. First, we construct a rich measure of reliability that explores both demographic dimensions of race and gender. This measure not only captures widely studied features such as identity representation, but also considers less prominent features such as the differential topic familiarity and interest across demographic groups. Second, we obtain our estimates from a large-scale standardized exam, providing us a direct measure of the impact of educational material on a student outcome with real-world stakes. Finally, we propose an innovative identification strategy suited to large-scale exams administered uniformly on a population, whereas other approaches rely on random assignment of exam booklets or focus on the impact of one test item out of many.

Lastly, our paper is complementary to the extensive psychometrics literature on differential item functioning (DIF) (see Zumbo (1999) for an overview). DIF models use exam outcomes to detect for item- or test-level differential performance across student characteristics, conditional on ability. However, distinguishing between benign and adverse differential item functioning is difficult (Douglas et al. 1996). There are no unambiguous quantitative methods for interpreting whether detected differences constitute bias, and thus it is left to test makers to look at the *ex post* exam *outcomes* to ask whether a test item flagged by a DIF model has desirable or undesirable properties. We make progress on this by developing two transparent, structured measures that indicate potential sources of bias based on the *ex ante* exam *content*. This allows test makers, in principle, to tailor exams to their institutional objectives prior to field testing or administration. Further, our approach can capture differential impacts across groups that might fall below conventional DIF thresholds.

We organize the rest of the paper as follows. Section 2 builds a simple conceptual

framework which drives our estimation. Section 3 describes the student test data and the time-use data. Section 4 discusses the topic relatability estimation strategy and results. Section 5 describes the identity relatability estimation strategy and results. We bring the topic relatability and identity relatability results together in Section 6. Section 7 extends our results to how relatability differentially affects students. Finally, section 8 concludes.

2 Conceptual Framework

Essential to our exercise is distinguishing between a student’s ability and the student’s relatability with the text. We present a simple model which illustrates how relatability influences test scores, followed by a practical discussion about what factors may affect relatability.

2.1 Model

The primary objective of exams is to measure the ability and progress of students. However, the signal observed via standardized testing may be a function of learning *and* other factors that the testing administrator may not want to consider. For example, consider a biographical excerpt about a sailor. Comparing two students who have identical reading comprehension ability but differ in exposure to boating and the sea, we may not be surprised to find that the student with greater exposure to the sea performs better on questions regarding this text passage. The passage topic may help or hinder the ability of students to infer the meaning of vocabulary words or identify the main arguments of the passage. Further, if reading takes mental effort, perhaps the cognitive costs decrease in topical familiarity. If this difference is systematic across demographic groups, this may affect the signals test administrators receive.

To formalize this idea, consider a model of passage-level, student testing outcomes given by

$$y_{ip} = \theta_i + \phi_p + \rho_{ip} \tag{1}$$

where i indexes a student and p indexes passages.⁵ Student performance is determined by three factors in this model: θ_i , individual student ability; ϕ_p , general passage difficulty; and ρ_{ip} , a passage-individual-specific term. Now, we parse ρ_{ip} into two parts: one that represents

⁵For ease of exposition, we omit from the model the possibility that each passage constitutes multiple questions, as is the case in most exams.

systematic variation, and the other that is idiosyncratic and uncorrelated with θ_i . That is,

$$\rho_{ip} = \underbrace{\vec{\varepsilon}_i' B \vec{\mu}_p}_{\substack{\text{systematic} \\ \text{or "reliability"}}} + \underbrace{\nu_{ip}}_{\text{idiosyncratic}},$$

if we model the systematic portion as linear interactions between some observed vector of student characteristics $\vec{\varepsilon}_i$ and an observed vector of passage characteristics $\vec{\mu}_p$. We assume that both $\vec{\varepsilon}_i$ and $\vec{\mu}_p$ are of dimension $T \times 1$, and B is $T \times T$. In the pursuit of tractability, we impose some additional assumptions such that B is zero off the diagonal.

Whereas educators only observe y_{ip} , they use it to draw inferences about individual student learning. For instance, educators will traditionally set $\tilde{\theta}_i \equiv \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} y_{ip}$ as their estimate of student ability, where $\mathcal{P}$ is a set of passages (e.g., the entirety of a single reading comprehension exam). Educators may also use test results to compare learning-levels across groups, whether that be classrooms, schools, regions, or demographic groups. It follows that educators may set $\tilde{\theta}_d \equiv \frac{1}{N_d} \sum_{i \in d} \tilde{\theta}_i$ as the group-level performance indicator for some group d (where N_d is number of students in group d). Then, $\tilde{\theta}_d - \tilde{\theta}_{d'}$ is the difference, or gap, between groups d and d' .

The testing outcomes model we propose suggests these simple estimates of student- and group-level ability based only on y_{ip} may be biased due to ϕ_p and the systematic components of ρ_{ip} ($\vec{\varepsilon}_i$ and $\vec{\mu}_p$). Bias due to ϕ_p is typically not an issue: If test administrators give the same passages to all students in a given grade and school year—as is often the case—there is no bias. The potential correlation between components of $\vec{\varepsilon}_i$ and θ_i , on the other hand, poses a threat to interpreting standardized test outcomes, $\tilde{\theta}$. Consider again the prior example of a biographical reading passage about a sailor. We represent the presence of sea and nautical themes in this passage as an element of $\vec{\mu}_p$, such that $\mu_{p,sea} = 1$. Student exposure to the sea and nautical activities may similarly be a component of $\vec{\varepsilon}_i$, thus student i has $\varepsilon_{i,sea} = 1$ while student i' has $\varepsilon_{i',sea} = 0$. Then, our model suggests a systematic wedge in observed performance between student i and i' of $\varepsilon_{i,sea} B_{sea,sea} \mu_{p,sea} - \varepsilon_{i',sea} B_{sea,sea} \mu_{p,sea} = B_{sea,sea}$. If we consider that attributes like exposure to topics are correlated with demographic groups, such as race or income, then group-level conclusions are also affected by this issue.⁶

This model framework and its implications drive our definition of content reliability and our estimation strategy. We refer to the interaction between $\vec{\varepsilon}_i$ and $\vec{\mu}_p$ as content reliability. We limit our focus to attributes of the text corresponding with elements of $\vec{\varepsilon}_i$ and $\vec{\mu}_p$ which are arguably outside the scope of evaluation for reading comprehension exam makers. In our

⁶Educational psychology research suggests indeed that student interests do diverge by demographic factors and they can meaningfully affect student test performance (Bray & Barron, 2004; Asher, 1979).

setting, this includes selecting topics which students are differentially exposed to that are orthogonal to reading comprehension such as sports or arts and crafts. This also includes the names of characters in the text which may evoke different relatability across students. Second, while we acknowledge the potential for variation in the presence of these topics to cause differential performance at non-demographic group levels (e.g., baking-interested students compared to their counterparts), we focus on how test attributes change test scores at the race/ethnicity-and gender-levels which is of key interest to educators and researchers. Given our interest in race- and gender-level estimation, we use race- and gender-level data when constructing estimates of elements of $\vec{\epsilon}_i$. We are also interested in defining different notions of how to design “fairer” tests with respect to this model and our findings. We return to this question in section 7.

2.2 Components of content relatability

We conceptualize content relatability to include three components: interest, familiarity, and identity. Each of these components can be represented as a “match” between student characteristics $\vec{\epsilon}_i$ and passage characteristics $\vec{\mu}_p$. For example, included in $\vec{\epsilon}_i$ is student i ’s interest in baking while $\vec{\mu}_p$ includes the degree to which baking appears in passage p . Similarly, $\vec{\epsilon}_i$ may include a term representing a student’s identity, with an analogous term in $\vec{\mu}_p$ representing whether characters with that identity appear in passage p . We collapse familiarity and interest in creating and defining our “topic relatability” measure (Section 4). We then propose a separate measure of “identity relatability” (Section 5).

Topic relatability is straightforwardly a function of personal interest in a topic, such as pets. For example, a student with high interest in a passage’s topic may find it easier to focus. However, a student with no interest in a topic may still be quite *familiar* with a topic if it is popular among her family members or others in her community. Even given no interest in football, she may still know football-specific terms and concepts if her siblings or parents are fans. A carefully designed laboratory experiment would be necessary to credibly disentangle these concepts and their mechanisms, and is thus outside of the scope of this paper, which is focused on quantifying these effects in observational data. There is a well-developed educational psychology literature that directly speaks to these ideas of familiarity (Singer and Alexander, 2016; Johnston and Pearson, 1982; Davey and Kapnius, 1985; Norris et al., 2003) and interest (Norris et al., 2003; Shirey and Reynolds, 1988; Schraw et al., 1995) as it relates to reading comprehension. Given the close relationship between interest and familiarity, we opt to collapse these concepts together into one measure. Then through our empirical strategy, we seek a representation of each student’s topical exposure (interest

and/or familiarity) and each exam passage’s topical salience (presence of topics in the text).

Similarly to topic relatability, a student who *identifies* with characters in a text would have high identity relatability. In this paper, we focus on identity relatability which arises from a student identifying with same-race or same-gender characters in the text. Identification with a character need not be limited to race and gender—for example, a student may identify with a character that is shy or the oldest sibling. Nonetheless, we restrict our attention to race and gender since they are relatively observable characteristics in both the administrative data and the passage text and they have been the focus of recent scholarly work on identity representation in educational material (Lucy et al., 2020; Adukia et al., 2023). Similar to our empirical strategy with topic relatability, we seek data on student demographic characteristics and passage character characteristics which interact to form identity relatability.

3 Data

3.1 Texas student and assessments data

Our primary dataset is from the Texas Education Agency (TEA), which administers standardized assessments to students in the state of Texas in the United States. We study the State of Texas Assessments of Academic Readiness (STAAR), mandatory end-of-year assessments taken by public school students in grades 3–8. Students in a common grade or common course receive the same test in a given year, barring two exceptions: the TEA offers Spanish-language tests and alternative tests for students with cognitive limitations. We only consider students who take the standard, English-language exam. To avoid concerns regarding COVID-19’s effect on education, we limit our attention to 2013–2019 reading comprehension exams. The reading exam format is standard across grades. Students read four to six text passages and answer multiple-choice questions regarding each passage, including the content, vocabulary used, and grammar.

We define a binary outcome measure using the item-level student responses on whether or not a student answered a question correctly. We match each item to its corresponding reading comprehension passage. If a set of items is associated with two separate pieces of text, both texts are combined and considered to be one passage for analysis purposes. The item-level data also include the TEA-designated reading standard associated with each item, which we use to define the genre of each passage. Our final dataset includes student responses and passage characteristics for 205 unique passages from 42 exams.⁷

⁷Seven out of 212 reading passages are unavailable due to copyright restrictions. Responses associated

Table 1: Summary statistics for student sample

	Mean	SD
Demographics		
Hispanic	0.46	
Non-Hispanic		
Asian	0.05	
Black	0.15	
White	0.34	
Female	0.49	
Test		
Passages	5.05	0.44
Test items	43.17	5.32
Score (0–1)	0.67	0.20
<i>N</i>	13,180,138	

Notes: The table displays sample means and standard deviations of characteristics in the student sample. The sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

We supplement the testing data with student demographic information, including their reported race, ethnicity, and sex. From the demographic information, we create four racial/ethnic categorizations: Asian, Black, Hispanic, and White. We remove from our sample students reporting as another race or multiple races, due to small sample sizes or limited ability to link them to external data. All Hispanic-identifying students are categorized as Hispanic, and the remainder of students are categorized according to their reported race. For brevity, we use “race” to mean both race and ethnicity for the remainder of this paper. Each student’s free- or reduced-price lunch status and school attended are also used in supplementary analysis.

Our analysis sample consists of 13,180,138 student-exam observations (henceforth, just called “students”). Table 1 gives a summary of this student population. The plurality of students in the sample are Hispanic (46%), much higher than the U.S. average.⁸ Among non-Hispanic students, the majority are white (34% of total) compared to Black (15%) or Asian (5%). As expected, students are roughly split by sex. Most exams contain 5 passages, corresponding to just over 43 test items. Students answer roughly two-thirds of these questions correctly.

with these test passages are removed from the analysis.

⁸According to U.S. Census estimates for 2019, 20% of individuals aged 5 to 19 were of Hispanic origin.

3.2 Time-use data

3.2.1 American Time Use Survey

We use the American Time Use Survey (ATUS) to construct measures of topic exposure at the demographic level. ATUS, which is sponsored by the Bureau of Labor Statistics and conducted by the U.S. Census Bureau, is a nationally representative survey which provides estimates on how Americans, 15 years and older, spend their time. Randomly selected individuals from a subset of households in the Current Population Survey report a diary of the prior day’s activities to a phone interviewer. The interviewer assigns each activity to one of 442 six-digit classification codes, nested within 17 major categories. Each code reflects 3 tiers of detail regarding the activity. For example, the activity of “sewing” would be classified as code 020103, with 02 representing the major category of “Household Activities,” 0201 representing an activity group of “Housework” within “Household Activities,” and 020103 representing the activity of “Sewing, repairing, and maintaining textiles.”

We reduce the dimensionality of the detailed activity data. We exclude codes unrelated to leisure or those reflecting life-maintenance activities performed by virtually all individuals. Appendix Table A1 summarizes the excluded activity codes. Although we exclude almost two-thirds of codes, most excluded minutes come from common activities such as sleeping, working, and eating, which account for 68% of excluded minutes. Next, we classify the remaining 140 activity codes into 24 mutually exclusive “topics,” including categories such as arts & crafts, animals, soccer, and winter sports (see Appendix Table A2 for examples). A detailed mapping of activity codes to topics is available in the online code repository.

Our ATUS sample includes all respondents from 2013–2019, the same time period as our student data. For each respondent, we observe minutes spent on each ATUS activity, as well as their race, household size, household income range, and other demographic characteristics. We follow the same the racial/ethnic categorization in this data as we do for the student data, by excluding multi-race respondents and categorizing Hispanic respondents into their own category.

Table 2 describes the individual time diaries and demographics of our 73,626 respondents. Each respondent is weighted by sampling weights provided by the U.S. Census Bureau. On average, a respondent reports 11 activities in their diary, spending 146 minutes per activity.⁹ Further, a respondent’s daily activities typically correspond to two to three topics. Activities in these topics correspond to 20% of an average respondent’s time. The ATUS sample is

⁹Dividing the number of minutes in a day by the average number of reported activities does not yield 146 minutes per activity. This occurs because the ratio of averages (average number of minutes divided by average number of activities) is not equivalent to the average of ratios (averaging minutes per activity across respondents).

Table 2: Summary statistics for ATUS sample

	Mean	SD	5%	95%
Activities				
Number of activities	11.34	4.22	5	19
Number of min spent per done activity	145.96	63.05	76	286
Number of topics	2.32	1.22	1	4
Share of time spent in topics	0.20	0.15	0.00	0.48
Demographics				
Asian	0.05			
Black	0.12			
Hispanic	0.16			
White	0.66			
Female	0.52			
<i>N</i>	73,626			

Notes: The table displays sample means, standard deviations, and the 5th/95th percentile value for each category. Respondent observations are weighted by ATUS sampling weights provided by the U.S. Census Bureau. The sample includes all ATUS respondents from 2013–2019.

predominantly non-Hispanic white, making up 66% of the sample compared to just 34% for the student sample. Much of this difference is due to disproportionately higher Hispanic population in the student sample. The share of Asian and Black respondents is very similar to the shares in the student sample.

For the main analyses in this paper, we opt to use data from all respondents in the ATUS sample period, which includes older adults and respondents outside of Texas. Incorporating data from a large sample reduces the potential for measurement error in the relatability measure biasing coefficient estimates. However, this larger group comes at the expense of the representativeness of the ATUS sample to the estimation sample: children residing in Texas. To address this concern, we consider more “relevant” ATUS subsamples based on age, household composition, and geography. We later consider children’s time use data from the Child Development Supplement of the Panel Study of Income Dynamics (PSID-CDS). More details on the subsamples and PSID-CDS data can be found in Appendix Section B.

3.2.2 Constructing demographic topic exposure

Using the ATUS sample, we construct a measure for each demographic group of expected exposure to topics. The key challenge is that we cannot directly observe test-takers’ interests or familiarity-levels. We proxy for this using reported time use across race in the ATUS data to construct measures of topic exposure. Although most ATUS respondents are not school-

aged, children are likely exposed to activities of adults in their households (e.g. parents, siblings) (Bandura 1977; Bisin and Verdier 2001).

Formally, we define topic exposure $e_{d,t}$ as the share of respondents in demographic group $d \in \mathcal{D}$ reporting any activity related to topic $t \in \mathcal{T}$, weighting respondents by ATUS sampling weights. We demonstrate differential exposure to activities across race in the ATUS data. In Figure 1, we present ratios of topic exposure by demographic group. We observe wide divergence in exposure to activities by race and gender. For example, in Panel A, Black respondents are significantly less likely than white respondents to have done a water sports-related activity, but much likelier to have participated in basketball. Hispanic respondents are not as exposed to winter sports, but much more likely to be exposed to soccer. We also observe that the topics to which white respondents have relatively higher exposure differ across Black and Hispanic respondents. In Panel B, we see that male respondents are much likelier to report participating in nature sports or football, while female respondents are much likelier to report participating in arts and crafts and animal sports. We provide the raw exposure ratios for all topics in Table A3.

3.3 Passage text data

We extract the passage text from publicly available PDFs of the STAAR exams. This corpus of text will allow us to generate measures of “topic salience” for use in topic relatability (Section 4) and construct identity relatability (Section 5). The topic salience measure and the categorization of character identity correspond to $\vec{\mu}_p$ in the conceptual framework: a set of passage characteristics that interacts with student characteristics to form relatability. We describe the construction of topic salience below.

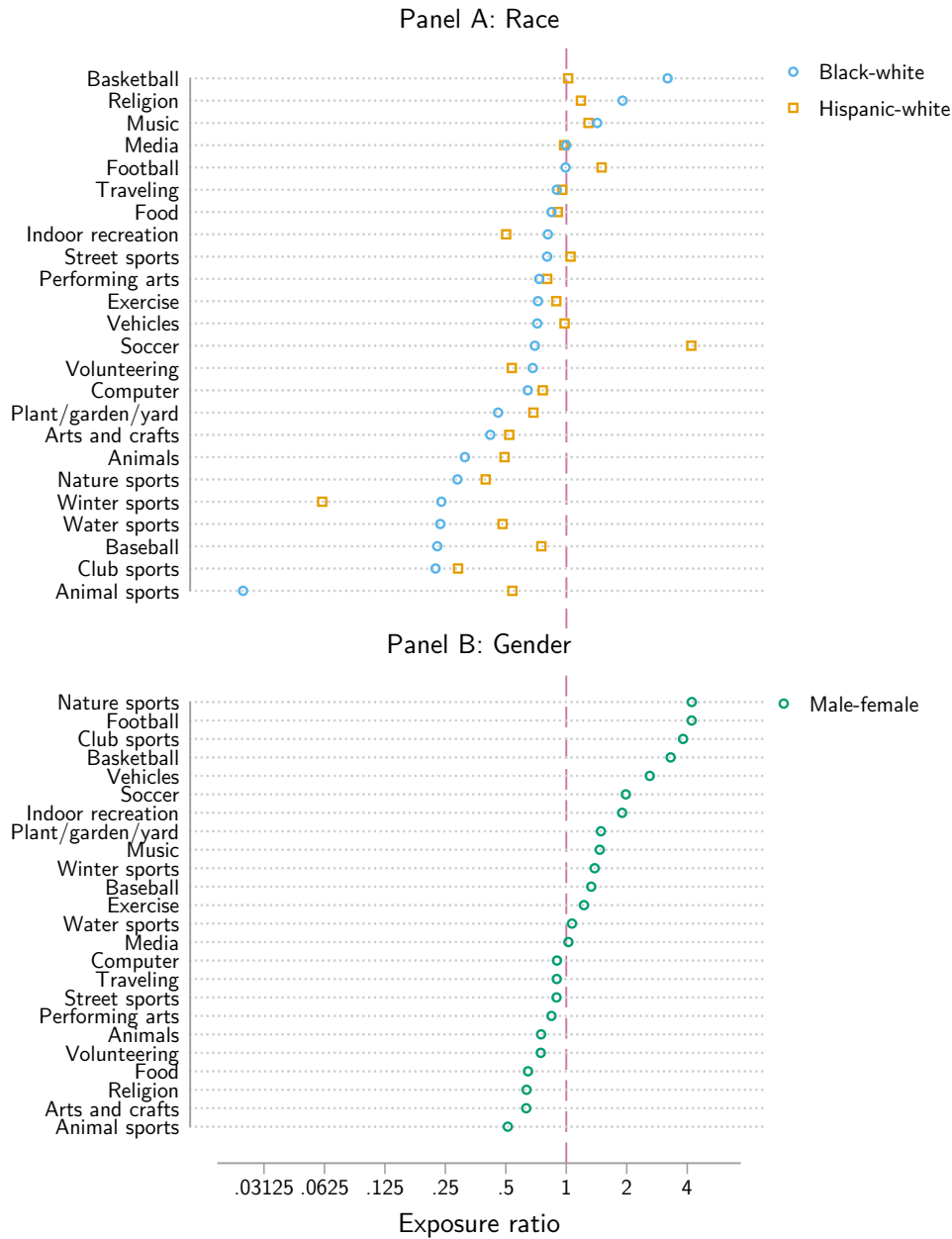
We score the content of each reading passage using an intuitive “dictionary method” standard in the natural language processing literature. For each topic-passage pair, we measure how salient a topic is in a specific passage. The algorithm operates as follows. First, determine a set of words which indicate the presence of a topic: a set B_t for each topic $t \in \mathcal{T}$. Second, for each passage $p \in \mathcal{P}$, calculate the share of words in a passage in a given topic dictionary as our measure of topic salience. This is commonly referred to as “term frequency” in NLP analysis.

We create a dictionary B_t for each topic. After grouping ATUS activities into the topics in $\mathcal{T}$, the authors separately listed as many terms related to the activities within a topic for each topic $t \in \mathcal{T}$.¹⁰ We then construct $B_t = B_{t,1} \cup B_{t,2}$.¹¹ On average, each topic’s dictionary

¹⁰The authors created two dictionaries $B_{t,1}$ and $B_{t,2}$ prior to any analysis and have not edited the dictionaries since

¹¹Dictionaries are provided in the online code repository.

Figure 1: Differences in topic exposure by race and gender



Notes: Graph plots the ratio of topic exposures between demographic groups. Panel A plots the ratio between Black and white respondents and the ratio between Hispanic and white respondents. Topics are ordered by the Black-white exposure ratio. Panel B plots the ratio between male and female respondents. Topics are ordered by the male-female exposure ratio. The vertical dotted line at 1 corresponds to the value at which both groups have identical exposure. Topic exposure is calculated separately for each racial group using the ATUS data. Topic exposure is the share of ATUS diaries for a demographic group reporting participating in any activity for a topic. The sample includes all ATUS respondents from 2013-2019.

has 29.7 words. To prepare the reading passages and dictionary terms for analysis, we follow standard steps for text data processing. This includes converting all words to lowercase, removing numbers, and removing stopwords (common words with no informational content in isolation). Our baseline specification also converts all words to their “stem,” such as converting *dogs* to *dog* or *running* to *run*.

Our baseline measure of topic salience, denoted $m_{t,p}$, is simply the share of words in a passage p that are in the dictionary B_t . In the distribution of this score displayed in Figure A1, the modal value is zero and is heavily right-skewed. We also consider other measures of topic salience and textual data cleaning, discussed further in Appendix Section B.2.

Considering a binary threshold of $m_{t,p} \geq 0.01$, we find the average passage contains 3.8 topics; further, a threshold of 0.02 suggests the average passage contains 1.6 topics. A majority (53%) of topic-passages are non-zero-valued, with the average passage having 12.6 (out of 24) non-zero topics. Histograms illustrating how many topics appear in a passage using each of these thresholds are found in Figure A2. We also present the frequency which a topic appears in the top 3 matches for any passage using our dictionary-based method in Figure 2. This chart indicates that that many passages relate to nature sports (hiking, climbing, fishing, etc); arts and crafts; animals; plants, gardening, and yards; music; and water sports (swimming, boating, etc). Lastly, we validate our topic salience scores with human coders to manually label the reading passages, which demonstrates that our dictionary-based scores are capturing the intended variation (see Appendix Section C).

4 Topic relatability estimation and results

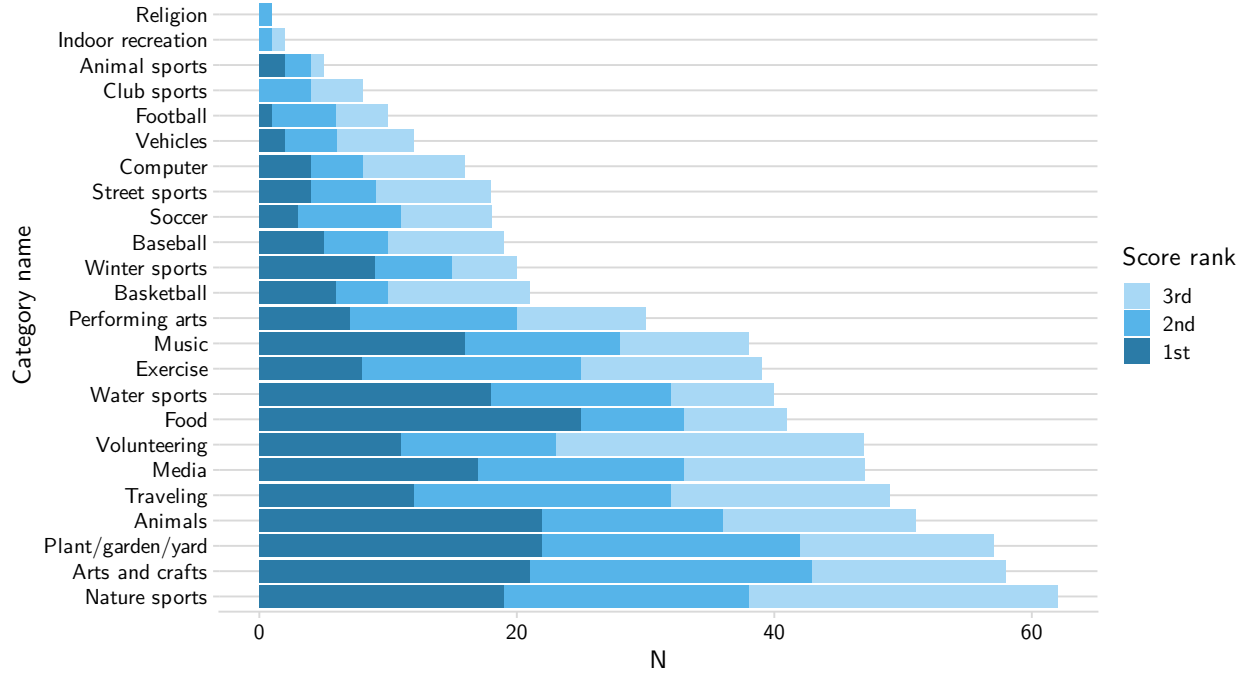
4.1 Constructing topic relatability

We construct a measure of topic relatability by joining the topic exposure measure $e_{d,t}$ described in Section 3.2 and the topic salience measure $m_{t,p}$ described in Section 3.3. We define relatability for demographic group d —defined over either race or gender—and passage p as

$$r_{d,p} = \sum_{t \in \mathcal{T}} e_{d,t} m_{t,p}, \quad (2)$$

where $e_{d,t}$ is the topic exposure of demographic group d to topic t and $m_{t,p}$ is the topic salience of topic t in passage p . This definition implies the following: (a) only the salience of topic t loads on exposure to t ; (b) topic loadings are homogeneous across topics; and (c) loadings are homogeneous across demographic groups and passages. Intuitively, $r_{d,p}$ is a weighted

Figure 2: Most frequent topics in STAAR reading passages



Notes: Reading passage measures are calculated for each topic-passage pair by the term-frequency metric discussed in Section 3.3. Keeping the three topics with the highest score for each passage, this histogram shows how frequently each topic is detected in the passages. The sample of passages include grade 3 to 8 STAAR reading comprehension tests from 2013–2019.

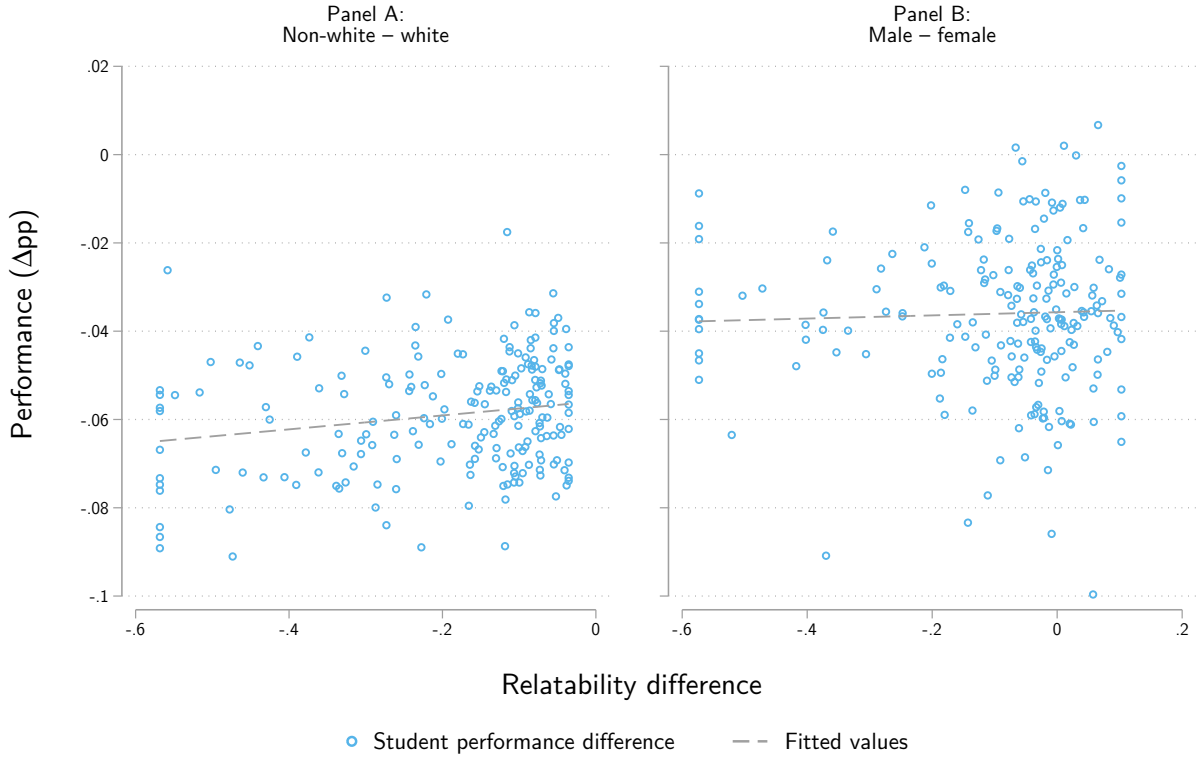
sum of topic salience for passage p weighted by the group d 's topic exposure. Finally, we standardize $r_{d,p}$ to unit variance across the estimation sample at the student-item level.

To show preliminary evidence that our relatability measure is predictive of test performance differences between demographic groups, we plot passage-level differential relatability and differential exam outcomes for non-white versus white students and male versus female students in Figure 3. Panel A shows a positive relationship between race topic relatability differences and race test outcome differences, which is suggestive of an impact of topic relatability on racial test gaps. However, we observe a less clear relationship between relatability and male-female test score differences in Panel B. We proceed to a formalized test of the relationship between topic relatability and test performance.

4.2 Estimating the causal effect of topic relatability

Given our relatability definition, consider the following data generating process and how it produces our identifying variation. Consider a test maker who is responsible for creating a

Figure 3: Demographic-level test outcome differences and reliability



Notes: Each graph plots average student test outcome differences between non-white and white students against reliability differences and the simple linear fit between these measures. Reliability differences at the passage-level are taken after the reliability measure is standardized. Panel A plots non-white – white student differences while Panel B graph plots male-female student differences. Observations are at the passage-level. Differential reliability is winsorized at the 5th and 95th percentiles. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

3rd grade exam each year which evaluates student competency on a fixed rubric. Due to state-mandated standards for third grade, she may need to include a poem, two fiction prose and two non-fiction prose passages each year with some fraction of vocabulary questions and comprehension questions. While she assembles five passages designed to appeal to third graders, each year there is some amount of residual differences in the reliability of the topics to the student population, which may differ by racial background. Thus, while the topic exposure levels of students are non-random, the topic salience measures within a grade do have an element of randomness across exam years (2013 to 2019, within a grade) and even passages within an exam. We then isolate this residual variation in our regression specification by selecting fixed effects which remove the mean expected reliability score for each demographic group at each grade-level. This approach follows a “shift-share” empirical

strategy, in which we leverage quasi-random variation in shifts (topic salience) conditional on endogenous shares (topic exposure).

Formally, our estimation strategy relies on conditional randomness of topic salience to identify the causal effect of content relatability on student performance. To illustrate this, consider a group-aggregated, potential outcomes model based on the conceptual model described in equation (1),

$$Y_d(p) = \bar{\theta}_d + \phi_p + \beta r_{dp} + \nu_{dp}, \quad (3)$$

where d indexes demographic groups and p indexes passages. $Y_d(p)$ is the share of questions answered correctly by demographic group d when faced with passage p . $\bar{\theta}_d$ represents passage-invariant, fixed test performance for the demographic group and ϕ_p represents fixed passage difficulty for all students. Finally, β is the impact of content relatability $r_{dp} = \sum_t e_{dt} m_{tp}$ on student performance.

Our goal is to identify β in equation (3) using the variation in topics across passages observed in our student testing data. The baseline estimation strategy may be to assert that topics across passages are exogenous and simply estimate $\tilde{\beta}$ an OLS regression of Y_{dp} on r_{dp} . We face three main challenges with this approach. First, $\bar{\theta}_d$ and r_{dp} may be correlated. Suppose topic t' is more salient in passages than topic t'' , group d' has higher exposure to t' than d'' , and group d' has higher achievement than d'' (i.e., $m_{t'p} > m_{t''p}$; $e_{d't'} > e_{d''t'}$; and $\bar{\theta}_{d'} > \bar{\theta}_{d''}$). It follows that $r_{d'p}$ is generally higher than $r_{d''p}$. Then, the estimate $\tilde{\beta}$ is biased upward, confounding the effect of r_{dp} with the effect of $\bar{\theta}_d$. Issues still arise with a stronger assumption that topic salience m_{tp} is random across topics and passages. Suppose students with high achievement $\bar{\theta}_d$ are also exposed to many more topics (i.e., $\text{corr}(\bar{\theta}_d, \sum_t e_{dt}) > 0$). For such students, we expect r_{dp} to be higher than that of their peers due to elevated $\sum_t e_{dt}$, even assuming total randomness in topics across passages. Then the estimate $\tilde{\beta}$ is still confounded by the effect of $\bar{\theta}_d$. The second challenge is that ϕ_p may be correlated with r_{dp} . A passage characteristic such as passage length or passage genre may be associated with higher topic salience, but these characteristics also impact student performance directly. Finally, r_{dp} may be correlated with ν_{dp} . For example, we return to our earlier example with topics t' and t'' and groups d' and d'' . Suppose now that topic salience and test performance are divergent only at lower grades. Now there is unobserved variation in test performance across grades that is correlated with r_{dp} .

To account for these potential sources of endogeneity, we identify the impact of relatability on test performance by leveraging only the differences in topic salience (m_{tp}), after conditioning on: (a) grade-specific topic salience means for each topic and (b) passage-

specific topic salience means across topics. That is, we account for the possibility that each grade differs in most salient topics and each passage differs in aggregate ATUS topic presence intensity. We assume that this variation is as-good-as-random with respect to other components in the model. Then, given the structure of r_{dp} as a linear interaction between “endogenous” shares (e_{dt}) and exogenous shocks (m_{tp}), we adapt a “shift-share” estimation framework which allows us to estimate a causal effect by controlling for conditional expected relatability (Borusyak, Hull, and Jaravel 2022).¹² Since the topic exposure shares are fixed, we effectively perform the conditional expectation over the object which varies: the topic salience shocks, m_{tp} .

In our setting, conditional expected relatability is spanned by controlling for (a) demographic-grade fixed effects $\delta_{d,g(p)}$ and (b) passage fixed effects π_p interacted with $E_d = \sum_t e_{dt}$. To see (a), consider first that topic exposure e_{dt} is passage-invariant. Thus, expected relatability within a topic-grade is simply given by interacting passage-invariant exposure e_{dt} with average topic salience $\bar{m}_{tg(p)}$, that is, $\sum_t e_{dt} \bar{m}_{tg(p)}$. This value only varies at the demographic-grade level. To see (b), we note that expected relatability conditional on a given passage p is equivalent to the average topic salience for a passage interacted with exposure shares, since $\mathbb{E}[\sum_t e_{dt} m_{tp} | p] = \sum_t e_{dt} \bar{m}_p = \bar{m}_p \sum_t e_{dt}$. This value varies at the passage-exposure sum level.

We formally estimate the following regression specification

$$Y_{dp} = \delta_{d,g(p)} + \pi_p E_d + \beta r_{dp} + \nu_{dp}, \quad (4)$$

where $g(p)$ indexes the grade level of passage p and β is the coefficient of interest. $\delta_{d,g(p)}$ are demographic group-grade fixed effects, and $\pi_p E_d$ are a full set of passage-specific slopes on E_d . Since we have aggregated Y_{ip} to the group-passage level, we estimate the equation with weights representing the number of students and test items which make up (d, p) . We obtain “exposure-robust” standard errors for equation (4) by estimating an analogous topic-passage level regression, following Borusyak et al. (2022). Further, since our identifying variation comes from the presence of topics in a passage, we must allow clustering of standard errors within passage. In practice, we more flexibly allow clustering of standard errors within an exam. This approach is motivated by two factors: (a) passages are the smallest unit which the exogenous “treatment” varies and (b) passages are expected to be correlated with other passages within an exam (e.g., test makers are unlikely to include three poems in a five-

¹²While analyses that have units with a vector of differential exposure “shares” and a vector of common shifts typically feature the resulting exposure-weighted average of shocks serving as an instrument in a IV/2SLS analysis, Borusyak, Hull, and Jaravel (2022) explicitly note that identification assumptions follow through if such objects are used in “reduced form” analysis as is the case in our setting.

passage exam). Further details on the exact procedure used to estimate the topic-passage level regression and obtain standard errors are available in Appendix section D.

We illustrate that group-grade fixed effects $\delta_{d,g(p)}$ and passage fixed effects interacted with exposure sums $\pi_p E_d$ are sufficient for identification using a simple example with two topics $t \in \{\text{soccer, animals}\}$, two groups $d \in \{A, B\}$, and two grades $g \in \{3, 4\}$. Suppose first that group A has higher θ than group B and these differences are greater in grade 3 than in grade 4. If group A has higher exposure to soccer compared to group B and soccer is more likely to appear on a grade 3 exam, then θ predicts relatability, leading to bias in estimating β . However, $\delta_{d,g(p)}$ purges variation in relatability arising from differential topic salience across topics and grade, such that relatability is no longer predicted by θ . Now suppose that passage 1 has higher ϕ than passage 2 and also is more about soccer *and* animals than passage 2. Further suppose that group A has higher exposure to both soccer and animals compared to group B . As in the prior situation, ϕ predicts relatability and this biases the estimation of β . Nonetheless, we can directly account for the higher prevalence of topics in passage 1 by including $\pi_p E_d$. Crucially, if we simply included passage fixed effects, ϕ would still predict relatability through differential exposure sums between group A and group B .

The interpretation of β in equation (4) as the causal estimate of demographic topic relatability r_{dp} on test performance requires three assumptions. The first assumption is that topic salience shocks are assumed to be quasi-random, conditional on topic exposure shares and controls. The second assumption is that the shocks are numerous and conditionally uncorrelated with one another. The final assumption relates to construct validity: topic exposure differences across demographic groups d , such as race, are actually attributable to differences across d . We discuss each assumption separately.

Conditional quasi-random shock assignment Assuming the topic salience shocks are quasi-random is akin to an orthogonality assumption between relatability and the unobserved error term after including demographic group- and passage-based controls. We identify three ways such an assumption may be violated. First, test makers may incorporate differential performance across groups when building a test for a given year. If the resulting adjustment jointly changes attributes of passages and the topics within passages, then our main estimates may be biased. Second, test makers may adjust the distribution of topics in response to changing underlying demographics of test takers. For instance, a projected increase in the number of Hispanic students in Texas may lead to more Hispanic-relatable passages in tests and improved in-classroom instruction for Hispanic students. Third, topic selection may be correlated with observable and unobservable passage characteristics which also affect outcomes.

We assess the plausibility of this assumption through a falsification test regressing potential confounders on our relatability measure for race using equation (4). We define these confounders consistent with the three violation examples discussed above. To test for test maker responsiveness to prior-year performance, we compute one-year lagged average performance for each group-passage dyad dp , matched by passage position. Lagged performance is calculated by fixing grade or fixing cohort.¹³ This allows for flexibility in how test makers may respond to observing differential performance on a particular exam: They may seek to correct (or enhance) these differences in the same-grade exam in the next year, or they may correct the exam for the affected cohort of students in the next year. We then either aggregate lagged performance at the exam level or match lagged passage performance to dp by passage position. Next, we consider as a confounder the share of test takers for passage p that identify as group d . Finally, we consider passage characteristics as confounders: calendar year, passage position, word count, and passage category (literary text or informational text).

Table A4 shows the results of our falsification tests using these potential confounders. We fail to reject the null hypotheses that relatability is uncorrelated with 10 of 11 covariates at the 99% confidence level. The coefficient estimates of the lagged performance confounders are substantially lower in magnitude than that of our main regression and we fail to reject the null hypothesis that the coefficient on relatability is equal to zero. Similarly, the student population characteristics and most passage characteristics are not predicted by relatability. We do find relatability is positively associated with the literary passage indicator, and the joint F -test rejects the null that all covariates are jointly uncorrelated with relatability. At the same time, we fail to reject no-association for 10 of the 11 covariates at the 1% level and a “predicted outcome” regression using these covariates yields a coefficient on relatability that is statistically indistinguishable from zero. We also test sensitivity to the inclusion of passage category FEs in a later analysis and find that coefficients are largely stable.

Many uncorrelated shock residuals To test for having too few salience shocks or highly concentrated exposure shares, we summarize the distribution of topic salience and aggregated topic exposure, both at the topic-passage level. Table A5 reports these summary statistics. Column (1) displays statistics without controls and column (2) displays the distribution with our specified controls. We find that there is significant remaining variation in topic salience even after including our specified controls. Next, we consider an inverse Herfindahl index

¹³For example, consider Black students taking the 6th grade exam in 2017. Lagged performance *within grade* would correspond to performance for Black students taking the 6th grade exam in 2016. Lagged performance *within cohort* would correspond to performance for Black students taking the 5th grade in 2016.

(HHI) of aggregated exposure shares. This metric serves as a measure of “effective” sample size for our estimation sample—a low value would indicate a high degree of concentration in exposure shares to a handful of topics. We find an effective sample size of 925 across topics and passages.

As discussed previously, the main source of correlation across topic-passage shocks is within passages. Passages contain a finite number of words so the presence of one topic may “crowd out” another topic. Topics that are thematically similar to one another can also be positively correlated within passages. Beyond the passage-level, since passages are drawn together to build an exam, we may also be wary of within-exam correlation of topics. We take a conservative approach and allow for clustering within exams, but our results are similar with either clustering approach.

Exposure shares attributable to group-level differences A remaining concern is that the group-level topic exposure measures are not actually attributable to exposure differences across groups $d \in \mathcal{D}$, but to a different set of groups $c \in \mathcal{C}$. We take racial groups as an example. In our setting and more broadly, there is a close correlation between race and a number of factors associated with both educational outcomes *and* topical interest/familiarity. For example, we observe wide disparities in socioeconomic status (SES) (e.g. income, wealth, and employment) across racial groups. Individuals differ in educational attainment and neighborhoods across race. While these individual and household characteristics have direct and indirect effects on students’ educational outcomes, they also plausibly divide households by the topics that they are familiar with and interested in. Exposure differences across racial groups could then be driven by these factors and misattributed to race.

Misattribution of $\mathcal{C}$ -level differences to $\mathcal{D}$ -level differences affect our estimates in two ways. Differences across c may correlate with both exposure e_{dt} and test performance Y_{dp} . This confound is addressed by the assumption of quasi-random shock assignment conditional on race-grade fixed effects and passage-by-exposure-sum fixed effects. For example, if high SES students have exposure to topic t' and topic t' is more salient across passages, our estimation with race fixed effects accounts for this, as described earlier. If high SES students have higher exposure to all topics, the passage-exposure sum fixed effects account for this. Even still, we later perform SES- and school-level heterogeneity analysis and show that our main estimates remain largely unchanged accounting for these factors. Misattribution also affects the interpretation of our estimates. If race topic relatability is closely linked to SES topic relatability, then our findings may be about SES differences and not about racial differences. We find qualitatively that this is not likely to be the case. To demonstrate, we repeat the exercise in figure 1, replacing race and gender differences in topic exposure with SES

differences in topic exposure for Black and white respondents in the ATUS. Individuals with high SES are defined as those that are above the 200% poverty line.¹⁴ Figure A4 displays these exposure ratios. Panel A orders the topic by white SES exposure ratios. We see some alignment between SES differences across race. The key comparison is between Panel A and Panel B, where we display the same exposure ratios but ordered by the Black-white exposure ratio. The SES differences seem to only been weakly correlated with Black-white differences. However, we argue that even if our race or gender topic relatability measures are closely linked to topic relatability for other groups, our results can still meaningfully represent disparate impact of test passage choices across race and gender.

4.3 Results for race topic relatability

Our baseline results of regressing test outcomes on relatability at the racial group-level are shown in Table 3. Column (1) shows results with race fixed effects and column (2) shows results with race-grade fixed effects, corresponding to equation (4). We find in both specifications that topic relatability has a positive impact on test performance. In our preferred specification, a standard deviation increase in relatability for a passage leads to a 1.87 percentage point increase in the share of items answered correctly for that passage, an effect that is significant at the 99%-level. As is standard in the education literature, we rescale our effect size in terms of standard deviation (SD) changes in student performance and find that our main effect is equivalent to a 0.08 SD increase in passage-level performance. To standardize at the exam-level, we take into consideration that relatability variation is larger across passages than across exams. We find that a SD increase in average relatability at the exam-level predicts a 0.05 SD unit increase in exam-level student performance. Table 3 also reports the results of estimating the two-way fixed effects model most analogous to our preferred specification. To do so, we re-calculate race topic relatability, normalizing the topic exposures for a demographic group to sum to one. This purges all variation in race topic relatability that is driven by differences in aggregate levels of topic exposure across demographic groups. We find slightly smaller coefficients, but the differences are not statistically significant.¹⁵

¹⁴Details on how SES is determined in the ATUS sample can be found in Appendix B.

¹⁵While there are appealing estimation properties of directly employing a commonly used “difference-in-differences” estimation strategy to our original topic relatability measure, we argue that the identification assumptions necessary for TWFE estimation are not sensible our setting. Crucially, treatment in our setting does not turn “on” and “off” consistently for any unit or passage (time), since r_{dp} is a passage-varying mixture of shocks. As such, we also do not observe a unit which experiences a consistent level of relatability which might serve as part of an appropriate comparison group. Further, as discussed earlier, only including passage fixed effects would allow the coefficient estimate of r_{dp} to be confounded by differences in exposure sums across race.

Table 3: Impact of race topic relatability on test performance

	(1)	(2)	(3)	(4)
Race topic relatability	0.0199*** (0.0055)	0.0187*** (0.0058)	0.0188*** (0.0047)	0.0176*** (0.0049)
Race FE	✓		✓	
Race-Grade FE		✓		✓
Passage FE			✓	✓
Passage-by-Exposure sum FE	✓	✓		
N of topic-passages	4,920	4,920	4,920	4,920
N of student-passages	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on race topic relatability. Standard errors reported in parentheses below coefficient estimates are (a) obtained using a shock-level regression following Borusyak et al. (2022) and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. For columns (1) and (2), race topic relatability is calculated as described in section 4.1. For columns (3) and (4), race topic relatability is calculated after normalizing topic exposure to sum to one for each racial group. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

4.3.1 Test score gaps

While our main specification returns a coefficient that represents the average impact of relatability on test performance, it is not directly informative of how relatability may contribute to observed racial test score gaps. We observe large average test score differences between non-white and white students on reading comprehension tests. Table A6 displays the Black-white and Hispanic-white test gap by grade, calculated as the percentage point difference in the percentage of items answered correctly. We find that Black students in the 3rd grade have a 13.6 percentage point lower share of questions answered correctly on an exam than white students, a difference which shrinks to 11.7 percentage points by grade 8. The difference for Hispanic students is slightly smaller: between 9 and 11 percentage points across grades. These gaps are economically meaningful, equating to 16–19% and 13–15% of the white student mean for Black students and Hispanic students, respectively.

We calculate the change in test scores if topic relatability were equalized for all students by predicting $\hat{Y}_{dp}$ from equation (4) and calculating a counterfactual $\tilde{Y}_{dp}$ after setting $r_{dp} = 0$. These variables represent outcomes at average relatability and equalized relatability, respectively. We then generate conditional means of each predicted outcome by race: $\hat{\mu}_d$ for $\hat{Y}_{dp}$ and $\tilde{\mu}_d$ for $\tilde{Y}_{dp}$. For groups d and d' we can then compute the share of average test score

gaps explained by relatability:

$$1 - \frac{\tilde{\mu}_d - \tilde{\mu}_{d'}}{\hat{\mu}_d - \hat{\mu}_{d'}}. \quad (5)$$

Using expression (5) we estimate what our regression results would imply about test score gaps if topic salience, and ultimately relatability, were set to zero. We find that given lower average relatability of Black and Hispanic students compared to white students, racial test gaps are potentially smaller than can be detected using raw student performance. Topic relatability accounts for 4% of the test score gaps between Black and white students and between Hispanic and white students. Further, relatability is a larger contributor to test score gaps in early grades. In 3rd grade the Black-White and Hispanic-White share of test gaps explained are 5% and 6%, respectively, while in 8th grade that falls to 2%.

The implications of these findings could differ based on the source of variation in relatability. If the primary driver of relatability differences is due to differences across groups in levels of exposure, then test makers would have to modify the overall levels of topic salience to mitigate the impact of relatability in exams. Further, this would imply that a planner seeking to purge relatability heterogeneity from test score gaps would need to change student exposure levels directly. However, if a portion of relatability differences cannot be attributed to exposure levels, this implies that a part of the issue is being exacerbated by the selection of topics more relatable to one group over the other. We discuss these issues further in section 7.2.

4.3.2 Heterogeneity

Our main estimates mask potential heterogeneity in topic relatability’s effect within racial groups. Certain characteristics or environments may mitigate or exacerbate the effects of relatability. For instance, very high-achieving and very low-achieving students may both be unaffected by relatability as their reading comprehension abilities may be greater than the effect of relatable content. Race topic relatability may also be correlated with household income, meaning that accounting for household income may reduce our estimated coefficient on race topic relatability. Students could also mitigate the effect of topic relatability through greater exposure to diverse peers at school. While their interests may be driven by household and cultural factors, exposure to peers of other racial groups may allow them to relate to a wider variety of topics. We consider heterogeneity along each of these dimensions.

Achievement We explore how student achievement mediates the effect of topic relatability on test performance. Since our main outcome measure is reading comprehension test scores,

we seek alternative proxies of reading comprehension ability.

We consider two proxies for reading comprehension ability. Our preferred measure is a student’s math score, taken from the same year as the reading comprehension test. While math and reading comprehension exams do not test for the same skills, they are highly correlated in practice. We also consider a student’s reading score from the prior year (if available). We prefer the math score proxy as it maintains a full sample across grades—since there is no prior-year test for third graders—and does not remove students who are held back in school. We use within-exam deciles of both proxies in the analysis.

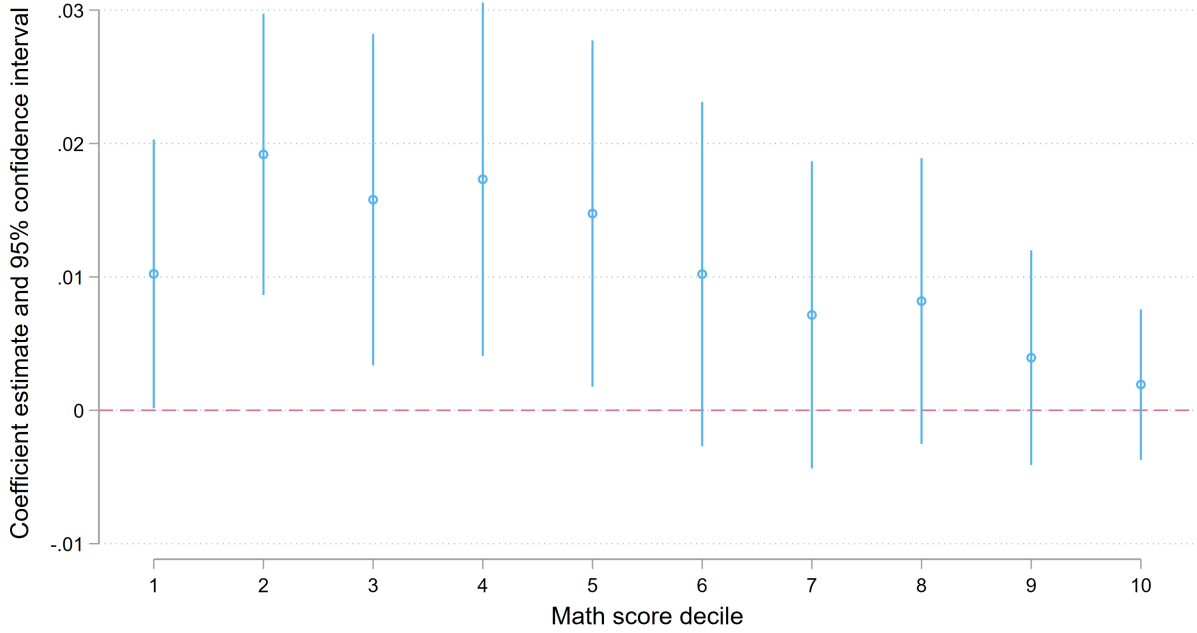
Figure 4 plots the coefficients of estimating equation (4) within each math score decile. We find that the coefficient on topic relatability is positive across math score deciles, but the effects are highest and statistically significant only for students in the bottom half of the achievement distribution. The point estimate at decile 2 is more than double that of the estimate at decile 7, and this difference is statistically significant, using a “stacked,” pooled specification. The difference in point estimates are also statistically significant for deciles 1 and 2, suggesting attenuation at both ends of the achievement spectrum. We find a similar pattern using past reading score as the achievement proxy, displayed in figure A3.

The achievement heterogeneity results are consistent with the interpretation that students with higher levels of reading comprehension or test-taking ability are better able to offset the influence of topic relatability on performance. This pattern can be particularly pronounced because the test performance measure is constructed from binary item responses: if students consistently answer nearly all items correctly (or incorrectly), the measure exhibits ceiling and floor effects, obscuring additional variation in performance at the tails of the ability distribution. For such high-achievement students, it may be appropriate to test the impact of relatability using exams with more difficult passages and harder questions.

These results also address potential concerns that student achievement differences across racial groups drive our main topic relatability findings. Ability may be directly or indirectly associated with exposure to certain topics. If high-ability students perform better on passages with such topics, it may be due to relatability or their reading comprehension ability. Given baseline test score differences across racial groups, this raises a concern that relatability is correlated with ability. However, our results demonstrate that topic relatability is still impactful when comparing students with similar levels of predicted achievement.

Socioeconomic status Race and SES are highly correlated in our student population. Over 56% of students in our sample are economically disadvantaged, defined as receiving free- or reduced-price lunch or being designated by the school to be experiencing economic disadvantage. Rates of economic disadvantage are much higher for Black (73%) and Hispanic

Figure 4: Heterogeneity of race topic relatability effects by math achievement

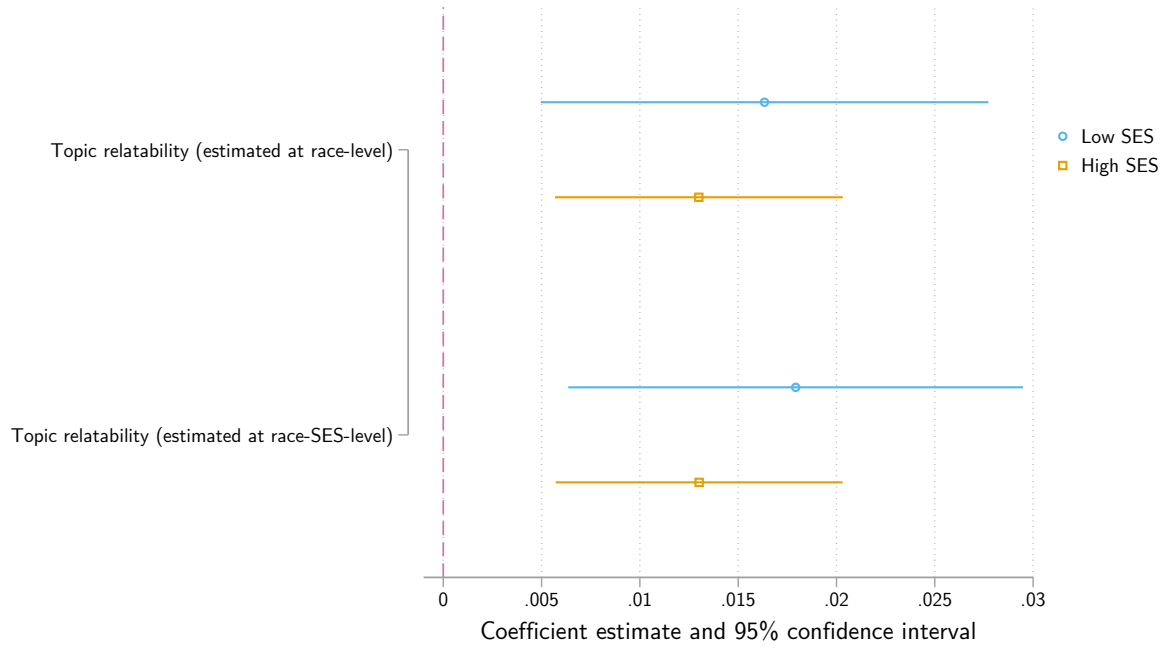


Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Each specification is run on a separate student subsample based on the decile of a student’s math test score. Each student’s math test score is from the standard STAAR exam for mathematics. Score deciles are formed within exam. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

students (75%) than Asian (29%) and white students (28%)

Given this relationship and the correlation between SES and school performance, we explore the extent to which topic relatability responses differ by SES. We classify students as being “low” or “high” SES based on their economically disadvantaged status in the administrative data. Next, we find an appropriate adult sample group for this population in the ATUS data. We classify ATUS respondents as being low SES or high SES based on whether their household income is above or below the 200% poverty line. We then estimate two versions of our main specification. First, we estimate within-SES impact of race topic relatability; that is, we estimate equation (4) separately for low SES and high SES students using race topic relatability. Second, we repeat this exercise but replace race topic relatability with race-SES topic relatability—that is, using race-SES instead of race for demographic group d . Details on SES classification and the construction of race-SES topic exposure measures

Figure 5: Heterogeneity of race topic relatability effects by SES



Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Specifications differ by sample and the way topic relatability is calculated. The top set of coefficients is estimated using the baseline topic relatability measure described in section 4.1, at the race-level. The bottom set of coefficients is estimated using race-by-SES topic relatability. The blue coefficients are estimated on the low SES student population, and the green coefficients are estimated on the high SES student population. Low SES is defined as students who are enrolled in free lunch, reduce-price lunch, or some other income-based program. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

can be found in Appendix section B.

Figure 5 displays the coefficient estimates on topic relatability by race and relatability estimation method. Focusing on the first set of coefficients, we see that the effect of racial relatability is slightly lower for high SES students than for low SES students, but we fail to reject the null hypothesis that the coefficient estimates are the same. The second set of coefficients show that shifting from race topic relatability to race-SES topic relatability does not significantly change the estimates.

These results show that race topic relatability matters for high and low SES students alike. We find some suggestive evidence that the effect is smaller for higher SES status students; we cannot rule out the possibility that observing more granular levels of SES

would firmly establish such a trend. Such a relationship may be possible due to higher SES students and families having more homogeneous interests and familiarity with topics or due to higher exposure to individuals outside their racial group. It may also reflect heterogeneity in the relatability effect across the achievement distribution. Since high SES students likely perform better on tests, there is less scope for improvement on test performance. We also note that a smaller race relatability effect for high-SES students would be inconsistent with the concern that high SES students’ exposure to certain topics or topics overall and higher test performance are confounding our main estimate. If this were the case, we should observe the opposite effect: our race relatability estimates should be higher for high SES students.

Schools and peers Peer-effects may weaken or intensify the impact of race topic relatability on test outcomes. We have only considered a coarse topic relatability measure that is constructed at the race level. Our approach is borne out of data limitations, whereas we are unable to directly observe the interest and familiarity of students in our sample. However, even from this crude starting point, we can make inferences about how the effect of topic relatability may differ within race. Students who live, play, and study in more racially homogenous settings are likely to have interests and familiarity, and thus topic relatability, similar to others of their race. Conversely, students living in racially diverse settings interact more with peers not like themselves. Even if their interests are not aligned with their school environment, this exposure to diverse peers may make them more familiar with a broad number of topics.

We proxy peer composition using school attended. First, we re-estimate equation 4 but saturate the fixed effects with a school indicator to estimate the topic relatability effect only variation from students within schools. In Table A8, column (2) shows the estimated impact of relatability are comparable to the main estimate.

We study heterogeneity along two peer diversity measures. First, we compare estimates between more and less racially integrated schools. To calculate school-integration for a given school, we use a Herfindahl index (HHI) based on the school’s population share of our four race categories, in which a higher value indicates more racial homogeneity. We estimate our main specification separately for each quartile of this index. Table A9 shows the results of the HHI analysis. Coefficient estimates are statistically indistinguishable from one another. These results are not particularly surprising, given that a school’s HHI masks significant heterogeneity in the experiences of each racial group. For example, if a school’s student body consists of 80% group A and 10% groups B and C, students of groups B and C in this school have a highly minoritized experience with high exposure to race A, but race A has a majority experience with low exposure to any other race.

Second, we conduct an analysis which separates the student population into subgroups based on the own-race share at their school. To do this, first we calculate for each student the share of their school that is their own race. Then, within each race group in the population, we divide students into quartiles relative to the distribution of this “own-race school.” Figure 6 shows the results of the school own-race share analysis. Students who are surrounded by more students of their race—indicated by a higher quartile—have stronger relatability effects. The point estimate on topic relatability is 45% smaller for quartile 1 compared to quartile 3. The coefficient differences across quartiles 1, 2, and 3 are statistically significant at the 5% significance level or more. This result suggests that higher exposure to individuals outside one’s own race moderates the impact of topic relatability.

Our findings suggest that peers can mitigate the topic relatability effect, but only up to a point. We found that topic relatability impacts test performance even when comparing students within schools. When we split schools by their HHI of racial concentration, we found little heterogeneity in relatability effects. At the same time, the size of a student’s racial group in a school does seem to matter: the relatability effect declines with a student’s own-race share.

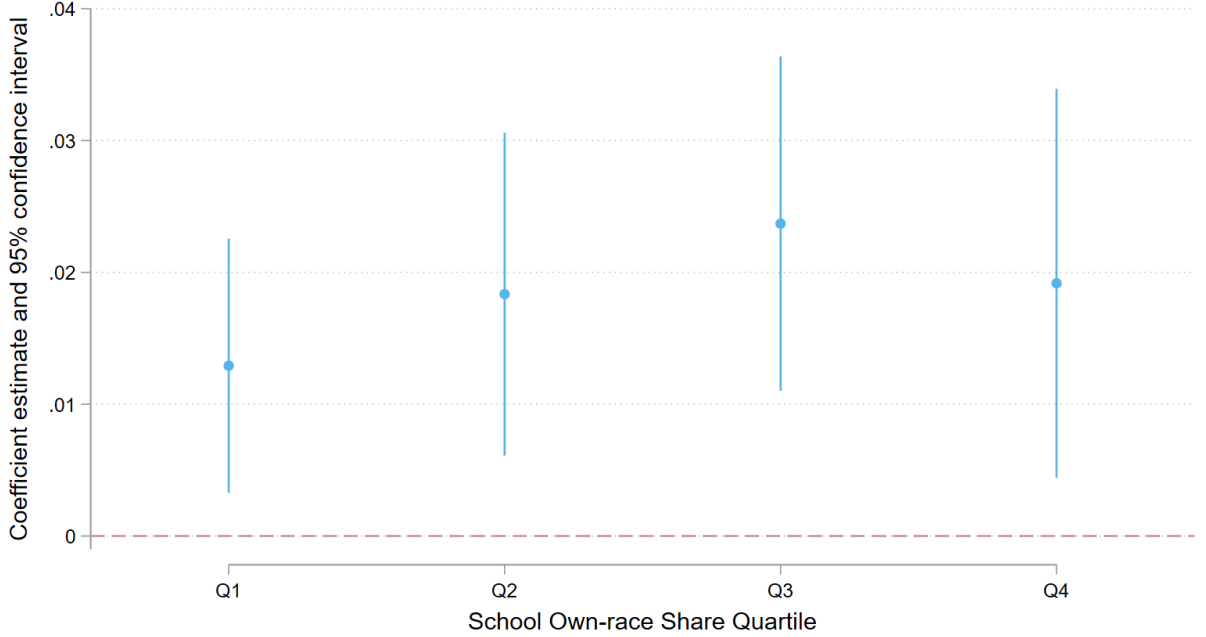
4.3.3 Sensitivity and robustness

We test if our results are sensitive to alternative specifications and alternative formulations of our main relatability measure. First, we re-run our main specification replacing race-grade fixed effects with alternative fixed effects. Second, we change our calculation of topic exposure e_{dt} by using subsamples of the ATUS data or by utilizing the time diary data differently. Third, we employ different NLP methods and measures to calculate topic salience m_{tp} and estimate our main specification with the resulting relatability measure. Finally, we drop each topic when calculating the topic relatability measure to test sensitivity to our construction of our topic set. We find our main result is robust throughout this battery of exercises. Details on the methodology and results of the robustness analysis is in Appendix Section E.

One remaining concern is the extent to which adult time use data is a good proxy for children’s interests and familiarity. We already showed through topic exposure sensitivity exercises in Appendix Section E that using the age 15–34 subsample of the ATUS produces slightly higher (but statistically indistinguishable) coefficient estimates on race topic relatability. We take this analysis one step further and directly compare our estimates to estimates based on children’s time use data.

We use data from the Child Development Supplement of the Panel Study of Income Dynamics (PSID-CDS) which includes relatively detailed time use data on children aged 0–17. The CDS surveys caregivers of children and/or older children of adults who are surveyed

Figure 6: Heterogeneity of race topic relatability effects by school own-race share



Notes: Each coefficient is an estimate from a quartile of the population using our baseline specification regressing share of items answered correctly on a passage on race topic relatability. We calculate for each student the share of their school that matches their race. Then, for each race, we split the population into quartiles. Thus, quartile 1 is a minority in their school and higher quartiles are relative majorities. Unreported controls include: (1) “unit” fixed effects at the level indicated in the legend and (2) exposure-sum-by-passage fixed effects. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

in the PSID, a panel survey of individuals and families in the United States since 1968. The CDS crucially includes a weekday and weekend time diary of each surveyed child. We use data from the 2014 and 2019 waves of the PSID-CDS, for children aged 8 to 17. Our final PSID sample includes 1,555 children. More details on the PSID-CDS data and our PSID empirical methodology is in Appendix Section B.

We use the PSID-CDS time diaries to construct a topic relatability measure analogous to our ATUS-based topic relatability measure, and we estimate its impact on test performance using our main specification. Table A7 displays the results for PSID and ATUS-based topic relatability. We show the standardized topic relatability point estimates as before, but we also show the effect sizes scaled by the Black-White and Hispanic-White topic relatability gaps. Columns (1) and (2) show that the effect of a SD increase in relatability is substantively

different based on the underlying survey used; the ATUS-based effect is almost 2 times greater than the PSID effect. However, we show that this is partially a result of the low sample sizes among certain groups in the PSID. Columns (3) and (4) show the same regressions estimated without Asian students, which tightens the difference in coefficients across the two measures.¹⁶ The scaled coefficient estimates corroborate this story. Across columns (1) and (2) and across columns (3) and (4), we fail to reject the null hypothesis that the Black-white scaled effects of the reliability estimates are the same.

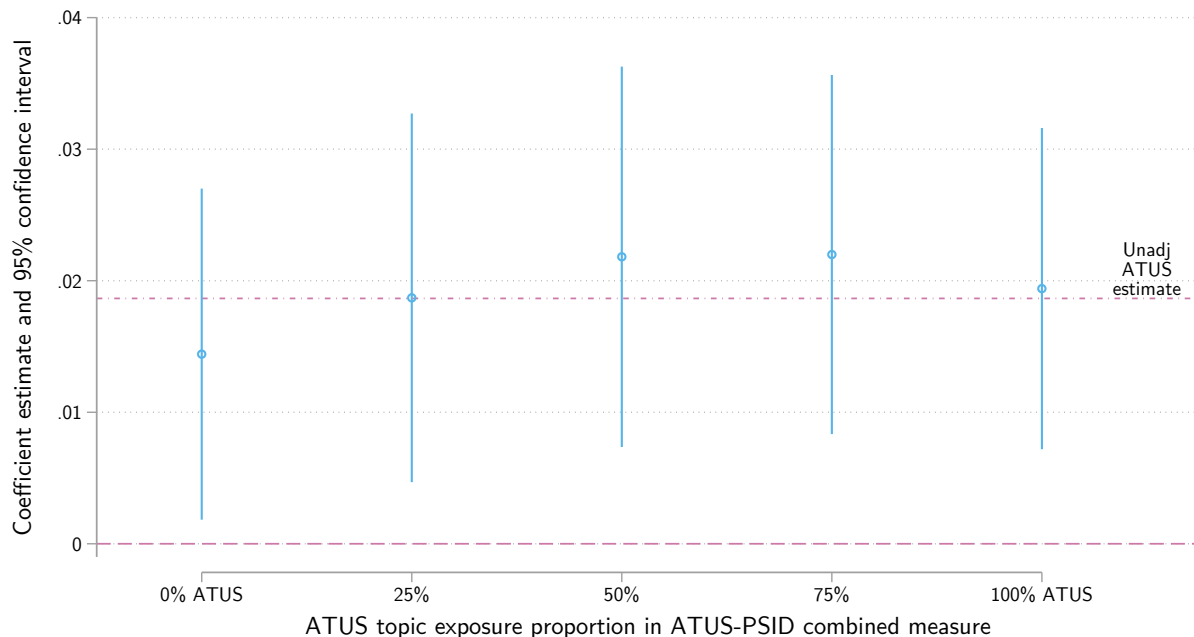
We test the extent to which different mixes of the ATUS and PSID produce different coefficient estimates. Given the concerns with low sample sizes in the PSID, we apply an empirical Bayes shrinkage estimator (Morris 1983, Walters 2024) to both the ATUS and PSID topic exposure shares, shrinking estimates to the population topic-level mean. Column (5) in Table A7 shows the effect of creating a composite reliability measure based on a simple average of the shrunk ATUS and PSID topic exposures. The point estimate is larger than our main ATUS-based estimate, but these differences are not statistically significant at the 5% level. Figure 7 plots coefficient estimates using different weighted averages of the ATUS and PSID measures. There is suggestive evidence that a composite measure may be more predictive of test performance by reducing measurement error from two sources: (a) using adult time use to approximate children’s interest and familiarity in the ATUS, or (b) using a survey with limited observations in the PSID. We find more evidence of the latter, as we find a statistically significant difference between the PSID-only measure and the 25% ATUS and 50% ATUS composite measures, when we use a “stacked” regression and use a *t*-test for equality between coefficient estimates.

Figure 7 shows two additional results. First, the effect of the shrunk PSID-based reliability measure is much closer to the ATUS measure than the raw PSID-based measure, again suggesting that measurement error due to lower observation counts in certain cells causes attenuation in the initial PSID-based estimates. Second, the effect of the shrunk ATUS measure is almost exactly the same as that of the ATUS measure, suggesting that classical sampling error in the ATUS does not appear to generate noticeable attenuation bias in our estimates.¹⁷

¹⁶Our PSID sample contains only 15 Asian respondents.

¹⁷We further assess the stability of our main estimates using a split-sample instrumental variables approach. We randomly split the ATUS sample into two halves and construct topic reliability measures separately for each subsample. These measures can be viewed as noisy but independent proxies for the true underlying reliability measure of interest. In a shift-share estimation framework with noisy proxies, consistency can be achieved by estimating a two-stage least squares regression of the outcome on one proxy, using the other proxy as an instrument. The resulting estimates are very similar to our main results, indicating that attenuation bias from sampling error in the ATUS has a negligible impact on our estimate.

Figure 7: Race topic relatability effect on test performance using different ATUS and PSID combinations



Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) “unit” fixed effects at the level indicated in the legend and (2) exposure-sum-by-passage fixed effects. Each specification uses a different average of race topic exposure based on ATUS and race topic exposure based on PSID. Estimates are ordered left to right based on the weight on the ATUS exposure. Race topic exposures are adjusted using an empirical Bayes shrinkage estimator which shrinks race topic exposure measures to the population topic exposure means. Our baseline coefficient estimate, based on the unadjusted ATUS topic exposure measure, is indicated by the dot-dash line. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

4.4 Results for gender topic relatability

We conduct the same analysis as before, but now focusing on gender. In Table 4, we fail to reject a null effect of gendered topic relatability on student test performance across all specifications. Our results point at interesting heterogeneity in the importance of different dimensions of topic relatability on race and gender. In Section 5, we discuss identity relatability which, interestingly, does yield comparable results for race and gender. In addition to our baseline ATUS measure, we generate topic relatability using the PSID children’s time-use data, which also fails to reject a null effect (see Table A10). Even if these point estimates are taken at face value, the share-of-gap-explained calculations are one or two or-

Table 4: Impact of gender topic reliability on test performance

	(1)	(2)	(3)	(4)
Gender topic reliability	-0.0009 (0.0057)	-0.0001 (0.0055)	-0.0023 (0.0053)	-0.0007 (0.0052)
Gender FE	✓		✓	
Gender-Grade FE		✓		✓
Passage FE			✓	✓
Passage-by-Exposure sum FE	✓	✓		
<i>N</i> of topic-passages	4,920	4,920	4,920	4,920
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on gender topic reliability. Standard errors reported in parentheses below coefficient estimates are (a) obtained using a shock-level regression following Borusyak et al. (2022) and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. For columns (1) and (2), gender topic reliability is calculated as described in section 4.1. For columns (3) and (4), gender topic reliability is calculated after adjusting topic exposure by exposure sums for males and females. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

ders of magnitude smaller than the race gap calculations. This is not only due to the smaller point estimates but also due to smaller gendered topic reliability differences compared to Black-White or Hispanic-White topic reliability differences.

In our discussion of the sub-components of reliability in Section 2.2, we outline that topic reliability bundles two distinct effects that cannot be precisely distinguished in our setting: interest in a topic and familiarity with a topic. This framework is a useful lens for thinking through the different results we observe across gender and race.

Suppose test outcomes are a function of both interest in the passage topic and familiarity with the passage topic. Demographic group differences are then driven by average differences in interest *and* familiarity. Holding personal interest fixed, a student’s topic familiarity should increase via family or community members engaging in topic-adjacent activities. However, students’ exposure across gender lines is effectively universal, whether in the classroom or in the home through opposite-gender siblings and parents. Due to well-documented residential sorting across racial and ethnic lines, many students will have much lower exposure to individuals of different races or ethnicities relative to cross-gender exposure. This suggests that cross-gender comparisons of test outcomes on the same passages should be driven more by variation in interest, whereas cross-race comparisons of test outcomes are likely further apart on both interest *and* familiarity.

This argument suggests, all else equal, our estimated coefficient will grow in the face of larger familiarity differences. The peer effects estimates in the race reliability setting

provide supportive evidence to this (see Figure 6). Students who are a relative minority in their school have smaller estimates than students who are a relative majority in their school. These effects are statistically distinguishable for the bottom three quartiles, demonstrating that students who are a relative minority, and thus have more cross-race familiarity, have a point estimate that is 45% smaller than students in a relatively more homogeneous environment.

5 Identity relatability estimation and results

When considering how a student might relate to text, it is natural that students may engage deeper with passages containing characters that share their gender or racial and ethnic background. Accordingly, we develop measures of identity-based relatability to complement our topic relatability measures. Our method is intuitive: first, identify each character in a reading passage; second, impute gender and race for each character from name (or context clues for gender, e.g. pronouns); third, calculate the predicted share of characters in this passage who are female/male or are Asian, Black, Hispanic, or White. We confirm the robustness of each of these steps. For gender, we also consider a dictionary-based, rather than character-based, measure for a passage’s gender congruence adapted directly from Adukia et al. (2023), which yields very similar results. We measure the test score impact of identity relatability separately for race and gender and calculate their contribution to race and gender test gaps.

5.1 Data and Methodology

Our core identity measures are based on the share of human characters in the reading passages that match the student’s identity. To start, we use large language models (LLMs) for named entity recognition (NER). For each reading passage, we instruct the LLM to read the passage and return structured output for each character. This structure has attributes including the character’s first name, last name, title, and the number of times they are mentioned by name in the text. We verify that all names appear in the text to guard against hallucinations and run the named-mention counting process in two distinct ways, which agree with each other over 96% of the time. For robustness, we consider both OpenAI’s GPT-4o-mini and GPT-5-mini models.

Next, we impute demographic characteristics for each character. Inferring gender is most straight-forward: we instruct the LLM to infer (binary) gender of characters from pronouns, titles, and other context clues, directing it to leave this field blank if unclear (0% missing).

To compare, we also use the US Social Security database of baby names by gender to impute gender for characters with first names. For this, we simply calculate the share of all babies born since 1950 with a given name that are [fe]male. Race and ethnicity is also imputed using names databases in three different ways. First, using last names matched to US Social Security databases with a Bayesian prior of US population shares (Imai & Khanna 2016, referenced as wru in tables). Second, supplementing the same last name data with first name data from six southern states voter files and a Bayesian prior informed by Texas demographic characteristics (Imai et al. 2022, referenced as wru+ tables). Third, using first names in a nationally-representative sample of mortgage applications (Tzioumis 2018). Some characters only have first names (22% of characters) or only have last names (12%), these characters are assigned to the Bayesian prior when using prediction methods that only use last or first names, respectively. Because we do not consider certain racial and ethnic groups such as Native Americans or multiracial, the race predictions we use may sum to less than unity.

Finally, we aggregate to the passage-demographic-level by averaging across characters. For gender, since the LLM yields universal labeling, it is as simple as taking the mean of binary gender indicators for each character in a passage. All race-based measures are predicted probabilities conditional on first and/or last names, so we aggregate over these predicted shares. Thus, our passage-level race score for race d represents the share of passage characters that are d “in expectation.” For robustness in this aggregation step, we also consider a mean weighted by the number of named mentions. This weighted mean measure gives more weight to protagonists and less weight to minor characters in a passage.

Our baseline race identity relatability and gender identity relatability measures use the simple mean aggregation method with wru+ and LLM-imputation, respectively. We plot the histograms of their distributions in Figures A5 and A6. For race identity relatability, we find that most passages have a substantial share of White-predicted characters. For each non-White racial group, we find that at least one passage predominantly features characters from a single non-White race. Passages are also more likely to be male skewed than female skewed in characters: an all-male character passage is 50% more likely than an all-female character passage. However, we also find that a substantial number of passages are balanced or near-balanced on gender. In contrast with our topic relatability measures, our identity relatability measures have a natural and interpretable scale: moving from zero to one on either gender or race identity relatability represents moving from a passage that contains no characters sharing your identity to entirely characters sharing your identity.

An alternative approach to the NER-centric methodology above is a dictionary-method computed directly at the passage level. Adukia et al. (2023) provides a list of gendered

terms that may appear in text. We use their most expansive list of gendered terms as well as their most narrow, comprised of just pronouns. For each set of words, we detect all gendered terms in a passage, and calculate the passage-gender level metric as the [fe]male word share of all gendered words, as defined by the dictionary.

With these identity relatability measures in-hand, we estimate the impact of identity relatability of test scores using a specification similar to equation (4)

$$Y_{dp} = \delta_{d,g(p)} + \pi_p + \beta^{identity} r_{dp}^{identity} + \nu_{dp}, \quad (6)$$

where $r_{dp}^{identity}$ is the identity relatability for group d on passage p . Since the econometric exercise for identity relatability differs from that for topic relatability, we do not utilize the shift-share empirical strategy. Most notably, we replace $\pi_p E_d$ from equation (4) with a simple passage fixed effect π_p . We cluster the standard errors at the exam-level (i.e., grade-year), as before.

Our choice of controls in the two-way fixed-effects specification in equation (6) follows from the general motivation of the controls in the shift-share specification in equation (4). Here, we worry first that overall test maker preferences for including some identity groups across passages is spuriously correlated with test performance. For example, passages in general feature more White characters while White students tend to have better educational outcomes. We also worry that unobserved passage attributes are correlated with both character selection and test performance. The $\delta_{d,g(p)}$ and π_p fixed effects account for both these issues.

5.2 Results

Identity-based relatability yields qualitatively and quantitatively consistent results for both race and gender measures, robustly maintaining conventional levels of statistical significance and coefficient stability across choices considered in the previous methodology section. The point estimates are larger for race than for gender, and have larger implications when scaled by the difference in average relatability.

Moving from all different race to all same race yields a 1.1 percentage point improvement in test performance, using our preferred race identity relatability measure, `wru+`. This corresponds to a 0.03 standard deviation improvement at exam-level (see Table 5). Further, both `wru` and `wru+` measures span nearly the entire range from 0 to 1 for all races, empirically suggesting that there are racially homogeneous passages for all four of our considered racial and ethnic groups. Average White identity relatability is near 0.65 whereas both average Black and average Hispanic relatability are approximately 0.12, suggesting that equalizing

Table 5: Impact of race identity relatability on test performance

Source	wru+		wru		Tzioumis	
	(1)	(2)	(3)	(4)	(5)	(6)
Race identity rel. (mean)	0.0106*** (0.0039)		0.0092** (0.0038)		0.0043 (0.0072)	
Race identity rel. (w.mean)		0.0109*** (0.0035)		0.0098*** (0.0033)		0.0052 (0.0055)
Reading Passage FE	✓	✓	✓	✓	✓	✓
Race-Grade FE	✓	✓	✓	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on race identity relatability. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Identity relatability is defined as share of characters matching the students race. Columns (1) and (2) calculate identity relatability using last names from social security data, first names from six Southern states, and imputes average Texas demographics if missing. Columns (3) and (4) calculate it only using the same last name data with imputation matching national demographics. Columns (5) and (6) use only first names from a nationally-representative sample of mortgage applications. Simple means give each character a weight of one, weighted means use number of named mentions for the weight. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

relatability would close both the Black-White and Hispanic-White test gaps by about 0.5 percentage points, which is 4% and 5% of the Black-White and Hispanic-White test gaps, respectively. Racial identity relatability results are also rather consistent across different imputation methods, some relying only on first name or last name information.

For our preferred gender measure using LLM-imputation, we find that moving from all opposite-gender to all same-gender characters in a passage results in a 0.76 percentage point improvement in test performance (see Table 6), which translates to a 0.02 standard deviation increase at the exam-level. The average female relatability is near 0.42 and average male relatability is near 0.58, suggesting that equalizing relatability would actually widen the male-female test gap by about 0.1 percentage points (3% of the 3.5pp Female-Male gap). Both of these are consistent across both NER-type and dictionary-type measures of gender identity relatability.

Additional robustness is reported in Tables A11 & A12, which suggests that race results are primarily a story about the share of own-race characters (as in the main results), whereas gender results are also responsive to the number of characters matching a student’s gender.

We also consider heterogeneous effects based on passage characteristics, such as whether the passage has historical figures, has celebrities, has any children, or is a memoir (see Tables A13 & A14). We classify characters into each category using the LLM (further details found in Appendix B.5). The 24% of passages with at least one historical figure have statistically

Table 6: Impact of gender identity relatability on test performance

Type	Character				Dictionary	
Source	LLM		SSN		Adukia	Pronouns
	(1)	(2)	(3)	(4)	(5)	(6)
Gender identity rel. (mean)	0.0076*** (0.0018)		0.0064*** (0.0019)		0.0076*** (0.0019)	0.0073*** (0.0018)
Gender identity rel. (w.mean)		0.0059*** (0.0017)		0.0046** (0.0017)		
Reading Passage FE	✓	✓	✓	✓	✓	✓
Gender-Grade FE	✓	✓	✓	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on gender identity relatability. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Identity relatability is defined as share of characters matching the students gender. Columns (1) and (2) calculate identity relatability using gender labels from a large language model, imputing from pronouns and other context clues. Columns (3) and (4) calculate it using first name data from nationally-representative social security data. Columns (5) and (6) follow the dictionary-based approach of Adukia et al (2023). Simple means give each character a weight of one, weighted means use number of named mentions for the weight. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

distinguishable larger effects for gender relatability: about 1.7pp relative to 0.4pp for the rest of the passages. A similar but weaker effect is found for individuals tagged as "celebrity" by the large language model plausibly recognizable by a grade school student. Results further suggest that gender effects may be muted for memoir passages. There is no corresponding heterogeneity in race.

6 Combined effect of relatability

To this point, we have taken a structured approach to studying the effect of content relatability on test performance. We have distinguished two sources of content relatability: topic relatability and identity relatability. This delineation transparently allows policymakers to compare the each measure’s impact on test performance, especially as the recent literature has focused on the latter factor. Importantly, these measures are quantified and could be applied to any exam. For topic relatability, we use a dataset that measures topic exposure via revealed preference (time spent in topic-related activities). For identity relatability, we use character names, for which we can leverage extensive data on identity associations. Our analysis focuses on content attributes orthogonal to targeted reading skills (e.g., leisure topics), rather than on constructs like vocabulary difficulty.

However, our structured approach has limitations. Our relatability constructs are not

exhaustive due to limitations on topics and identities we observe. There are likely attributes of test passages, not captured by our relatability estimates, which cause differential test performance across demographic groups. Our estimated effects are also not necessarily mutually exclusive of each other. Many passages are drawn directly from existing texts, so character identities and passage topics may naturally be tightly correlated.

We address these issues by supplementing our relatability measures with an external survey directly asking the general population how they assess the relatability of different passages. This offers a more flexible, less prescriptive measure than our structured approach based on time-use data, NLP, and named-entity recognition. Additionally, it serves as a robustness check for these existing measures. Then, we consider the combined impact of all three relatability measures on test scores.

6.1 Survey-based relatability

We survey 640 adults residing in the United States. Participants are recruited from Prolific, a widely used online platform to host surveys for social science research. Respondents are selected to be representative of the U.S. population by race, gender, and age ranges. In addition to these demographic characteristics, we collect information on respondents’ state of residence, their parental status, and the frequency with which they interact with children. Further details on the participant recruitment, participant characteristics, and survey procedures not described here can be found in section F.

Each respondent read ten passage summaries and assessed children’s relatability to each text. For each passage in our estimation sample, we ask an LLM to produce an approximately 220 words summary at an eighth-grade reading level. We rely on these passage summaries rather than full-length text, for both pragmatic and methodological reasons. Passages in our corpus take at least three minutes to read and comprehend, which risks significant drop-off in study participation. Longer texts also restrict how many relatability assessments we can elicit. This is important in a setting where repeated assessments may improve response quality. Methodologically, summaries also reduce sensitivity to textual features like length or vocabulary level; similarly, genre cues may persist but are dampened. The set of passages and the order in which they are delivered are randomized for each respondent.

We construct a survey-based measure of relatability by aggregating the relatability assessments across respondents. Respondents were provided a general definition of relatability which includes topic familiarity, content interest, and identity alignment. We elicit race-specific rankings of relatability and a separate gender assessment. For race, respondents must rank the relatability of the text from most relatable to least relatable. For gender,

respondents simply answer whether they think the text is more relatable to girls or boys. Our preferred measure of survey relatability comes after fitting a rank-ordered logit choice model (“exploded logit”) by passage, using respondents’ rankings. We use the predicted probability that a group ranks first as that passage’s relatability for the group.¹⁸

We estimate regressions of test performance on survey relatability including demographic group-grade and passage fixed effects, as in equation (6). Table A15 shows the results of these regressions. A one-unit increase in the predicted probability that a passage is most relatable to a racial group is associated with a 0.9 percentage point increase in test performance. We observe slightly larger coefficients for gender, but the difference is not statistically significant. We scale coefficients by the between-group relatability gaps to obtain gap-relevant effect sizes. We find the survey relatability effect explains less of racial differences in test scores than topic and identity relatabilities. We also find that, similar to identity relatability effects, estimated survey-identity relatability effects would widen, not close, test differences between male students and female students.

6.2 Correlation across relatability measures

The interpretation of the estimated results across our three measures depends on the correlation among our three measures. If the topic and identity relatability measures are closely related, that may imply that the estimated effects of topic relatability may not necessarily be attributable to familiarity or interest but to a correlation with identity. Given that survey relatability predicts performance, if it is uncorrelated with topic and identity relatability, this would indicate that there are other factors embedded in exams which influence performance differences across demographic groups.

We first consider the relationship across our three measures by regressing one measure on the other measures using a version of equation (6). We standardize each measure by its standard deviation for ease of comparability across estimates. Tables A16 and A17 show the result of these cross-correlations for race and gender, respectively. We find little correlation between the topic and identity relatability measures (columns (1, 3, 5) in both tables). However, we find that the survey measure explains 7% of the residual variation in the racial identity measure and 32% in the gender identity measure (column (4) in both tables). These results yield two insights. First, the presence of relatable topics for a group does not necessarily imply that characters from that group appear in a passage. This finding is consistent with a test-making process that prioritizes including diverse characters across

¹⁸Our results are robust to using a simpler measure: the share of respondents who ranked a demographic group first for a passage.

many contexts rather than diversifying the contexts themselves. Second, survey respondents appear to anchor more strongly to identity than to topics when assessing relatability of passages. Identity cues such as names and pronouns may be more salient and easier to detect, while topics require deeper thinking and interpretation. Similarly, identity cues may be easier to perceive than cross-group differences in topic interest. The respondents' behavior may also reflect an underlying belief that identity is the most important driver of relatability for a piece of text.

6.3 Joint estimation of relatability effects

We next compare the effect of all three measures of relatability together, to explore whether each relatability measure estimates a separate effect and assess the magnitudes of the effects. We run this joint specification following our main topic relatability specification (equation (4)), where we include demographic group-grade fixed effects and passage-exposure sum fixed effects and all three relatability measures.¹⁹ We continue to standardize each relatability measure by standard deviation to ensure comparable magnitudes. Table 7 displays the results of these regressions separately for race and gender. Column (1) shows that topic, identity, and survey relatabilities all have a positive and statistically significant effect on test performance for race. Topic relatability has the largest standardized coefficient. This likely reflects greater cross-passage dispersion in the construction of the topic measure relative to that of the identity and survey measures. When scaled by within-passage variation, the standardized topic and identity coefficients become similar in magnitude. Column (2) shows that only survey relatability has a positive and statistically significant effect on test performance for gender. Given the strong survey-identity correlation for gender, multicollinearity attenuates the identity coefficient once survey relatability is included.

As before, we also consider the total effect sizes scaled by the relatability gaps between demographic groups. For both race and gender, we find that accounting for all three measures leads to a larger scaled effect than simply considering the estimated topic relatability and identity relatability effects. These differences are statistically significant, suggesting that the relatability differences across demographic groups identified by survey respondents have a meaningful impact on demographic test score differences. However, given the correlation between the identity and survey relatability measures, we proceed with considering only topic and identity relatabilities for counterfactual policy analysis. Closing gender relatability gaps would widen the male-female test score gap, implying that content relatability is not a first-order driver of the observed gender gap.

¹⁹Results are largely similar when replacing passage-exposure sum fixed effects with passage fixed effects.

Table 7: Joint estimation of reliability effects on test performance

	(1) Race	(2) Gender
Topic reliability	0.0190*** (0.00701)	-0.00444 (0.00529)
Identity reliability	0.00328** (0.00139)	0.000539 (0.000620)
Survey reliability	0.00197** (0.000961)	0.00381*** (0.000744)
Race-Grade FE	✓	
Gender-Grade FE		✓
Passage-by-Exposure sum FE	✓	✓
N of topic-passages	4,536	4,536
N of student-pasages	59,305,495	59,305,495

Notes: Each specification is a regression of the share of items answered correctly on various measures of reliability. Standard errors reported in parentheses below coefficient estimates are clustered by exam. Standard errors for topic reliability are obtained using a shock-level regression. * $p < .10$, ** $p < .05$, *** $p < .01$. Column (1) reports estimates for race and column (2) reports estimates for gender. The full estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test which contain estimates for topic, survey, and identity reliability.

7 Extensions

7.1 Reconsidering student test standards

Our analyses demonstrate that content reliability systematically affects reading comprehension scores, inflating measured Black-White and Hispanic-White gaps. Beyond shifting average scores, reliability may also directly disadvantage students. We consider whether Black and Hispanic students are misclassified as meeting key test performance standards.

Meeting performance standards can have an important impact on future student learning. For students in all grades, misclassification through the reading comprehension test can lead to students being placed in an intensive program or a remedial class for that subject. During our time period, Texas law required school districts to provide “accelerated instruction” for students due to non-satisfactory performance on a standardized test (Texas Education Code § 28.0211 2013, 2019). School districts are also required to create “intensive programs of instructions” for such students (Texas Education Code § 28.0213 2013, 2019). At the other end of the spectrum, reaching higher test standards helps determine eligibility for gifted programs. These test standards also aggregate to the school and district level, where the share of students meeting standards feeds directly into accountability indices. Misclassification

due to exam performances can lead to larger consequences for students in grades 5 and 8. In addition to all of the previously mentioned potential impacts, exams are used as a grade promotion requirement during our sample period. In practice, we observe relatively few students repeating a grade. However, the requirement to retake an exam is close to universally binding. We identify clean discontinuities at the expected score cutoff; answering just one additional question incorrectly at the cut-off increases the probability of a retaken test from 0.0% to 97.8% (see Figure A7).

We test what share of students would have been categorized for a higher standards category if topics and characters—and, therefore, reliability—for a given test had been different. For each grade, we identify the test in our sample that minimizes Black-White and Hispanic-White reliability differences. We use these tests as reliability “benchmarks” and adjust observed scores based on the reliability coefficient estimates obtained using equations (4) and (6). Using these adjusted scores, we assign students to new performance standards categories. Further details on the institutional setting, standards categories used, and the new performance standards assignment procedures can be found in Appendix Section G.

Tables A18 and A19 summarize the results of applying this exercise to the sample of Black students and Hispanic students, respectively. Each row corresponds to a different performance standards category and the columns are organized by grade. Each cell shows the mean difference in the share of students labelled in the performance category between the adjusted and unadjusted exams. We find that the benchmark test would have led to fewer Black and Hispanic students being classified as not meeting each of the three standards, but the magnitudes differ across grades. Across all standards, 1.5% more Black students in grade 3 would have been placed in a higher performance category with reliability adjustments, but 0.4% fewer Black students in grade 6 would have been placed in a higher performance category. For Hispanic students, we find 0.9% under-classification for grade 6 on the high end and 0.4% under-classification for grade 5 on the low end. In total, we estimate that almost 11,000 Black students and almost 37,000 Hispanic students may have achieved a higher reading comprehension standard if reliability had been more equalized across groups.

In short, reliability affects not only average scores but also classification into consequential categories. Even modest shifts in reliability are large enough to push thousands of students into different instructional tracks. Further, even though our results suggest smaller reliability-based distortions for students in higher grades, our single-year analysis masks potential cascading effects due to misclassification in earlier years. Improper classification in earlier grades could meaningfully change the learning trajectory of students.

7.2 Counterfactual policies

We consider a few policies that the test maker may implement in the context of our model and estimates, focusing on test score differences across race. In light of our discussion thus far, the most obvious goal a policymaker may want to consider is to reduce the influence of content relatability on test scores, thereby more precisely measuring student ability, θ_i . Given racial differences in relatability, these changes to exams would ensure that estimates of θ_i are not systematically divergent by race.

We briefly formalize the decision the policymaker needs to make, borrowing notation from section 2. Let $\vec{\mu}_p = (\vec{\mu}_p^{topic}, \vec{\mu}_p^{identity})$, which represents the relatability-relevant passage attributes of passage p consisting of two subvectors representing topic salience, $\vec{\mu}_p^{topic} = (\mu_p^1, \dots, \mu_p^t, \dots)$, and character identity composition, $\vec{\mu}_p^{identity} = (\mu_p^1, \dots, \mu_p^d, \dots)$. For each student in d , there is an analogous vector $\vec{\varepsilon}_d = (\vec{\varepsilon}_d^{topic}, \vec{\varepsilon}_d^{identity})$, giving the average degree of relatability of group d to topics and character identities. Taking the inner product and gathering terms yields our relatability measures: topic relatability, $\sum_t \varepsilon_{dt}^{topic} \mu_p^t$, and identity relatability, $\varepsilon_d^{identity} \mu_p^d$.²⁰ Our empirical strategy yields the test score effect of topic relatability $\hat{\beta}^{topic}$, and the test score effect of identity relatability, $\hat{\beta}^{identity}$.

Given these estimates and relationships between objects, the policymaker selects passage attributes across a set of passages $\mathcal{P}$. Formally, they choose $\vec{\mu}^*$, which is the average of $\vec{\mu}_p$ across the set of passages $\mathcal{P}$. We suppose in this case the policymaker is concerned about differential contribution of relatability across two racial groups d and d' . Then they are selecting $\vec{\mu}^*$ such that

$$\vec{\mu}^* = \arg \min_{\vec{\mu}} \left| \underbrace{\hat{\beta}^{topic} \sum_t (\varepsilon_{dt}^{topic} - \varepsilon_{d't}^{topic}) \mu^t}_{\text{difference in topic relatability}} + \underbrace{\hat{\beta}^{identity} (\mu^d - \mu^{d'})}_{\text{difference in identity relatability}} \right|. \quad (7)$$

That is, the policymaker selects topics and character identities such that they minimize the differential impact of relatability on test scores across racial groups.

We consider what would happen to test scores if a policymaker embarked on this minimization with various constraints.

7.2.1 “Unconstrained” equalization across groups

Equation (7) shows that equalizing the effect of relatability means selecting passages such that differences in group exposure and character identity effects cancel out. In theory, if

²⁰The latter is true because we assume that a student in group d only relates to characters of group d .

the set of feasible $\vec{\mu}_p$ is unconstrained, test makers can achieve this by strategically selecting passages that generate the same average relatability across racial groups and ensure a similar proportion of character identities. This is also always trivially possible if the test maker sets $\vec{\mu}^* = (0, \dots, 0)$. We find that if relatability is equalized in this way, policymakers could observe 8.7% smaller test score gaps between Black and white students and 9.7% smaller test score gaps between Hispanic and white students.

7.2.2 Equalization based on existing exams

In reality, however, test makers may face constraints on the available combinations of passage attributes, $\vec{\mu}_p$. For instance, our analysis thus far does not take into account the “supply” of passages from which test makers can choose when constructing exams or the presence of additional political or educational considerations which influence topic selection or character identity selection in exams.²¹ We can add an additional constraint to equation (7) that $\vec{\mu}^*$ must be based on the observed $\vec{\mu}_p$ in our passage sample. Practically, we repeat a version of the exercise in section 7.1. First, we search for the exam within each grade which minimizes the nonwhite-white gap in average content relatability and designate this race-level relatability as the best feasible relatability for their respective grades. Second, we predict counterfactual test scores under the scenario where students in all years received their grade’s best feasible relatability. We estimate only 2.4% smaller Black-White and 3.6% smaller Hispanic-White test gaps through this procedure. Since the relatability adjustments in this exercise reflect observed relatability measures from our sample, this demonstrates a lower bound of what policymakers may achieve even in the presence of existing constraints on passage attribution selection.

We find differences across Black and Hispanic students in the primary source of the feasible gap closure. For Black students, much of the relatability differences stem from topic relatability, indicating large variance within grade on the topics that are relatively more relatable to Black students. For Hispanic students, much of the differences stem from identity relatability, indicating that some tests feature significantly more Hispanic representation than others.

²¹Authors of children’s stories or books may themselves be from a selected population. Their exposure to topics as a child or an adult may influence the topics they write in the work, which may then mean the corpus of selectable passages is already biased prior to selection by test makers. This may further be exacerbated (mediated) if there are considerations which prioritize Texan writers or stories that are based primarily in Texas.

7.2.3 Equalizing topic distributions

Alternatively, a test maker could object to the previous goal of equalizing relatability as it may “disadvantage” students with higher aggregate topic exposure. While a policymaker who prefers the earlier approach may claim that different levels of aggregate exposure are due to differential constraints which should be corrected (e.g. differences in household financial resources), this test maker may want to be agnostic about that heterogeneity. Thus, their ideal topic distribution would be one where all topics are equally likely: $\vec{\mu}^{topic*} = (\mu^*, \mu^*, \dots)$. Under this policy, content relatability would be given just by the difference in aggregate topic exposure across racial groups.

We can illustrate what this means in practice by decomposing the empirical topic relatability differences across groups in our sample into a portion that is due to the topic distribution as opposed to differences in topic exposure. For this exercise, we use the objects m_{tp}, e_{dt} (see section 3) which are the empirical analogues to $\mu_{tp}, \varepsilon_{dt}$, respectively. We can decompose cross-group differences in average relatability as

$$\bar{r}_d - \bar{r}_{d'} = (E_d - E_{d'})\bar{m} + \frac{\sum_p \sum_t (e_{dt} - e_{d't})\tilde{m}_{tp}}{|\mathcal{P}|}, \quad (8)$$

where $\bar{m}$ is average topic salience across all topics and passages, $\tilde{m}_{tp}$ is the residual of m_{tp} from $\bar{m}$, $|\mathcal{P}|$ is the number of passages, and, as before, E_d and $E_{d'}$ are the sums of exposures across topics. The first term of the right-hand side of equation (8) represents differences in relatability due to differences in overall levels of exposure. The second term of the question represents differences in relatability due to selection of differentially favorable topics in passages.²² Intuitively, we can think of the first term as reflecting the fact that if overall exposure is higher for one group than another, then any randomly selected topic will lead to some baseline difference in relatability on average. However, if the second term is non-zero, any differences in relatability over and beyond this baseline difference must be due to topic selection being skewed toward one group over the other. It follows that setting $\vec{\mu}^{topic*} = (\mu^*, \mu^*, \dots)$ is equivalent to keeping only this first term in impacting test performance.

We can apply equation (8) to our sample to see the extent to which topic selection is impacting topic relatability differences across students. We find that the topic salience residuals $\tilde{m}_{tp}$ contribute to around one-third of Black-white (34%) and Hispanic-white (33%) differences in average relatability. Said differently, if test makers had counterfactually set $\vec{\mu}^{topic*} = (\bar{m}, \bar{m}, \dots)$, topic relatability differences would be almost a third smaller than

²²We note since $\tilde{m}_{tp}$ is a residual, it is possible for this term to be negative even if $e_{dt} - e_{d't} > 0, \forall t$.

currently observed relatability differences by race. This implies that 1.3% of Black-white gaps and 1.2% of Hispanic-white gaps can potentially be explained by which topics are selected. Combined with our estimate of the effect of adjusting identity relatability, adjusting tests in this way could close roughly 6–7% of the nonwhite-white test gap.

We collect all counterfactual estimates in Figure 8, including estimates of the gender gap explained. Relatability explains a larger share of the test score gap between Hispanic and white students than the share of the gap between Black and white students. As discussed earlier, relatability does not explain the male and female test score gap. While identity relatability has a statistically significant impact on test performance, the under-representation of female characters in test passages suggests that the male-female gap is potentially *understated*.

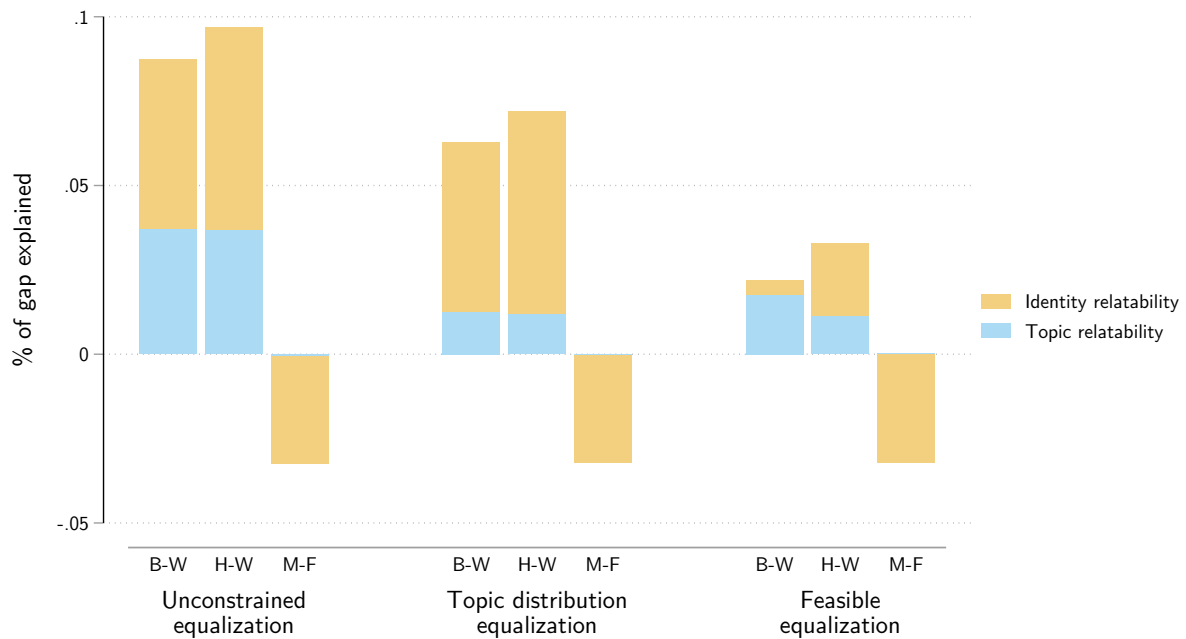
8 Conclusion

We study the extent to which the sociocultural content in standardized tests impact performance using end-of-year reading comprehension exams for 3rd to 8th grade students in the state of Texas. We develop a measure of “relatability” between a reading comprehension passage and a demographic group using natural language processing methods and time use data from the American Time Use Survey. We find that our measure is predictive of students’ standardized test performance—a standard deviation higher race-based relatability for a passage leads to a 1.9pp increase in probability of answering questions for that passage correctly. Given differences in average content relatability across race and ethnicity, we find that relatability accounts for 4% of the Black-white and Hispanic-white test gaps.

Our results have implications both for test writers and education policymakers. First, it highlights that in order to write balanced assessments, test makers should take into account not only the identities of characters, but also the general content of the passage or question itself as we show that this may influence performance. Second, when policymakers consider outcome differences along demographic dimensions, one additional component to examine might be the standardized tests used to calculate those differences. However, we also note that the contribution of test construction to the gaps we find are both non-negligible and modest; that is, they cannot explain a substantial portion of why Black and Hispanic students on average have lower performance on tests than white students.

Finally, an alternative interpretation of our results may be that adaptability to different environments and new concepts is an important part of student learning. To test this ability in standardized tests, students should read passages on topics with which they are unfamiliar, captured in our case by “relatability.” We contend our findings still have meaningful

Figure 8: Race topic relatability effect on test performance using different ATUS and PSID combinations



Notes: Each bar represents the share of the test score gap explained by topic relatability (lower bars) and identity relatability (upper bars). “B-W” indicates the Black-white test gap, “H-W” indicates the Hispanic-white test gap, and “M-F” indicates the male-female test gap. “Unconstrained equalization” shows the share of the test gap explained if relatability were equalized across the two groups. “Topic distribution equalization” shows the share of the test gap explained if residual relatability (after accounting for the average topic salience and character identities) were balanced across the two groups. “Feasible equalization” shows the share of the test gap explained if relatability for each grade were set to the smallest observed exam-level relatability effect difference between the two groups. Since topic relatability and identity relatability are determined jointly within a passage and exam, the relatability effect difference is determined accounting by averaging the effect stemming from the topic relatability gap and the effect stemming from the identity relatability gap. The coefficient estimates from equation (4) and equation (6) are used for topic relatability and identity relatability, respectively. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

policy-relevant implications in this scenario. Recast in this light, our main result demonstrates that students on average are not fully “adaptable” given that on average we are able to predict they will perform worse on topics with which they are less familiar. This suggests that education curriculum should put more emphasis on teaching students skills to be “adaptable.” Further, our racial gaps results demonstrate that if test makers seek to test the ability of students to be interested in concepts and settings unfamiliar to them, they must incorporate the fact that familiarity differs by demographic group; an unfamiliar topic to one group may be a familiar topic to another. Ultimately, our preferred interpretation of

our results stems from the fact that the stated goals of most reading comprehension exams do not include testing for breadth of topic knowledge or whether students are adaptable to unknown topics. Insofar as stated testing standards reflect the knowledge and skills educators truly expect students to have, we take these standards seriously in our conclusions and set aside any ancillary skills that educators would like to test.

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A Appendix tables and figures

Table A1: Summary of ATUS activity codes not selected for analysis

	Mean Min./Respondent	# of activity codes
All activities	1427	442
Excluded activities	1113	302
Sleeping	534	2
Personal care	46	4
Child care	39	66
Work	156	22
Education	14	22
Shopping	24	10
Eating	66	4
Telephone	7	11
Traveling (non-leisure)	58	51
Other	170	110
Included activities	314	140
<i>N</i>	73,626	73,626

Notes: This table displays an overview of ATUS activity codes which are not included in our topic set. Each row represents the statistics for each group of activity codes. Column 1 represents the average reported minutes per respondents for the group of activities. Column 2 represents the total number of activity codes for the group of activities. Respondent observations are weighted by ATUS sampling weights provided by the U.S. Census Bureau. The sample includes all ATUS respondents from 2013–2019.

Table A2: Examples of ATUS activities in each topic

Topic	Example activities	# of six-digit activity codes
Animal sports	(<i>equestrian, rodeo</i>)	4
Animals	(<i>caring for pets, going to the vet</i>)	9
Arts and crafts	(<i>sewing, decorating</i>)	5
Baseball		4
Basketball		2
Computer		1
Exercise	(<i>running, lifting</i>)	10
Food	(<i>baking, cooking</i>)	4
Football		2
Indoor recreation	(<i>billiards, bowling</i>)	4
Media	(<i>movies, TV</i>)	2
Misc. sports		21
Music		2
Nature sports	(<i>kayaking, fishing, climbing</i>)	6
Performing arts	(<i>musicals, dancing</i>)	4
Plant/garden/yard	(<i>gardening</i>)	3
Religion	(<i>attending church</i>)	7
Club sports	(<i>golf, tennis</i>)	6
Soccer		2
Street sports	(<i>skateboarding, scootering</i>)	6
Traveling		2
Vehicles	(<i>fixing car</i>)	3
Volunteering		25
Water sports	(<i>swimming, water polo</i>)	5
Winter sports	(<i>skiing, ice skating</i>)	4

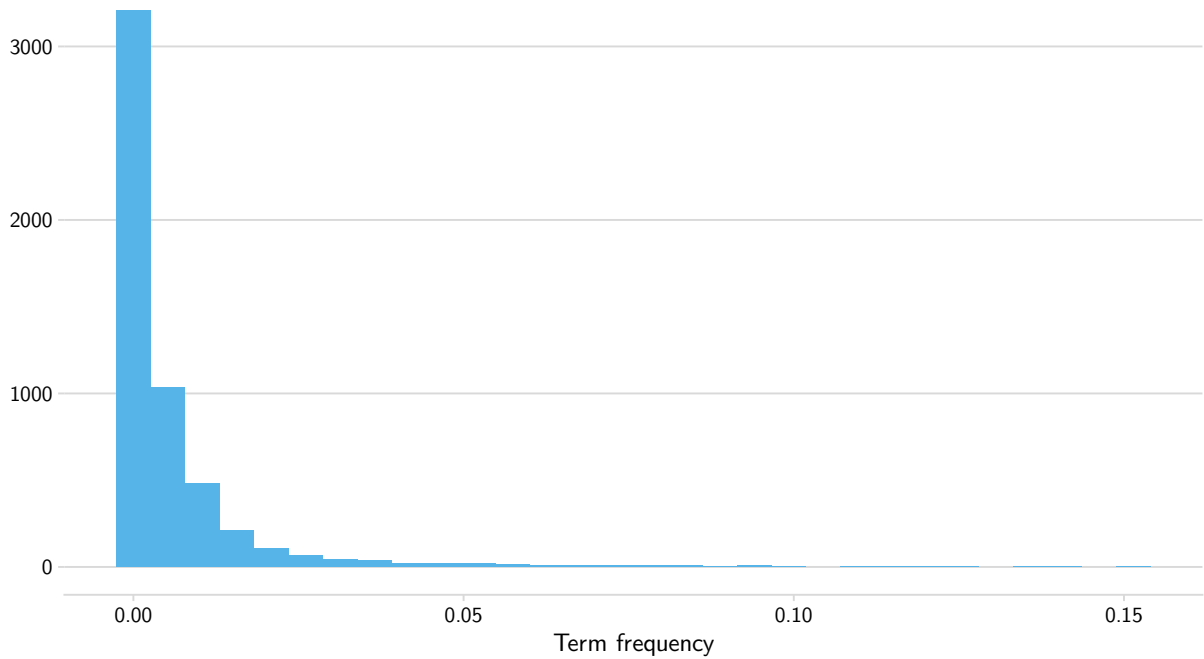
Notes: This table provides an overview of the ATUS activities which make up each topic. Each topic is associated with a set of mutually exclusive ATUS activities/activity codes. A full mapping of activities to topics are available in the online code repository.

Table A3: Relative exposure of groups to topics

	Asian-white	Black-white	Hispanic-white	Male-female
Animal sports	0.000	0.025	0.538	0.511
Animals	0.328	0.312	0.492	0.749
Arts and crafts	0.573	0.418	0.519	0.632
Baseball	0.310	0.228	0.751	1.330
Basketball	0.991	3.189	1.020	3.308
Club sports	0.851	0.223	0.289	3.813
Computer	1.491	0.642	0.763	0.899
Exercise	1.381	0.722	0.891	1.224
Food	1.023	0.843	0.906	0.645
Football	0.305	0.990	1.497	4.208
Indoor recreation	0.558	0.809	0.501	1.896
Media	0.883	0.997	0.975	1.024
Music	1.147	1.423	1.289	1.467
Nature sports	0.261	0.287	0.396	4.214
Performing arts	0.796	0.732	0.802	0.844
Plant/garden/yard	0.675	0.457	0.685	1.487
Religion	1.218	1.903	1.181	0.634
Soccer	1.371	0.697	4.184	1.979
Street sports	1.639	0.802	1.048	0.894
Traveling	0.858	0.895	0.952	0.896
Vehicles	0.598	0.716	0.979	2.604
Volunteering	0.507	0.679	0.535	0.746
Water sports	0.854	0.236	0.481	1.067
Winter sports	0.704	0.239	0.061	1.384

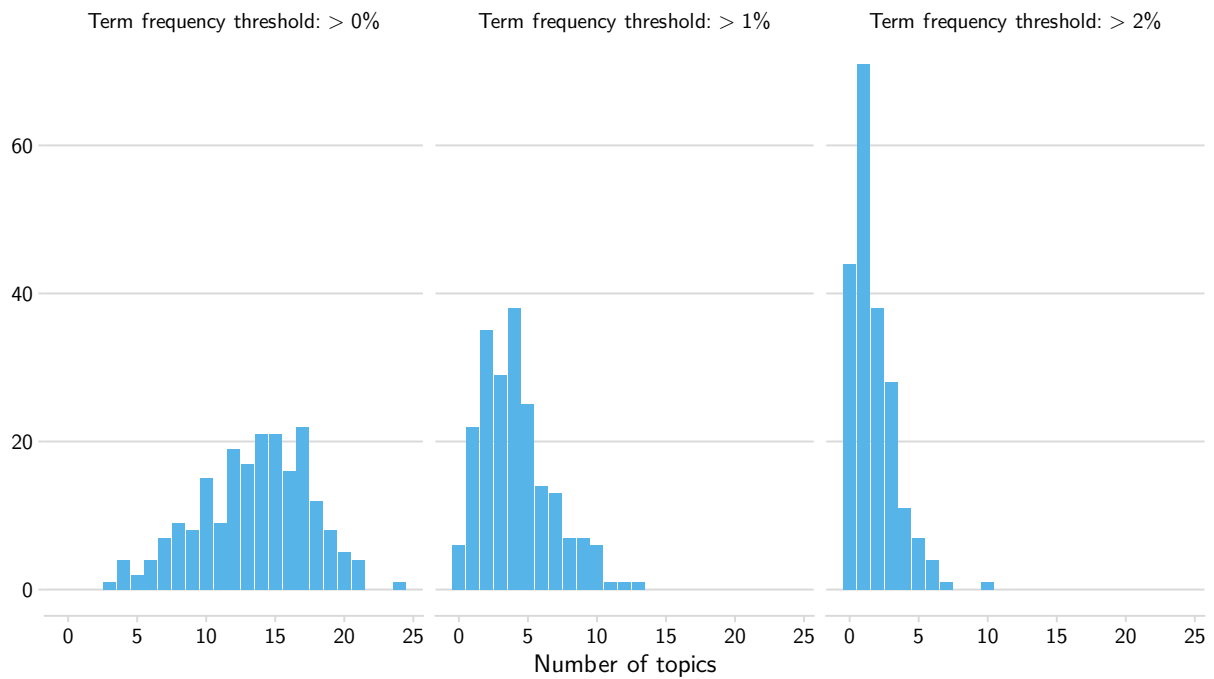
Notes: Each cell in this table represents a ratio of topic exposures between two demographic groups. Exposure is the share of ATUS diaries for a demographic group reporting participating in any activity in the topic. For racial groups, exposure for white respondents is used as the comparison group. Topics are formed combining six-digit ATUS activity codes. Respondent observations are weighted by ATUS sampling weights provided by the U.S. Census Bureau. The sample includes all ATUS respondents from 2013–2019.

Figure A1: Histogram of the topic salience score



Notes: This histogram is constructed using topic-passage level observations. Details on the term-frequency metric used here can be found in Section 3.3. The sample of passages include grade 3 to 8 STAAR reading comprehension tests from 2013–2019.

Figure A2: Number of topics appearing in each passage



Notes: The unit of observation in this figure is a passage. Reading passage measures are calculated for each topic-passage pair by the term-frequency metric discussed in Section 3.3. Using different thresholds, this histogram illustrates how many topics are detected in each passages. The sample of passages include grade 3 to 8 STAAR reading comprehension tests from 2013–2019, for which passages are not copyright restricted.

Table A4: Effect of relatability on falsification outcome variables

	Coeff.	SE	<i>N</i> (race-passage)
Prior year performance...			
by grade-race	-0.0031	(0.0035)	700
by cohort-race	0.0033	(0.0032)	592
by grade-race-passage position	0.0041	(0.0067)	688
by cohort-race-passage position	0.0059	(0.0091)	568
Previous passage perf.	0.0048	(0.0081)	640
Subsequent passage perf.	-0.0006	(0.0066)	640
Population of race by exam	-0.0039	(0.0024)	820
Exam year (continuous)	-0.0747	(0.1049)	820
Passage position (continuous)	0.0450	(0.0675)	820
Passage word count	6.2689	(9.6022)	820
Literary passage	0.0716***	(0.0178)	820
Predicted performance (all variables)	0.0067	(0.0073)	340
Predicted performance (non-missing variables)	0.0019	(0.0032)	820
	Statistic	<i>p</i> -value	
Joint <i>F</i> -test of covariates	3.901	0.001	

Notes: Each row reports the coefficient from a regression of the specified outcome variable on race topic relatability. Unreported controls in each specification include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Standard errors reported in parentheses below coefficient estimates are (a) obtained using shock-level regression following Borusyak et al. (2022) and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Prior-year performance is the test performance for the listed group in the previous year. Previous and subsequent passage performances are scores on the preceding and subsequent passages, respectively. Population of race by exam is the share of test takers who are the same race as the student. Literary passage describes one of two passage categories that make up exams. Observations are at the race-passage level and are weighted by the number of student-item. The estimation sample differs for each specification; it includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test for which the outcome measure is available.

Table A5: Topic-passage-level summary statistics

	(1)	(2)
Topic salience summary		
Mean	0.007	-0.000
Standard deviation	0.013	0.011
Interquartile range	0.009	0.007
Inverse HHI of topic exposure weights		
Across topics and passages	925.461	925.461
Across exams	41.504	41.504
Largest topic exposure weight		
Across topics and passages	0.003	0.003
Across exams	0.029	0.029
Residualized on $tg(p)$ and p FEs		✓
N of topic-passages	4,920	4,920
N of passages	205	205
N of topics	24	24

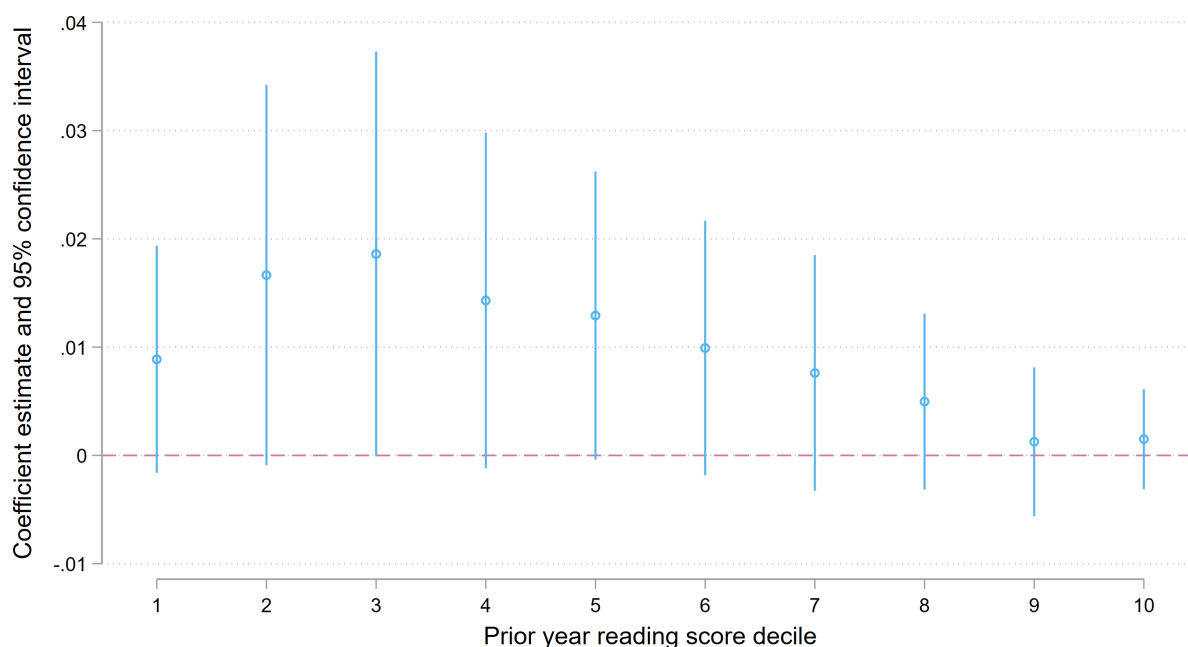
Notes: The table summarizes the distribution of topic salience shocks g_{tp} and the aggregated topic exposure weights at the topic-passage level. Topic salience shocks are developed using NLP over the set of passages. Aggregated topic exposure weights are developed based the topic exposure shares calculated at the demographic group and passage-level. Topic exposure shares are averaged within a topic-passage, with each demographic group topic exposure share weighted by the number of student-items for that demographic group and passage. Then these aggregated values are normalized so that the weights sum to one across all topic-passages. Statistics are weighted by these topic exposure weights. Column (1) displays statistics with no controls. Column (2) displays statistics after residualization on topic-grade ($tg(p)$) and passage (p) FEs. The inverse HHI of topic exposure weights is calculated as the inverse of the sum of squared aggregated exposure weights. Exposure weights are aggregated in two ways, within topic-passage or within exam.

Table A6: Black-white and Hispanic-white test score gaps by grade

	Grade					
	3	4	5	6	7	8
Black-white	-0.136	-0.137	-0.128	-0.127	-0.122	-0.118
Hispanic-white	-0.094	-0.093	-0.094	-0.109	-0.104	-0.100
N of students	2,155,843	2,175,995	2,202,413	2,232,987	2,220,076	2,192,824

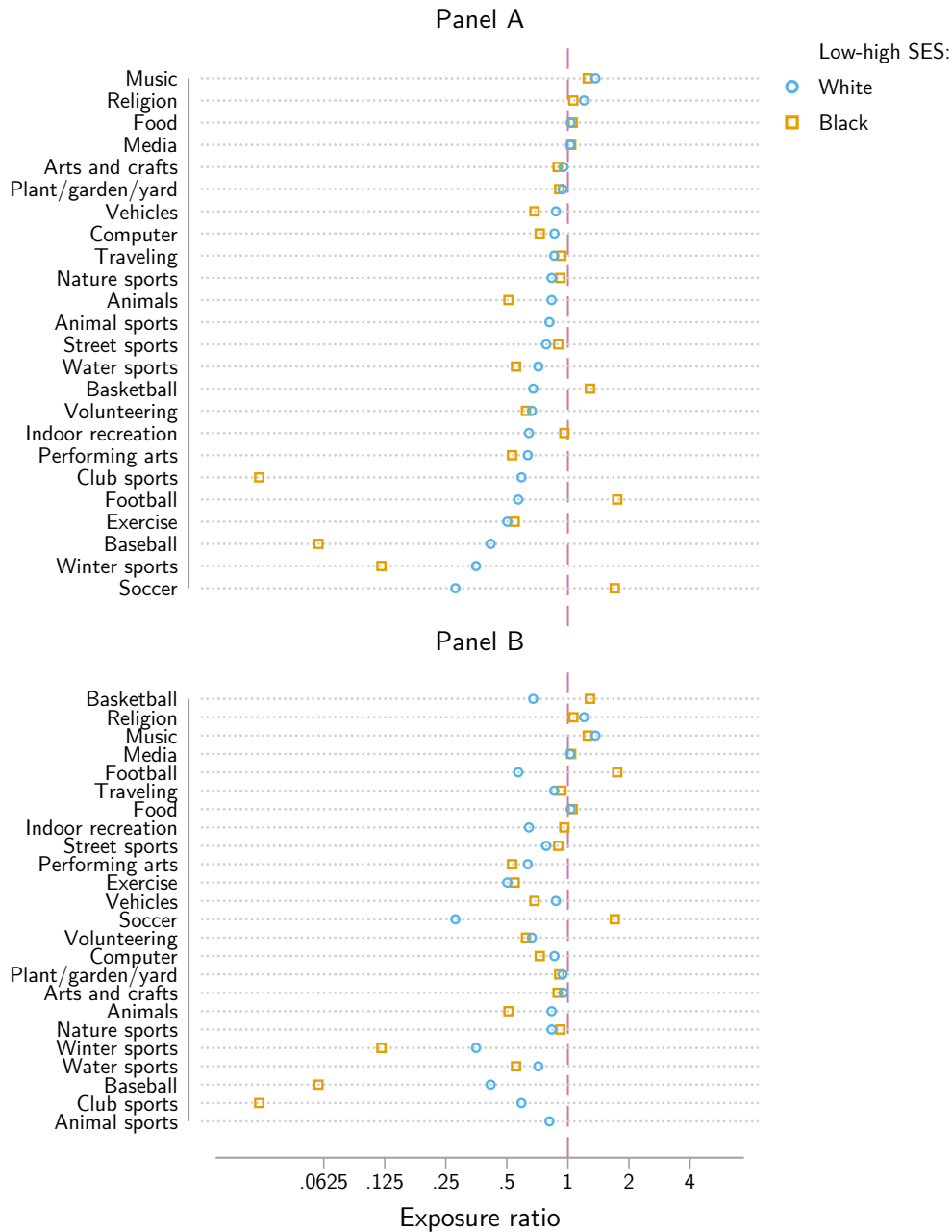
Notes: Each column displays the test score gap between different racial groups for a given grade. Test score gaps are calculated as the average difference between two groups' average share of questions correct at the test-level. The sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Figure A3: Heterogeneity of race topic relatability effects by past reading achievement



Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Each specification is run on a separate student subsample based on the decile of a student's prior-year reading test score. Each student's prior-year reading test score is from the standard STAAR exam for reading comprehension. Score deciles are formed within exam. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 4 to 8 from 2014–2019 taking the standard STAAR reading comprehension test. This differs from the estimation sample in the rest of this manuscript, as prior-year test scores are not available for 2013 and grade 3.

Figure A4: Differences in topic exposure by SES



Notes: Graph plots the ratio of topic exposures between low SES and high SES individuals separately by race. Panel A orders topics by SES exposure ratio for white individuals. Plot B orders topics by the Black-white exposure ratio. The vertical dotted line at 1 corresponds to the value at which both groups have identical exposure. Topic exposure is calculated separately for each racial/ethnicity group using the ATUS data. Topic exposure is the share of ATUS diaries for a demographic group reporting participating in any activity for a topic. The sample includes all ATUS respondents from 2013-2019.

Table A7: Impact of race topic relatability on test performance by time-use sample

	(1)	(2)	(3)	(4)	(5)
Race topic rel. (PSID)	0.0103*** (0.0034)		0.0162*** (0.0059)		
Race topic rel. (ATUS)		0.0187*** (0.0058)		0.0204*** (0.0071)	
ATUS-PSID average					0.0218*** (0.0072)
Black-white rel. diff.	-0.0045 (0.0015)	-0.0048 (0.0015)	-0.0069 (0.0025)	-0.0052 (0.0018)	-0.0068 (0.0022)
Hispanic-white rel. diff.	-0.0014 (0.0005)	-0.0037 (0.0011)	-0.0022 (0.0008)	-0.0040 (0.0014)	-0.0037 (0.0012)
Race-Grade FE	✓	✓	✓	✓	✓
Passage-by-Exposure sum FE	✓	✓	✓	✓	✓
Sample	All	All	Excl. Asian	Excl. Asian	All
<i>N</i> of topic-passages	4,920	4,920	4,920	4,920	4,920
<i>N</i> of student-passages	64,352,860	64,352,860	61,120,843	61,120,843	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on race topic relatability. Standard errors reported in parentheses below coefficient estimates are (a) obtained using a shock-level regression and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. PSID race topic relatability uses the Panel Study of Income Dynamics-Child Development Supplement to calculate race-level topic exposures. ATUS-PSID average uses race-level topic exposure calculated as an average of the ATUS-based topic exposure measure and the PSID-based topic exposure measure, after both topic exposure measures are adjusted using an empirical Bayes shrinkage estimator which shrinks race topic exposure measures to the population-level topic exposure means. Coefficient estimates (and standard errors) alternatively scaled by racial differences in topic relatability are displayed in the bottom rows. This scaling is done by dividing the coefficient estimates by the average Black-White and Hispanic-White relatability gap. Columns (3) and (4) use data only from non-Asian students. The full estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A8: Robustness of baseline race topic relatability results on test performance with school fixed effects

	(1)	(2)
Race topic relatability	0.0152*** (0.0045)	0.0142*** (0.0043)
School-Race FE	✓	
School-Race-Grade FE		✓
School-Passage-by-Exposure sum FE	✓	✓
<i>N</i> of topic-passages	4,920	4,920
<i>N</i> of student-passages	66,494,672	66,494,672

Notes: Each specification is a regression of the share of items answered correctly on a passage on race topic relatability. Standard errors reported in parentheses below coefficient estimates are (a) obtained using a shock-level regression following Borusyak et al. (2022) and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. For columns (1) and (2), race topic relatability is calculated as described in section 4.1. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A9: Heterogeneity of race topic relatability effects along school diversity index

	(1)	(2)	(3)	(4)
Race topic relatability	0.0182*** (0.0050)	0.0193*** (0.0055)	0.0210*** (0.0061)	0.0161* (0.0087)
HHI Quartile	1	2	3	4
Race-Grade FE	✓	✓	✓	✓
Passage-by-Exposure sum FE	✓	✓	✓	✓
<i>N</i> of topic-passages	4,920	4,920	4,920	4,920
<i>N</i> of student-passages	16,623,668	16,623,668	16,623,668	16,623,668

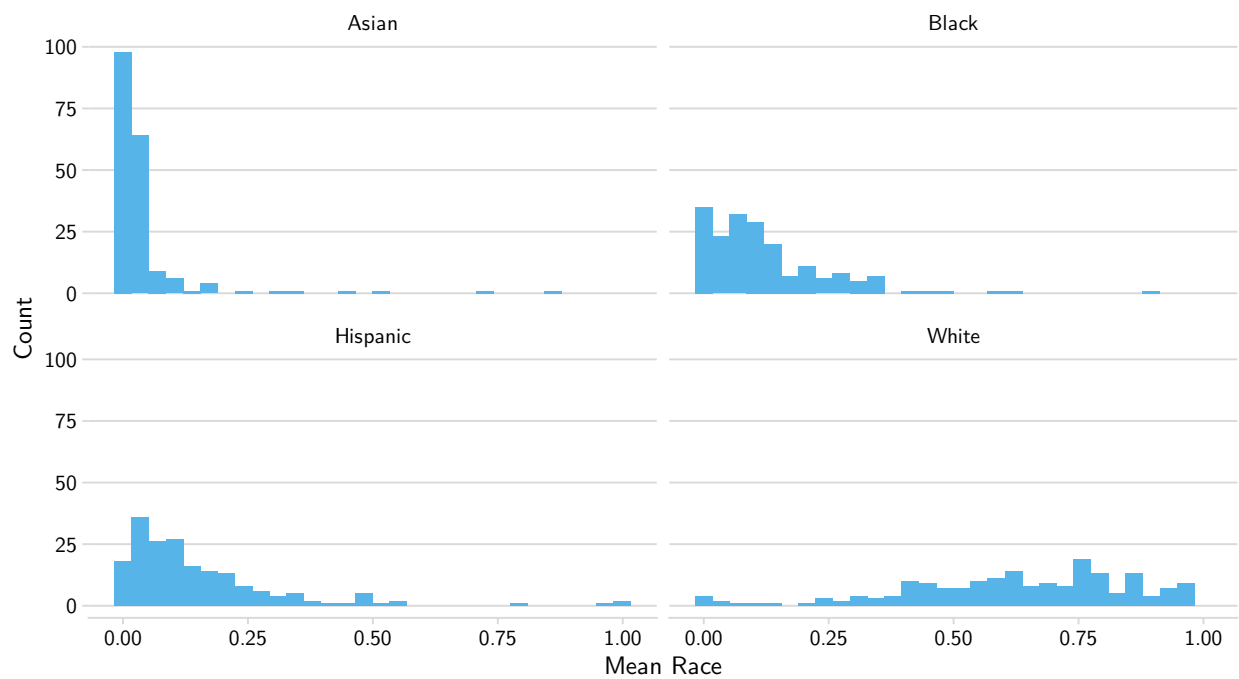
Notes: Each specification is a regression of the share of items answered correctly on a passage on race topic relatability on a subsample of the data. Subsamples are determined by splitting schools into quartiles of a school-integration index (HHI), and thus lower quartiles are more integrated. Standard errors reported in parentheses below coefficient estimates are (a) obtained using a shock-level regression following Borusyak et al. (2022) and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A10: Impact of gender topic relatability on test performance by time-use sample

	(1)	(2)	(3)
Gender topic rel. (PSID)	0.00590 (0.00437)		
Gender topic rel. (ATUS)		-0.000148 (0.00554)	
ATUS-PSID average			0.00427 (0.00726)
Gender-Grade FE	✓	✓	✓
Passage-by-Exposure sum FE	✓	✓	✓
N of topic-passages	4,920	4,920	4,920
N of student-passages	64,352,860	64,352,860	64,352,860

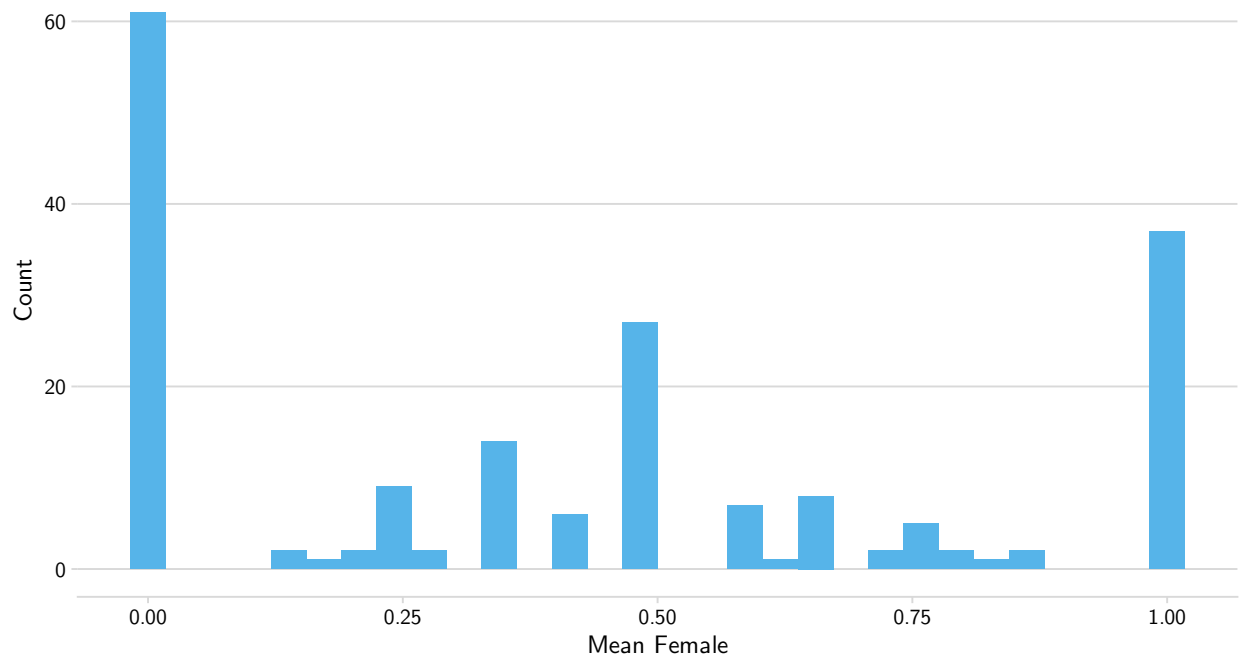
Notes: Each specification is a regression of the share of items answered correctly on a passage on gender topic relatability. Standard errors reported in parentheses below coefficient estimates are (a) obtained using a shock-level regression and (b) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. PSID gender topic relatability uses the Panel Study of Income Dynamics-Child Development Supplement to calculate gender-level topic exposures. ATUS-PSID average uses gender-level topic exposure calculated as an average of the ATUS-based topic exposure measure and the PSID-based topic exposure measure, after both topic exposure measures are adjusted using an empirical Bayes shrinkage estimator which shrinks gender topic exposure measures to the population-level topic exposure means. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Figure A5: Histogram of race identity relatability



Notes: This histogram is constructed using passage level observations. Each observation is the mean predicted race for all characters in a passage, giving each character equal weight. Further details on the race identity relatability can be found in Section 5.1. The sample of passages include grade 3 to 8 STAAR reading comprehension tests from 2013–2019.

Figure A6: Histogram of gender identity relatability



Notes: This histogram is constructed using passage level observations. Each observation is the mean predicted gender for all characters in a passage, giving each character equal weight. Further details on the gender identity relatability can be found in Section 5.1. The sample of passages include grade 3 to 8 STAAR reading comprehension tests from 2013–2019.

Table A11: Additional robustness of impact of race identity relatability on test performance

	Levels		Above Race Median	
	(1)	(2)	(3)	(4)
Race id. rel. (# Characters)	0.0010 (0.0008)			
Race id. rel. (# Mentions)		0.0000 (0.0001)		
Above median race id. rel. (mean)			0.0037*** (0.0010)	
Above median race id. rel. (w.mean)				0.0038*** (0.0011)
Reading Passage FE	✓	✓	✓	✓
Race-Grade FE	✓	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on race identity relatability. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Identity relatability here is defined in levels rather than shares for columns (1) and (2): column (1) has number of characters matching the students race, whereas column (2) counts the total number of mentions of characters matching the (2). Note that since racial predictions are between 0 and 1, this more precisely corresponds to the “expected value” level, rather than true level. The racial prediction model is wru+, which uses last names from social security data, first names from six Southern states, and imputes average Texas demographics if missing. Columns (3) and (4) use the same identity relatability share values as the main regression, but further dichotomize the score to be an indicator for above the median identity relatability score within race. Thus, if the median Hispanic share of passages is 20%, passages predicted to have more than 20% of Hispanic characters are given a 1 for Hispanic students and others are given a 0. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A12: Additional robustness of impact of gender identity relatability on test performance

	Levels		Above Gender Median	
	(1)	(2)	(3)	(4)
Gender identity rel. (# Characters)	0.0025*** (0.0006)			
Gender identity rel. (# Mentions)		0.0002* (0.0001)		
Above median gender id. rel. (mean)			0.0065*** (0.0014)	
Above median gender id. rel. (w.mean)				0.0043*** (0.0014)
Reading Passage FE	✓	✓	✓	✓
Gender-Grade FE	✓	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on gender identity relatability. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Identity relatability is defined as share of characters matching the students gender. All columns calculate identity relatability using gender labels from a large language model, imputing from pronouns and other context clues. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A13: Heterogeneity in the impact of race identity relatability on test performance

	(1)	(2)	(3)	(4)	(5)	(6)
Race identity rel. (mean)	0.0114*** (0.0040)	0.0116*** (0.0040)	0.0125*** (0.0039)	0.0112*** (0.0040)	0.0097** (0.0039)	0.0114** (0.0042)
Race identity rel. (mean) $\times$ Historic	-0.0048 (0.0035)					
Race identity rel. (mean) $\times$ Celebrity		-0.0046 (0.0059)				
Race identity rel. (mean) $\times$ Famous			-0.0062 (0.0043)			
Race identity rel. (mean) $\times$ Any children				-0.0040 (0.0033)		
Race identity rel. (mean) $\times$ Memoir					0.0019 (0.0030)	
Race identity rel. (mean) $\times$ No author						-0.0019 (0.0033)
Reading Passage FE	✓	✓	✓	✓	✓	✓
Race-Grade FE	✓	✓	✓	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on race identity relatability. Details of additional variables are described in Section B.5. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Identity relatability is defined as share of characters matching the students race. The racial prediction model is wru+, which uses last names from social security data, first names from six Southern states, and imputes average Texas demographics if missing. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A14: Heterogeneity in the impact of gender identity relatability on test performance

	(1)	(2)	(3)	(4)	(5)	(6)
Gender identity rel. (mean)	0.0041** (0.0018)	0.0060*** (0.0018)	0.0039* (0.0019)	0.0076*** (0.0023)	0.0108*** (0.0025)	0.0012 (0.0029)
Gender identity rel. (mean) $\times$ Historic	0.0133*** (0.0044)					
Gender identity rel. (mean) $\times$ Celebrity		0.0075* (0.0039)				
Gender identity rel. (mean) $\times$ Famous			0.0114*** (0.0039)			
Gender identity rel. (mean) $\times$ Any children				-0.0001 (0.0034)		
Gender identity rel. (mean) $\times$ Memoir					-0.0086** (0.0040)	
Gender identity rel. (mean) $\times$ No author						0.0116*** (0.0041)
Reading Passage FE	✓	✓	✓	✓	✓	✓
Gender-Grade FE	✓	✓	✓	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on gender identity relatability. Details of additional variables are described in Section B.5. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Identity relatability is defined as share of characters matching the students gender. All columns calculate identity relatability using gender labels from a large language model, imputing from pronouns and other context clues. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A15: Impact of survey reliability on test performance

	(1)	(2)	(3)
Race survey reliability	0.0092** (0.0038)		0.0092** (0.0038)
Gender survey reliability		0.0137*** (0.0021)	0.0137*** (0.0021)
Black-white rel. diff.	-0.0031 (0.0013)		-0.0031 (0.0013)
Hispanic-white rel. diff.	-0.0030 (0.0012)		-0.0030 (0.0013)
Male-female rel. diff.		0.0025 (0.0004)	0.0025 (0.0004)
Race-Grade FE	✓		
Gender-Grade FE		✓	
Race-Gender-Grade FE			✓
Passage FE	✓	✓	✓
<i>N</i> of student-passages	64,352,860	64,352,860	64,352,860

Notes: Each specification is a regression of the share of items answered correctly on a passage on survey reliability. Standard errors reported in parentheses below coefficient estimates are clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Race survey reliability and gender survey reliability are created using separate survey questions eliciting assessments of reliability for a passage. Coefficient estimates (and standard errors) alternatively scaled by racial and gender differences in survey reliability are displayed in the bottom rows. The full estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

Table A16: Correlation across race reliability measures

	(1)	Time-use (2)	(3)	Identity (4)	(5)	Survey (6)
Time-use reliability					-0.337 (0.383)	0.481 (0.325)
Identity reliability	-0.00734 (0.0135)		-0.0108 (0.0129)			0.302*** (0.0816)
Survey reliability		0.00733 (0.00957)	0.0115 (0.00777)	0.223*** (0.0618)	0.226*** (0.0614)	
Within FE R^2	0.002	0.002	0.007	0.067	0.070	0.072

Notes: Each specification is a regression of one race reliability measure on other race reliability measures. Each specification includes race-grade fixed effects and passage fixed effects. Standard errors reported in parentheses below coefficient estimates are (a) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Within fixed effect R^2 are displayed for each specification. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test which contains characters.

Table A17: Correlation across gender reliability measures

	(1)	Time-use (2)	(3)	Identity (4)	(5)	Survey (6)
Time-use reliability					0.297 (0.437)	0.207 (0.307)
Identity reliability	0.0133 (0.0142)		0.00956 (0.0140)			0.564*** (0.0557)
Survey reliability		0.0145 (0.0114)	0.00656 (0.0109)	0.560*** (0.0450)	0.556*** (0.0464)	
Within FE R^2	0.008	0.009	0.009	0.317	0.319	0.318

Notes: Each specification is a regression of one gender reliability measure on other gender reliability measures. Each specification includes race-grade fixed effects and passage fixed effects. Standard errors reported in parentheses below coefficient estimates are (a) clustered by exam. * $p < .10$, ** $p < .05$, *** $p < .01$. Within fixed effect R^2 are displayed for each specification. The estimation sample includes students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test which contains characters.

Table A18: Effect of balancing reliability across tests on share of Black students meeting performance standards

	Grade 3	4	5	6	7	8
Below Approaches (pp)	-0.005	-0.002	-0.002	0.001	-0.003	-0.001
Below Meets	-0.006	-0.002	-0.002	0.001	-0.004	-0.002
Below Masters	-0.005	-0.002	-0.002	0.001	-0.003	-0.002

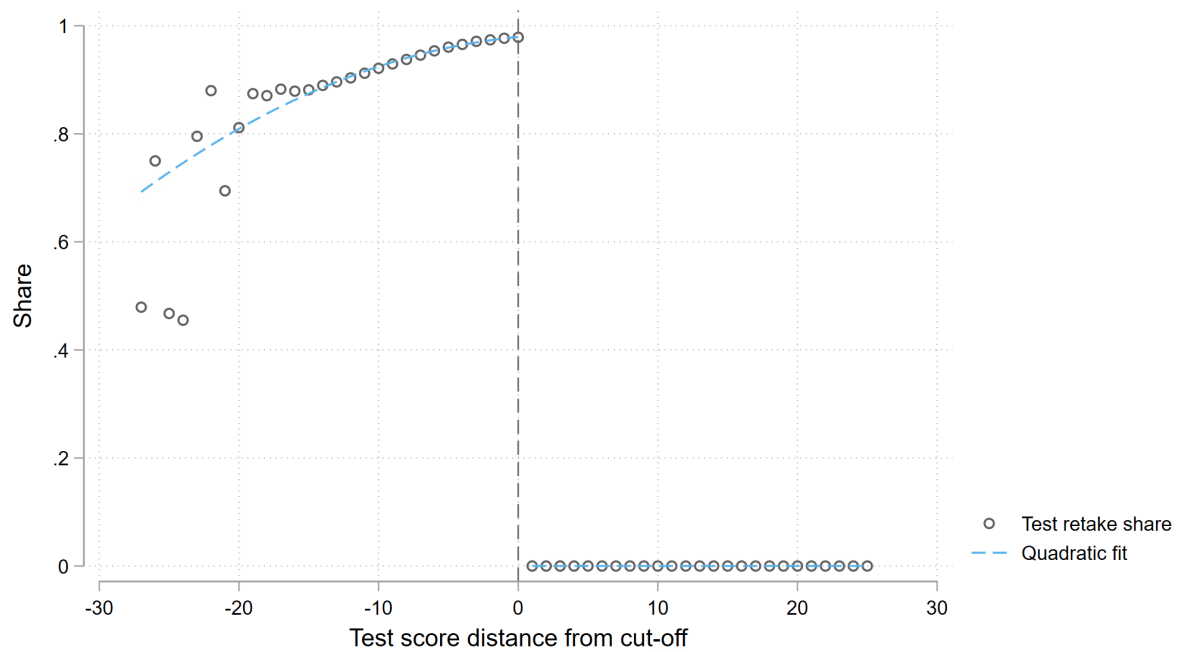
Notes: Numbers show the change in the share of students below a given performance standards category by rescaling reliability to a benchmark test within the grade. Observations are at the test-level. The sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test. Students taking the benchmark test within each grade is excluded.

Table A19: Effect of balancing reliability across tests on share of Hispanic students meeting performance standards

	Grade 3	4	5	6	7	8
Below Approaches (pp)	-0.002	-0.002	-0.001	-0.003	-0.002	-0.002
Below Meets	-0.003	-0.003	-0.001	-0.003	-0.003	-0.003
Below Masters	-0.003	-0.003	-0.001	-0.003	-0.003	-0.003

Notes: Numbers show the change in the share of students below a given performance standards category by rescaling reliability to a benchmark test within the grade. Observations are at the test-level. The sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test. Students taking the benchmark test within each grade is excluded.

Figure A7: 5th and 8th test retake rates by distance from cut-off



Notes: This graph plots the share of 5th and 8th grade students within each score that retook the reading comprehension exam. A student is considered to retake an exam if they take the initial reading comprehension exam and subsequently takes the next available reading comprehension exam in the same school year. The score is defined as the number of questions the student got correct on the first exam minus the TEA-designated cut-off for satisfactory performance on the exam, such that a positive score means the student performed satisfactorily on the exam and a negative score means the student did not perform satisfactorily on the exam. The dashed line plots a quadratic fit on either side of the cut-off. The sample includes all students grades 5 and 8 from 2013–2019 taking the standard STAAR reading comprehension test, excluding years and grades where test retakes were not offered.

B Construction of additional variables

B.1 Additional race topic exposure measures

In addition to our standard measure of race topic exposure, we construct additional measures through various changes to the process described in section 3.2.2. We first make various restrictions to the ATUS sample. We then aggregate data from time use diaries differently.

We create subsamples of the ATUS sample across five dimensions: (a) weekend/weekday survey response, (b) U.S. Census region, (c) state, (d) age, and (e) parental status. First, since ATUS respondents give a diary of a single day, if there is heterogeneity in activities between the weekend and weekday, this may affect our relatability measure. We split the sample by whether the respondent’s diary day is for Saturday or Sunday (weekend) or Monday through Friday (weekday). Second, while we leverage data respondents from across the country for predictive power, Texans may behave substantially differently than other Americans. To account for this possibility, we split the sample by whether or not respondents are in the “South” U.S. Census Region which includes Texas.²³ We also create a Texans-only sample which has low respondent counts but potentially even higher alignment to Texans interest and familiarity. Third, there may be differences across age in activity participation. We create the following four subsamples by age: 15-34, 35-49, 50-64, and 65-85. The four subsamples are mutually exclusive and almost completely exhaustive age ranges that roughly split the sample into quartiles. Finally, if we are using adults to proxy for the familiarity of children, our prediction may be more accurate for adults with children. Thus, we identify respondents with children and respondents without children.

Ultimately, after creating exposure measures for the subsamples described above, we recalculate topic relatability measure, r . We ultimately use 5 alternative relatability measures:

$$r^{weekend}, r^{south}, r^{texas}, r^{parent}, r^{age15-34}.$$

²³Other states in the South Census Region include Oklahoma, Arkansas, Louisiana, Mississippi, Tennessee, Kentucky, Alabama, Florida, Georgia, North Carolina, Virginia, South Carolina, West Virginia, Maryland, and Delaware, as well as Washington, D.C.

We next create new measures of race topic exposure aggregating time use data differently. Previously, we calculated exposure e_{dt} simply as the fraction of respondents of demographic group $d \in \mathcal{D}$ who report any activity related to topic $t \in \mathcal{T}$. We can think of this as an extensive margin measure of topic exposure. Alternatively, we can make use of the intensity of the time spent related to a particular topic.

We create four alternative “intensive margin” measures of topic exposure. Our first measure uses an average of minutes spent in a particular topic for a demographic group

$$e_{dt}^{int_simp} = \frac{1}{W_d} \sum_{j \in d} W_j time_{jt},$$

where W_j is the survey weight for respondent j and W_d is the total survey weight for respondents in group d . Our next two measures transform minutes spent using an inverse hyperbolic sine function and a square root function prior to computing group-level averages:

$$e_{dt}^{int_sine} = \frac{1}{W_d} \sum_{j \in d} W_j \operatorname{arcsinh}(time_{jt})$$

$$e_{dt}^{int_sqrt} = \frac{1}{W_d} \sum_{j \in d} W_j \sqrt{time_{jt}}$$

Finally, we construct a measure based on the average percentile of minutes spent in a particular topic category:

$$e_{dt}^{int_tile} = \frac{1}{W_d} \sum_{j \in d} W_j \operatorname{PercentileRank}_t(time_{jt})$$

where $\operatorname{PercentileRank}_t(time_{jt})$ is the population-level percentile rank of time spent in topic t for respondent j .

This process yields four additional race topic relatability measures when combined with our baseline topic salience measures.

B.2 Measures of topic salience

Formally, our baseline measure is calculated as follows. We count the number of times some word w is in passage p and denote it $\text{count}(w, p)$. Then, we calculate *term-frequency* as $\text{tf}(w, p) = \frac{\text{count}(w, p)}{|W_p|}$, where W_p is the set of words in passage p (including repeated words). As our main measure of topic salience, we define:

$$m_{t,p} \equiv \sum_{w \in B_t} \text{tf}(w, p) .$$

While term frequency is our preferred definition of topic salience, we also consider two other definitions. The simpler being the *count* metric, which is simply $m_{t,p} \equiv \sum_{w \in B_t} \text{count}(w, p)$ where $\text{count}(w, p) \equiv \sum_{v \in W_p} \mathbb{1}[v = w]$ and W_p is the set of words in passage p (including repeated words). Another we consider adds a weight for “uniqueness” of a word. We can multiply the term frequency by an *inverse document frequency* measure, defined as $\text{idf}(w) \equiv \log \left(\frac{|P|}{\sum_{p \in P} \mathbb{1}[w \in W_p]} \right)$, which is zero for words that appear in all passages and emphasizes words that are less commonly used across passages. Here, we define $m_{t,p} \equiv \sum_{w \in B_t} \text{tf}(w, p) \cdot \text{idf}(w)$.

We also consider many variations of the dictionary method such as “stemming” the words, using only nouns, and removing words from the dictionary.²⁴ Further, while our empirical strategy leverages the intensive margin variation in the methods described so far, we do consider discrete shocks of topics by setting a topic shock to 1 if it is in the top 1/25th of m_{tp} within the grade-level and 0 otherwise. Thus, on average each passage will be about one topic but some passages will be about nothing and others will have multiple topics.

B.3 PSID topic relatability

We use the Child Development Supplement (CDS) of the Panel Study of Income Dynamics (PSID) to create an alternative measure of topic relatability. The PSID is a longitudinal

²⁴We stem the words using Porter’s stemming algorithm (Willet 2006) to collapse all instances of a word to a shared stem. For example, this algorithm would transform *running*, *runs*, and *ran* to the stem *run*.

survey of U.S. families. Families were selected and interviewed in 1968, and every subsequent wave has mostly included individuals from these families and their descendants. The CDS component of the PSID, which focuses on information regarding children, was launched in its initial form in 1997 and updated in 2014. The CDS includes children’s daily time diaries from one weekday and one weekend, filled out by the caregiver or the child, depending on the age of the child.

We collect data from the 2014 and 2019 waves of the PSID-CDS. As our measure of a child’s race/ethnicity, we primarily use child’s reported race from the parent(s). If both parents’ reports are aligned, the child is assigned that race/ethnicity. If there is a report from only one parent, the child is assigned the race/ethnicity that is reported by that parent. If parents disagree, they are coded as mixed race. We use the child’s report of their race only for children that are missing race/ethnicity characterizations. We define race as we have throughout this paper, with Hispanic as its own “race” and all other racial groups being non-Hispanic. We restrict our analysis to children between the ages of 8 and 17.

We construct the topic exposure much in the same as we do for the ATUS data. We assign each PSID activity code to a topic category t . We observe a single weekday diary and a single weekend diary for the vast majority of children in the CDS. We opt to combine these minutes for each activity. Then, we calculate PSID topic exposure as the fraction of children of demographic group who report any activity related to topic $t \in \mathcal{T}$, using sampling weights provided by the PSID-CDS.

Table [A20](#) provides summary statistics for the PSID sample. Comparing the PSID sample to the ATUS sample summary (displayed in Table [2](#)), we find that children on average do a higher number of activities than adults do. This may reflect the difference between the nature of children’s time use and adult’s time use, or this may reflect differences in activity coding across the PSID and the ATUS. Children engage in more leisure activity than adults; on average, children in the sample are exposed to one additional topic compared to adults and spend 30% of their time in these topics compared with 20% for adults. We find under-

Table A20: Summary statistics for PSID-CDS sample

	Mean	SD	5%	95%
Activities				
Number of activities	17.33	4.73	9	25
Number of min spent per done activity	88.74	30.38	55	144
Number of topics	3.42	1.59	1	6
Share of time spent in topics	0.30	0.16	0.08	0.60
Demographics				
Asian	0.02			
Black	0.14			
Hispanic	0.16			
White	0.58			
Female	0.47			
<i>N</i>	1,555			

Notes: The table displays sample means, standard deviations, and the 5th/95th percentile value for each category. Respondent observations are weighted by PSID-CDS sampling weights provided by the Panel Study of Income Dynamics. The sample includes PSID-CDS respondents from 2014 and 2019 aged between 8 and 17.

sampling of white and female respondents compared to the ATUS sample.

The demographic distribution in the weighted PSID sample masks significant imbalance across racial groups in the underlying observations. The weighted ATUS sample and the unweighted ATUS sample are fairly similar in its racial composition, with deviations fewer than 2 percentage points for all racial groups. By contrast, the unweighted PSID sample deviates significantly from the weighted PSID sample. Only 15 Asian children are included in the PSID sample, making up 1% of the observations. Hispanic children are 9% of the raw sample, totaling 134 observations. By contrast, Black children make up 42% of the sample. These differences reflect the structure of the PSID, which uses descendants of the 1968 initial cohort as its main study sample. While recent waves of the PSID have added additional racial groups, they still do not fully reflect the experiences of recently immigrated populations to the U.S.

B.4 Leveraging non-race- and non-gender-based topic exposure

The ATUS data and the student data have additional measures of individual demographic characteristics beyond race and gender such as economic disadvantage. We outline below a process which allows us to leverage socioeconomic status variation to construct new measures of exposure and relatability.

We start by bringing SES definitions in the ATUS in close concordance to the level of variation that exists in the student data. For economic disadvantage, the item-level testing data contains an indicator for whether a student is on free lunch, on reduced-price lunch, or is on another social insurance programs provided by the state or the federal government. We collapse this measure into a binary variable of economic disadvantage indicating whether or not the student participates in any program. We do not observe directly in the ATUS data whether the respondent lives in a household which participates in any social insurance program or has a child on free or reduced-price lunch. Instead, we observe a respondent’s household income range and household size. Since household participation in state or federal assistance programs—free and reduced-price lunch included—is often tied to being at or below federal poverty line thresholds, we use the income range, household size, and federal poverty line tables to determine a respondent’s distance to the poverty line. Respondents whose household income is at or below 200% of the federal poverty line are classified as low SES. Since income is reported as falling within a range, it is not possible to classify some respondents using this method. We drop such respondents from the data when calculating exposure by economic disadvantage due to this ambiguity. This may result in sharper topic exposure differences across SES than may be true in the actual population.

In our analysis, we use a measure of race-SES topic relatability. The topic exposure basis for this variable is generated in a procedure much like before. We categorize respondents into groups $c \in \mathcal{C}$ which is the race-SES status of each respondent. Then e_{ct} is simply the share of respondents in group c which reported any number of minutes participating in activities related to topic t . Once we obtain $e_{ct} \forall c, t$, we can calculate both E_c , the sum of

topic exposure for c , and r_{cp} , the relatability of passage p to group c .

B.5 Measures of identity relatability

We ask the LLM for a few additional attributes of each character, which we leverage for the heterogeneity analyses in Tables A13 & A14. We ask the LLM whether the character is a historical figure, whether they might be considered a celebrity to a grade-school student, whether the character is a child or adult, and whether the character is an author. Passages are then classified as Historic or Celebrity if they contain *any* character flagged with that attribute, and Famous is the union of the two. A passage is labeled to have “Any Children” if at least one character is classified as a child. The Memoir and No author labels are for passages that either have an author that is mentioned by name more than once or do not have any author detected by the LLM at all.

C Topic salience validation

We validate our topic salience metric with manual topic labeling by two student research assistants. The labeling task is designed to capture the relative salience across different topics within passage as well as the intensive margin variation in a topic’s presence across passages. That is, the ordinal and cardinal properties of passage topics. To do this, each research assistant was instructed to read the entirety of a passage (ignoring the contents of the related question items) and perform two labeling activities. First, having the list of topics we consider for our analysis, the research assistants may select between 0 and 3 topics that appear in the passage and rank them ordinally in terms of relative salience.²⁵ Second, if they reported at least one topic as appearing in the passage, they would categorize the topic they ranked as number one to be either high, medium, or low salience in this passage.

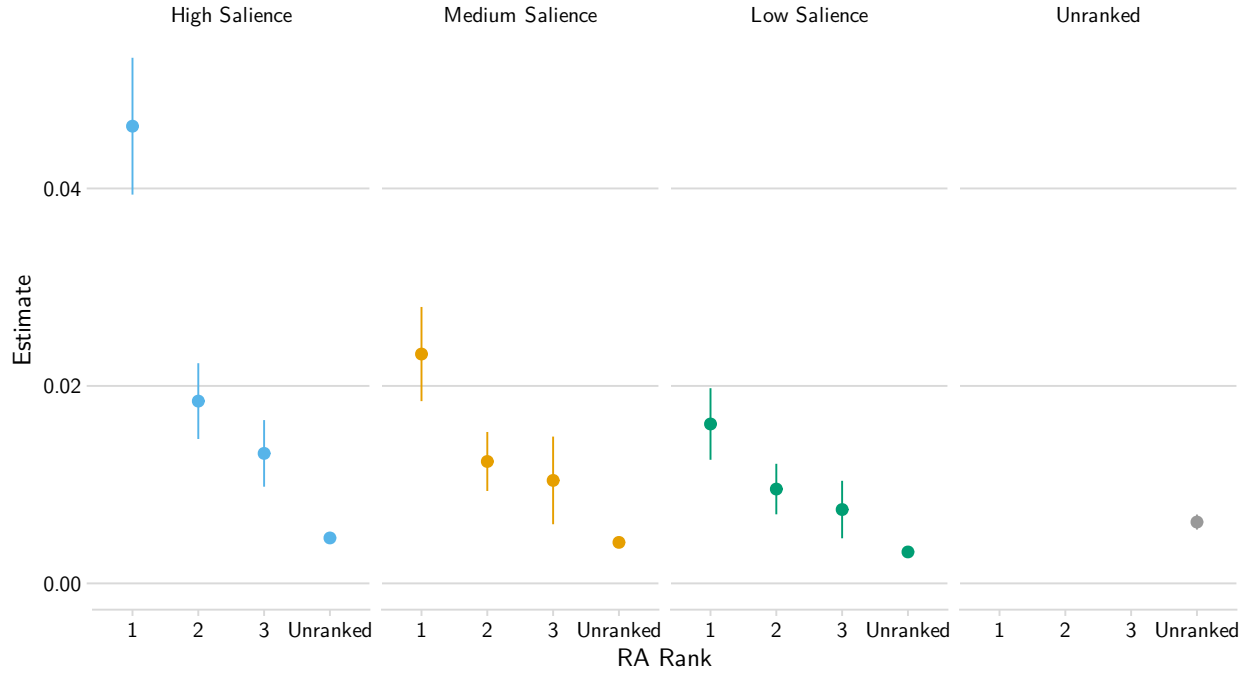
Now, consider how our topic salience metric m_{tp} should relate to the human labeling results. We would expect that when a research assistant ranks topics t, t', t'' , respectively, as the 1st, 2nd and 3rd most salience topics in a passage, that $m_{tp} \geq m_{t'p} \geq m_{t''p}$. Further, if three passages p, p', p'' are labeled as a high, medium and low salience passage with respect to top-ranked topic t , we would expect that $m_{tp} \geq m_{tp'} \geq m_{tp''}$.

To test whether our topic salience metric aligns with these expectations, consider the regression

$$\begin{aligned}
 m_{tpi} = \sum_{k \in \{1,2,3,\text{unranked}\}} & \left(\alpha_k \times \mathbb{1}[\text{rank}_{tpi} = k] \times \mathbb{1}[\text{salience}_{pi} = \text{high}] \right. \\
 & + \beta_k \times \mathbb{1}[\text{rank}_{tpi} = k] \times \mathbb{1}[\text{salience}_{pi} = \text{medium}] \\
 & + \gamma_k \times \mathbb{1}[\text{rank}_{tpi} = k] \times \mathbb{1}[\text{salience}_{pi} = \text{small}] \Big) \\
 & + \mu \times \mathbb{1}[\text{salience}_{pi} = \text{unranked}] + \varepsilon_{tpi}
 \end{aligned}$$

²⁵The research assistants received no information about the methodology we used to classify the passages nor do they know about the dictionaries used in our NLP approach.

Figure C1: Regression of the topic salience scores m_{tp} on research assistant labeling



Notes: Standard errors clustered by passage are used to construct the 95% confidence intervals. Observations are at the topic-passage-RA level, while the outcome only varies at the topic-passage level. The estimation sample is all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

where $rank_{tpi}$ is the rank research assistant i assigned topic t in passage p and $salience_{pi}$ is the research assistant i 's assigned salience level of the topic with the highest ranking in passage p . We first test whether $\theta_1 \geq \theta_2 \geq \theta_3 \geq \theta_{\text{unknown}}$ for each $\theta \in \{\alpha, \beta, \gamma\}$. To verify our NLP scores capture the intensive margin variation well, we next test $\alpha_1 \geq \beta_1 \geq \gamma_1$. The estimated coefficients displayed in Figure C1 exhibit the expected properties, suggesting that our dictionary-based methodology accurately reflects how a typical person may describe the relevant characteristics of a passage. We conduct this same analysis separately for each research assistant, and the results are qualitatively similar.

D Proper estimation of standard errors

Our estimation strategy, which closely follows Borusyak et al. (2022), relies on quasi-random variation in topic salience to identify the causal effect of relatability on student performance. We present our main specification as a regression in “standard” form, that is, at the race-passage level in which we observe our data. However, in actuality, given that our true, identifying variation occurs at the topic-passage level, we calculate standard errors in a regression at the topic-passage level. This is because since students receive common topic salience “shocks,” we must account for the possibility that relatability and the residual may be correlated across racial groups. We detail below exactly how we obtain these “exposure”-robust standard errors using our main specification.²⁶ While we illustrate this approach for our main specification, the approach straightforwardly applies to all additional regressions relying on our core identification assumptions laid out in section 4.2.

We first define N as the number of student-item dyads which make up our underlying estimation sample and N_{dp} as the number of student-item dyads which correspond to race d and passage p . This allows us to formally define our regression weights $w_{dp} \equiv \frac{N_{dp}}{N}$.

Next, we separately residualize Y_{dp} and r_{dp} on the same controls variables in equation (4) through a regression with w_{dp} weights, obtaining residuals of $Y_{dp}^\perp$ and $r_{dp}^\perp$, respectively. In order to convert the outcome and independent variables from the race-passage level, for each topic t in passage p , we calculate a weighted average of each variable weighted by number of underlying observations w_{dp} and exposure e_{dt} :

$$\bar{v}_{tp}^\perp = \frac{\sum_d w_{dp} e_{dt} v_{dp}^\perp}{\sum_d w_{dp} e_{dt}} \quad (9)$$

with $v \in \{Y, r\}$.

²⁶As discussed, the approach for calculating “exposure”-robust standard errors was formalized in Borusyak et al. (2022) and adapted for our specific setting.

Finally, we estimate an IV model with second-stage equation

$$\bar{Y}_{tp}^{\perp} = \varepsilon + \beta \bar{r}_{tp}^{\perp} + q'_{tp} \psi + \zeta_{tp} \quad (10)$$

where in the first-stage equation we instrument $\bar{r}_{tp}^{\perp}$ using m_{tp} , $q'_{tp} \psi$ includes topic-grade fixed effects and passage fixed effects, and the equations are weighted by $\sum_d w_{dp} e_{dt}$. Standard errors are clustered at the exam-level, which reflects potential positive and negative correlation between topic salience within passage and across passages for the same exam. Borusyak et al. (2022) shows the coefficient on $\bar{r}_{tp}^{\perp}$ using this specification is equivalent to the coefficient obtained from estimating the “standard” form regression.

To make sense of the form of this final specification, we intuitively explain each step of this calculation process. First, we need to purge all non-identifying variation from both student performance and reliability. Then, we effectively “unpack” our data by recognizing that (t, p) underlies (d, p) . After disaggregating the data to the (d, t, p) -level, we collapse the variables across race, but emphasizing observations with more underlying student and item data (which contribute more to our “standard” regression estimates) and racial groups with higher exposure to the topic. The two-stage IV strategy isolates just the variation in reliability that is driven by topic salience. Further, the topic-grade fixed effects and passage fixed effects are exactly analogous to the race-grade fixed effects and passage-by-exposure sum fixed effects in the “standard” regression; when disaggregating from the race-passage level and aggregating to the topic-passage level, race fixed effects become topic fixed effects and passage-by-exposure sum effects collapse to passage fixed effects. Crucially, while we have already orthogonalized Y_{dp} and r_{dp} with race-grade fixed effects and passage-by-exposure sum fixed effects, we require topic-grade fixed effects and passage fixed effects to purge this variation in m_{tp} .

E Robustness

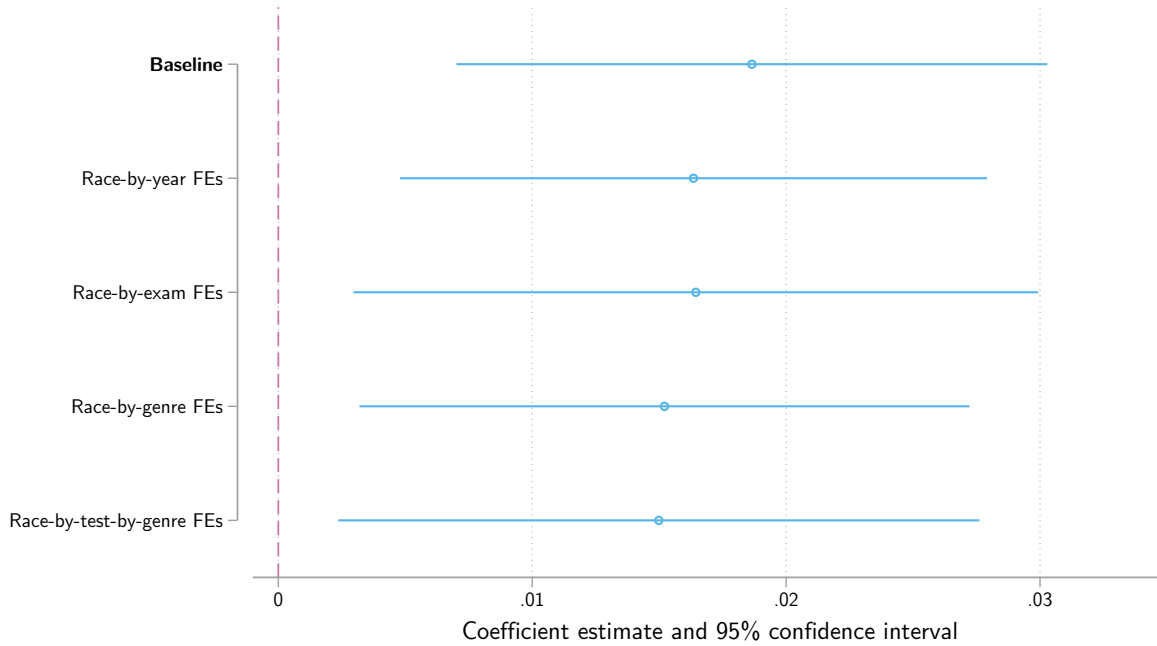
Our estimates are downstream of modeling and data choices that are baked into our race-based relatability measure and model specification. First, we show that the estimates are qualitatively the same with alternative fixed effect specifications. Second, we show that our results are not sensitive to specific choices pertaining to the relatability measure, considering topic categories to include from the ATUS, the natural language processing algorithms or metrics, and the ATUS respondent sample construction.

Fixed effects Given our empirical strategy, intuitively, the model’s fixed effects are intended to condition sufficiently such that we have plausibly exogenous topic salience measures drawn for each passage. Accordingly, our baseline specification uses race-by-grade fixed effects. We do consider more and less granular specifications (see Figure E1). Using race-by-exam fixed effects (i.e. race-by-grade-by-year) gives similar results to baseline with a slightly attenuated coefficient. Using race-by-genre fixed effects, which is conditioning on passage category (“literary”, “informational”, or “mixed”), also results in a qualitatively similar coefficient, while making the comparison set less restrictive.

Leave-one-out Examining the relatability measure, we consider the choice of which topic categories we include. The baseline estimate has 25 topic categories, and in Figure E2 we demonstrate that the coefficient is relatively stable to the removal of any individual category. Quantitatively, the outliers that swing down our estimates the most (relative to the baseline) are a maximum of a 30% decrease in the point estimates, and these differences are not statistically significant.

Topic salience Next, we consider using alternative measures of topic salience. We deviate from the baseline metric of term frequency and consider the term frequency-inverse document frequency (tf-idf) measure (see Appendix Section B.2 for details). Further, we consider a variety of changes to the NLP data processing steps such as leaving the words unstemmed,

Figure E1: Robustness of baseline effect to different levels of saturation in the specifications of fixed effects

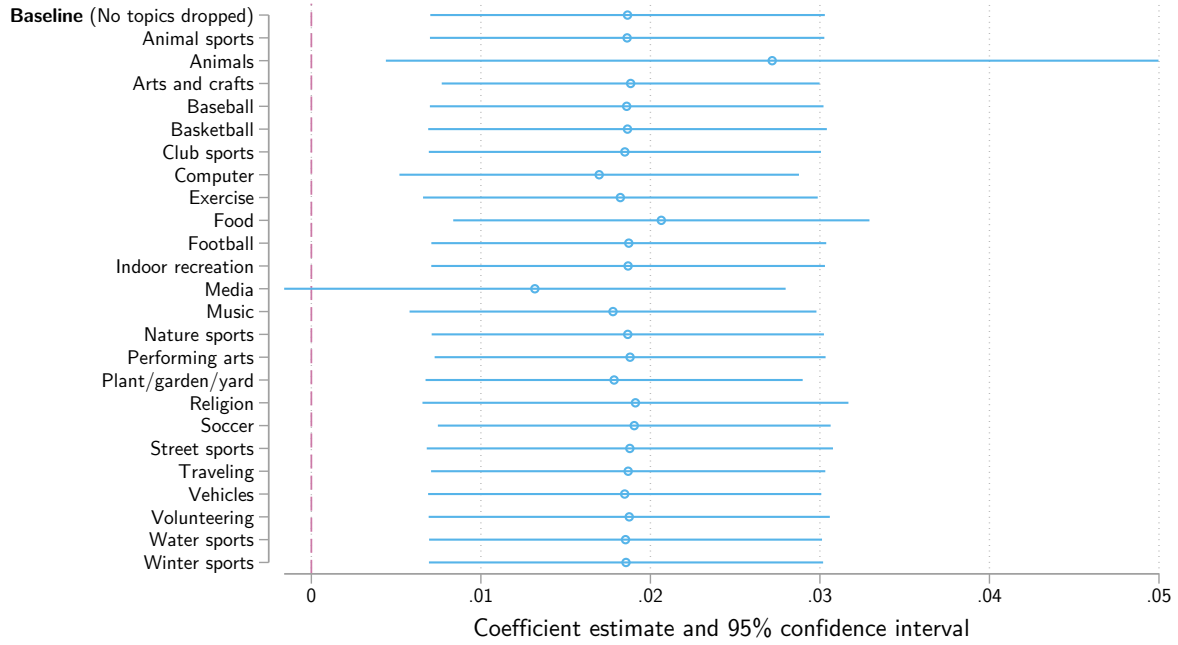


Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) “unit” fixed effects at the level indicated in the legend and (2) exposure-sum-by-passage fixed effects. The baseline specification includes unit fixed effects at the race-by-grade level. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

using only the nouns, and discretizing the shocks. We see in Figure E3 that the results are qualitatively similar, with all significant except for the discretized shocks which is relying only on the extensive margin variation and ignoring the intensive margin variation that the NLP methods allow us to leverage. We also demonstrate that our results are not reliant on specific words in any of the topic dictionaries by generating 1000 permutations of the dictionary set, leaving out one word at random for each topic in each permutation. These estimates are compared to our baseline estimate in Figure E4.

Topic exposure Lastly, we consider how we construct the race-based exposure measure. In our main specification, we use the entire ATUS data to construct exposure, but here

Figure E2: Leave-one-out topic relatability estimates



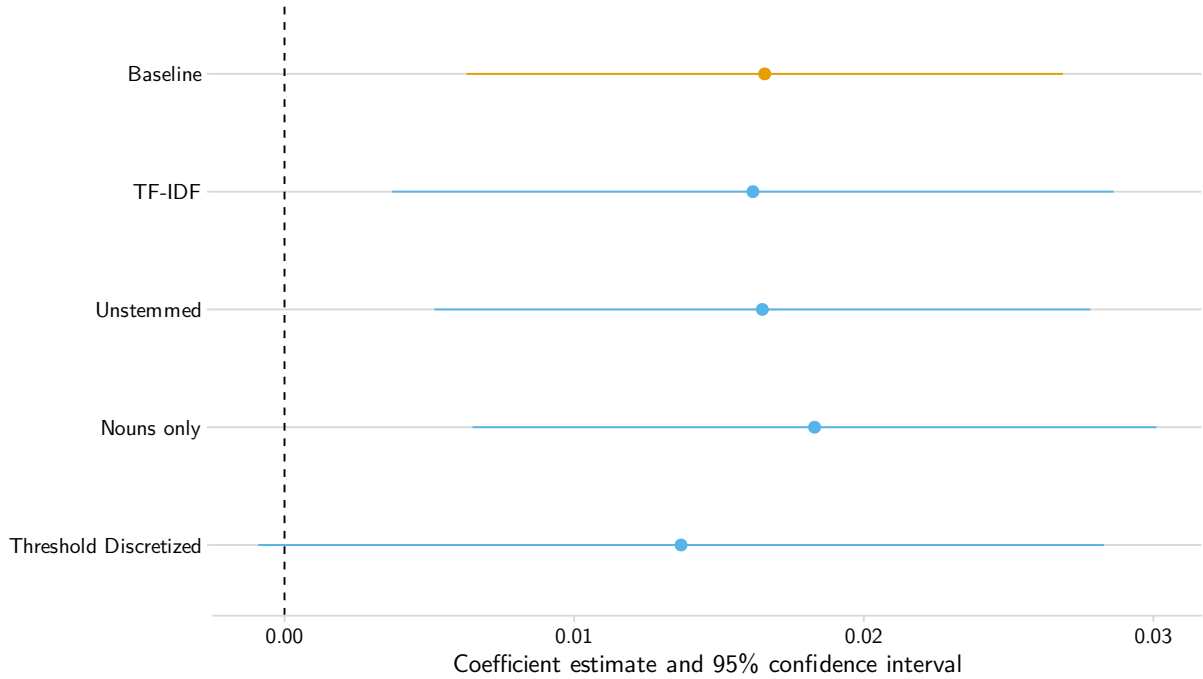
Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Each estimate, aside from the baseline estimate, uses a topic relatability measure which excludes the indicated topic in the calculation process. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

we consider filtering down to subgroups that may be more representative (but have smaller sample size), discussed in section B.1. We see in Figure E5 that using the youngest age group and respondents with children provides a more predictive point estimate.²⁷ In particular, we find the difference in coefficients between our baseline estimate and the estimate from respondents with children is statistically significant. Further, restricting to Southerners or Texans seems to slightly attenuate the estimate, and only using weekend responses has close to no difference. Nonetheless, these all these differences are relatively minor and collectively point to affirming the baseline estimates that just use the ATUS data as-is.

We also consider using “intensive margin” topic exposure measures when constructing

²⁷Results for other age groups are qualitative similar to the baseline estimate.

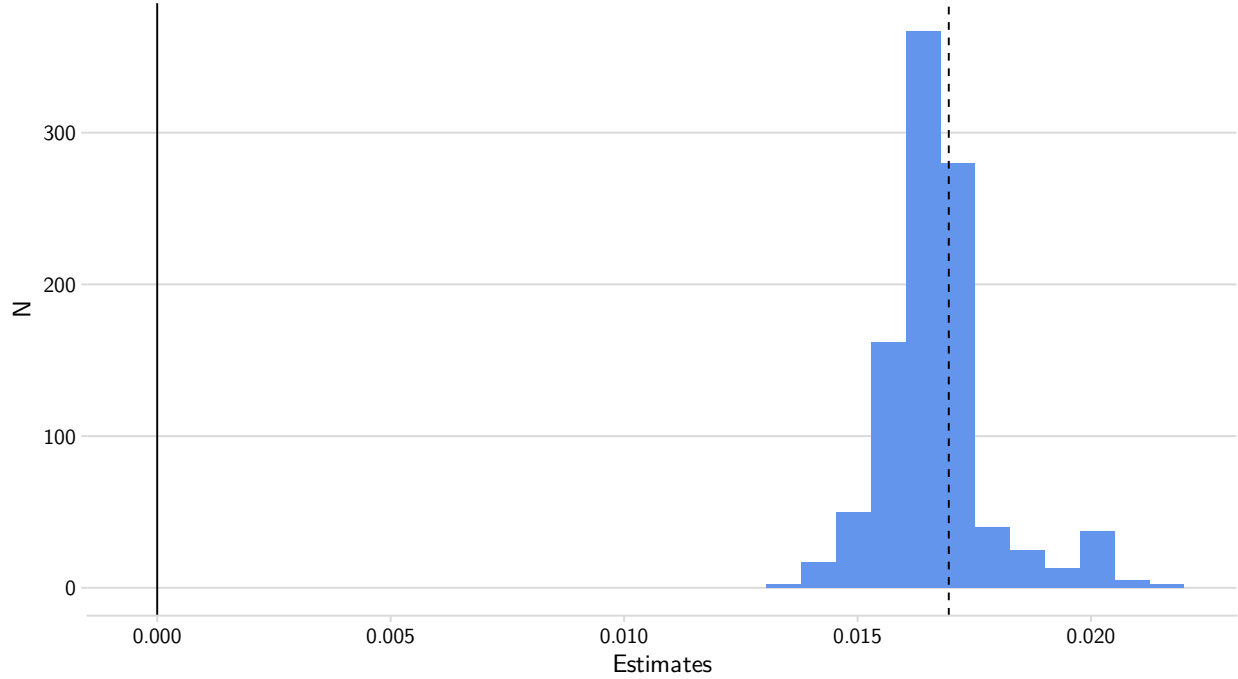
Figure E3: Robustness to different NLP measures for topic salience



Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. The “TF-IDF” specification uses inverse document frequency weights to calculate the topic salience. The “stemmed” specification uses stemmed words when matching between the dictionaries and passages. The “Nouns only” specification filters down to nouns when matching between the dictionaries and passages. Lastly, the “Threshold Discretized” specification sets the top 1/25th topic salience values to 1 and the rest to zero, within grade-level. Further details on variable construction can be found in Appendix Section B.2. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

relatability to make full use of the entire distribution of time use data available in the ATUS data. We exercise appropriate caution in creating these measures, as we believe these measures capture a different attribute of individual relationships to topics. First, while more time spent in an activity is positively correlated with familiarity or interest in that activity, the precise relationship between those objects is unlikely to be linear or follow a standard functional form. For example, a basic level of interest and familiarity may be associated with a wide range of minutes spent, with differentiation only occurring at the tail ends of minutes spent. Second, the time spent distribution will vary widely across topics. An activity such

Figure E4: Histogram of estimates after removal of one word from each topic’s dictionary

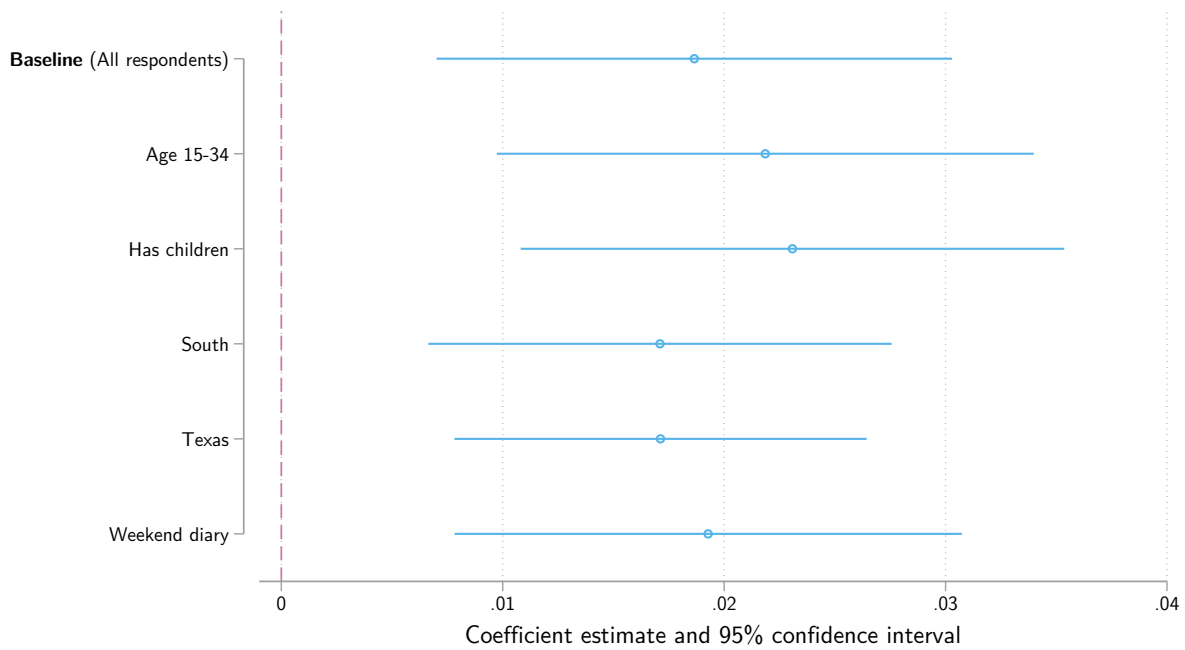


Notes: Each of the 1000 observations is a separate regression of the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Each regression uses a different estimate of topic relatability obtained after removing a single word at random from each topic’s dictionary when constructing topic salience measures. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

as golf will be characterized by extreme values of time spent (0 minutes and 180+ minutes, for example) while an activity such as TV viewing will follow a smoother distribution. This raises scaling concerns across activities.

With these points in mind, we construct four alternative measures of topic exposure, which is detailed in section B.1. Figure E6 shows the coefficient estimates using these new measures. We see that all intensive margin measures show smaller coefficients than the main relatability measures. However, the effect sizes are dependent on the functional form which converts minutes spent into topic exposure. As suspected, a similar linear aggregation of minutes spent introduces noise in our topic exposure estimates and downwardly biases our coefficient estimates. Utilizing a more concave functional form results in estimates that are

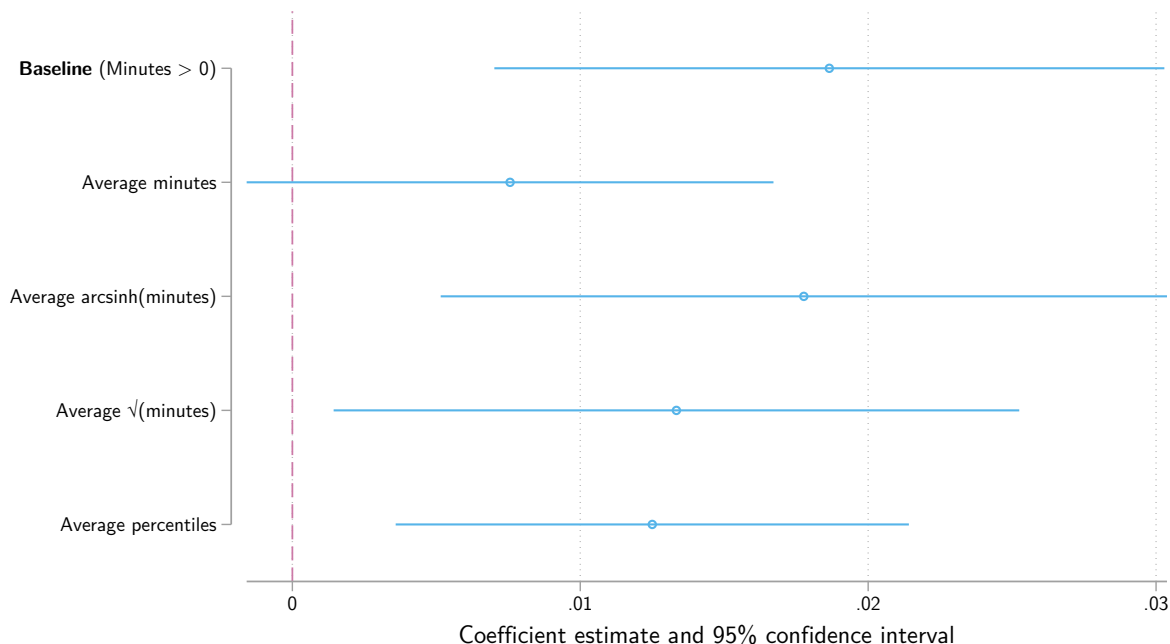
Figure E5: Robustness to different ATUS samples for calculating topic exposure



Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Each estimate uses a different ATUS sample in calculating topic exposure. The baseline measure makes no restriction to the ATUS sample. “Age 15-34” corresponds to respondent age. “Has children” corresponds to respondents which have a child in the household. “South” corresponds to the Southern U.S. Census Region. “Texas” corresponds to respondents in Texas. “Weekend diary” corresponds to using only the ATUS sample by which respondents’ diary day is on the weekend. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

much more similar in magnitude to our main results.

Figure E6: Robustness to different ways of aggregating ATUS time diaries



Notes: Each coefficient estimate is from a separate specification which regresses the share of items answered correctly on a passage on race topic relatability. Unreported controls include: (1) race-grade fixed effects and (2) exposure-sum-by-passage fixed effects. Each estimate uses a different methodology to aggregate ATUS when calculating race topic relatability. The baseline estimate uses the share of a racial group having non-zero minutes. “Average minutes” uses an average of minutes within a racial group. “Average arcsinh(minutes)” uses an average of the inverse hyperbolic sine of minutes within a racial group. “Average $\sqrt{\text{minutes}}$ ” uses an average of the square root of minutes within a racial group. “Average percentiles” first calculates the sample-level percentile of topic minutes and computes the average of the percentile within a racial group. Exposure-robust standard errors clustered by exam are used to construct the 95% confidence intervals. Observations are at the race-passage level and are weighted by the number of student-items. The estimation sample includes all students grades 3 to 8 from 2013–2019 taking the standard STAAR reading comprehension test.

F Details on survey reliability

This appendix section contains detailed information on participant recruitment, participant characteristics, and procedures for the survey which generates our survey reliability measure.

F.1 Recruitment

Respondents were recruited on the online study platform Prolific. The study was titled “Relatability of different text,” with a listed base payment rate of 3 U.S. dollars for approximately 20 minutes of their time. The participants saw the following recruitment text:

The survey consists of reading text and answering a handful questions. We estimate it takes no longer than 20 minutes to complete.

Participants can earn up to \$4 by completing the survey and correctly answering simple comprehension questions.

There are no requirements for taking part in this study, simply answer the questions as honestly as you can.

Thank you for your interest.

Once they agreed to participate in the study, participants were informed of all payment and participation rules at the beginning of the survey. In particular, participants were told they could voluntarily leave the survey at any point. Ultimately, 640 respondents provided partial or full responses to our survey.

The respondent pool is selected to be representative of the United States adult population. We rely on the Prolific platform’s built in representative sample screener. Prolific uses age by sex by race proportions calculated from U.S. Census Bureau estimates from 2021. The age groups are: 18–24, 25–34, 35–44, 45–54, and 55–100. The races are: Asian, Black, Mixed, Other, and White. The platform did not allow screening for Hispanic respondents.

F.2 Survey procedures

Respondents were asked a series of demographic questions at the beginning of the survey. Respondents were asked to provide basic information on their gender, race, ethnicity, and age. We also asked respondents to provide their state of residence and whether they received free- or reduced-price lunch while in school. Finally, respondents were asked about their interactions with children in their daily lives. This included questions about whether they are a parent or guardian, the frequency with which they interact with children other than their own direct children, and the demographic characteristics of the children with which they interact.

Respondents were asked to read and answer questions regarding 10 pieces of text. Each piece of text corresponded to a summarized passage in our test passage sample. Passages were summarized using Claude 3.5 Sonnet, an LLM developed by Anthropic. The LLM was told to act as an “an expert literary analyst tasked with creating informative summaries of various passages.” For each passage, the LLM was prompted to carefully read the passage, analyze the passage, and create a summary of the passage. Summaries were required to meet the following criteria:

- Be approximately 220 words long
- “Capture the essence of the passage without revealing every detail”
- Do not include commentary on the passage
- Retain the style and tone of the original passage
- Write at an 8th-grade reading level

Each summarization criteria was developed through testing the model with a variety of prompts and sample passages. We set a word count target to make sure that passage length is not a significant contributor to respondents’ assessments of relatability. However, we are not prescriptive about this target as (a) we wanted to allow for word count flexibility in

creating a high-quality summary, and (b) we found that our LLM was not generally well-equipped to hit a hard word count target. We also found that sometimes the LLM would over-emphasize certain details in a passage over others. While we observed no overall pattern in this emphasis, our final prompt prevents the model from becoming overly detailed given the short summary length. In piloting, the LLM had a tendency to include commentary on the passage itself, commenting on the morals or lessons that could be gleaned from the story. Our prompt focuses the LLM to provide only direct summarization gleaned from the text. Finally, we give the LLM a target, uniform reading level so that the perceived difficulty of the passage also does not sway respondents when assessing relatability.

After reading a summary, respondents were asked to assess the relatability of the passage for racial groups and gender separately. To focus respondents' attention, we provided a basic relatability definition at the beginning of the survey. The definition reads as follows:

We will primarily ask you to assess these summaries based on how ***relatable*** *you think they are to children*. A summary or story is considered relatable to a child if he/she (a) is familiar with topics in the text, (b) has interest in the text content, and/or (c) is represented by characters in the text.

The respondents were also told that “when answering these questions, feel free to consider children you know, children that you grew up with, or your own experiences as a child.” Respondents were asked two separate questions, one for race and one for gender. For race, respondents were asked:

Please rank the following groups by how relatable this summarized story would be to members of each group. Assign a unique rank from 1(most relatable) to 4 (least relatable).

The four options provided were “Asian children”, “Black children”, “Hispanic children”, “White children.” For gender, respondents were asked:

Which group would find this summarized story more relatable: boys or girls?

Respondents also were asked to provide a personal rating of relatability to “children with similar childhood experiences as mine.”

With surveys, one may be concerned about low quality responses due to respondent fatigue or low effort. These concerns are especially true in our current survey structure where respondents are asked to perform the same task multiple times. We address these concerns in several ways. First, we randomized both the set of passages given to a respondent and the order in which these passages are presented. We also randomize the order in which the racial groups are presented in the race question. Together, these question randomization adjustments ensure that if respondent quality declines throughout the survey or if respondents provide low effort responses, they are distributed randomly throughout passages and demographic group assessments. Second, we include an incentive for respondents based on successfully answering a reading comprehension question related to the summary. Respondents answering this question correctly receives an additional \$0.10, which means respondents can earn up to \$1 if they successfully answer all 10 comprehension questions. Reading comprehension questions are created using the same LLM and manually validated for any errors.

F.3 Respondent and response characteristics

Table F1 summarizes the respondent sample. In comparison to U.S. Census Bureau estimates, Hispanic adults are slightly underrepresented in our sample (12% vs. $\sim 16\%$). The non-Hispanic sample is largely representative of the adult population. This is not surprising as the Prolific representative sample screener does not account for ethnicity. We find that the majority of our adult sample is between the ages of 35–64. This results in a over-representation for this age group, at the expense of individuals aged 65 or above, who are under-represented. This may also reflect differences between the age ranges collected and the Prolific screener: the oldest category in the platform’s screen was age 55 and above. The Texas resident share and eligibility for free- and reduced-price lunch as a child are both roughly in line with national statistics. We also find that a large proportion of our adult respondents interact regularly with white children (71%). Proportions are significantly lower

Table F1: Survey respondents summary statistics

	Mean	SD
Race and gender		
Asian	0.05	
Black	0.12	
Hispanic	0.12	
White	0.62	
Female	0.50	
Age		
18–34	0.30	
35–64	0.56	
65+	0.13	
Other characteristics		
Texas resident	0.09	
Free- and reduced-price lunch as child	0.46	
Has children	0.48	
Demographics of children interacted with		
Asian	0.17	
Black	0.33	
Hispanic	0.29	
White	0.71	
Boys	0.76	
Girls	0.73	
Survey response statistics		
Number of passages answered	9.70	1.26
10/10 passages answered	0.89	
Minutes	27.58	13.40
Comprehension score	8.59	2.00
<i>N</i>	640	

Notes: The table displays sample means and standard deviations of respondent characteristics in the survey sample. Race includes individuals who identify as Hispanic. Asian, Black, and White exclude individuals who identify as Hispanic. “Free- and reduced-price lunch as child” includes all individuals who reported being eligible for free- and reduced-price lunch when they were children. A passage is considered as “answered” if a respondent answers all reliability-related questions related to that passage. Survey minutes are winsorized at the 1% and 99% levels, as a small portion of respondents did not formally quit our survey and allowed the survey to time out. The sample includes all respondents who answered at least one reliability-related question.

for all other racial groups.

We also show basic summary statistics on the survey responses elicited. On average, close to 90% of respondents answered questions for every passage and more than 95% of respondents

answered questions for at least 9 passages. Respondents on average took slightly longer than expected to finish the survey (28 minutes). The average comprehension score was 8.59 out of 10. The median score was a 9, with over two-thirds of the respondent sample receiving this score or higher.

We take the survey responses to create survey reliability measures. Our preferred measure comes from the predicted probabilities after fitting an rank-ordered logit model for each passage. Suppose we observe a set of rankings $d_{ip1} \succ d_{ip2} \succ \dots d_{ip4}$ of racial groups for individual i and passage p . We assume latent reliability assessment is given by: $R_{ipd} = V_{ipd} + \varepsilon_{ipd}$, where ε_{ipd} is assumed to be Type I extreme value. We assume $V_{ipd} = \delta_{pd}$ where δ_{pd} are reliability assessment levels for group $d \in D$ for passage p . Then the probability of observing the ranking is

$$\Pr(d_{ip1} \succ d_{ip2} \succ \dots d_{ip4} \mid \delta_p) = \prod_{m=1}^3 \frac{\exp(\delta_{pd_{ipm}})}{\sum_{h \in D \setminus \{d_{ip1}, \dots, d_{ip, m-1}\}} \exp(\delta_{ph})},$$

that is, the probability of observing the ranking is the probability of “preferring” d_{ip1} over all alternatives, preferring d_{ip2} for all alternatives excluding d_{ip1} , etc. We estimate δ separately for each passage, meaning the log-likelihood estimator maximizes for a given p

$$\mathcal{L}_p(\delta_p) = \sum_i \sum_{m=1}^3 \left[\delta_{pd_{ipm}} - \log \left(\sum_{h \in D \setminus \{d_{ip1}, \dots, d_{ip, m-1}\}} \exp(\delta_{ph}) \right) \right]. \quad (\text{F1})$$

We estimate the δ_p that maximizes $\mathcal{L}_p$ given the rankings we observe from respondents. For identification, we impose one constraint per passage, choosing a reference group d^* and setting $\delta_{pd^*} = 0$. We note that δ_p for gender is trivially obtained since there are only two alternatives. After obtaining δ_p from maximizing equation (F1), we calculate

$$\hat{\Pr}(d \text{ is rank } 1 \mid p) = \frac{\exp(\hat{\delta}_{pd})}{\sum_{h \in D} \exp(\hat{\delta}_{ph})}, \quad (\text{F2})$$

which is the predicted probability that racial group d is assessed to relate most to passage

Table F2: Survey relatability summary

	Asian	Black	Hispanic	White	Female	Male
Rank	2.771 (0.495)	2.727 (0.385)	2.731 (0.384)	1.771 (0.473)	1.587 (0.312)	1.413 (0.312)
Rank = 1	0.156 (0.169)	0.135 (0.139)	0.115 (0.143)	0.594 (0.219)	0.413 (0.312)	0.587 (0.312)
$\hat{\text{Pr}}(d \text{ is rank } 1 \mid p)$	0.168 (0.138)	0.165 (0.111)	0.170 (0.110)	0.498 (0.206)	0.413 (0.312)	0.587 (0.312)
N passages	205					
N respondents	640					

Notes: The table displays means of relatability rankings provided by survey respondents for each demographic group. Standard deviations are reported in parentheses. Means are calculated at the passage-level after responses are averaged within passage. For example, the mean rank for Asian is calculated after first calculating the mean rank for Asian children for each passage and then calculating the mean of this mean rank across passages. “Rank” is the raw rank provided by respondents, where 1 indicates most relatable and 4 indicates least relatable. “Rank = 1” is the share that said a passage was most relatable for a particular race. “ $\hat{\text{Pr}}(d \text{ is rank } 1 \mid p)$ ” is the predicted probability that racial group d is assessed to relate most to passage p from maximizing the rank-ordered logit estimator specified in equation (F1). The sample includes all respondents who answered at least one relatability-related question.

p . We define $r_{dp}^{\text{survey}} = \hat{\text{Pr}}(d \text{ is rank } 1 \mid p)$.

Table F2 provides a summary of relatability responses across passages. Respondents on average rate passages to be a full one relatability rank higher than other racial groups. Passages are four times more likely to be rated as most relatable to white children than for other racial groups. However, the aggregate differences across racial groups are more muted for the predicted probability measure. For gender, we find that passages are generally rated as being more relatable to boys than to girls.

G Details on STAAR standards analysis

Every year, the TEA sets STAAR performance standards which are meant to link STAAR results to state-mandated curriculum standards. Students are then assigned to standards categories based on their results on the STAAR tests. These categories allow educators, parents, and students to anchor better make sense of their test performance. The categories are also used to direct extra resources to students and in certain cases to prevent them from being promoted to the next grade. Figure A7 demonstrates one such outcome based on the standards categories. In Figure A7, we plot the shares of 5th grade and 8th grade reading comprehension test takers who have to retake the test. We organize the horizontal axis by initial test score, normalizing across tests based on the prevailing “satisfactory” performance cut-off for that test. We observe a stark discontinuity around this cut-off, with 98% of students just below the cut-off retaking the exam and virtually 0% of students just above the cut-off retaking the exam.

Over the course of STAAR testing, the TEA has continually changed the overall framework for determining performance standards. From the 2012–13 to 2015–16 academic year, students were in one of three categories: “Unsatisfactory,” “Satisfactory,” and “Advanced.” The TEA originally intended to gradually raise the threshold for “Satisfactory” to a long-run, pre-announced level (“Satisfactory: recommended”), but only did so for the 2015–16 academic year. Starting in the 2016–17 academic year, the TEA instead switched to a four-tier system: “Did Not Meet,” “Approaches,” “Meets,” and “Masters.” Students who would have been classified in the lowest and highest categories would continue to do so across the three-tier and four-tier system. However, the “Satisfactory” category was split into two categories, for students who were below the “Satisfactory: recommended” threshold and students who were above it. Common across these regimes is the conversion of raw test scores to scale scores to the assignment of performance standards categories.

In order to make consistent predictions across years, we create four categories across all years. For the 2016–17 to 2018–19 testing years, we maintain the TEA-designated categories.

For the 2012-13 to 2015-16 years, we split the “Satisfactory” category into two groups based on the “Satisfactory: recommended” threshold. While this adjustment may not exactly reflect how students were actually classified in these years, we argue that this approach is reasonable. First, this procedure reflects precisely how the TEA modified the three-tier system to the four-tier system and allows for easier comparisons across time. Second, the TEA had already announced the threshold for “Satisfactory: recommended” before the 2012–13 school year, meaning that it could have potentially been used as an unofficial benchmark by parents, teachers, and schools.

We now consider the extent to which content reliability affects the misclassification of students to standards categories. Given that our estimation approach assumes that topic salience may be determined jointly within a given exam, we adjust tests for passage reliability that hews to this assumption. First, we calculate the average topic reliability and average identity reliability for each test by race. Then, we use $\hat{\beta}$ from estimating equation (4) and $\hat{\beta}^{identity}$ from estimating equation (6) to estimate the “content reliability benefit” of each test

$$rel_benefit_{ghd} = \hat{\beta} \cdot \bar{r}_{ghd} + \hat{\beta}^{identity} \cdot \bar{r}_{ghd}^{identity}$$

where h indexes exam years and $\bar{r}_{ghd}$ is the average topic reliability for each test gh and race d . We identify a “benchmark” test within each grade with (a) the lowest Black-white difference in $rel_benefit$ and (b) the lowest Hispanic-white average reliability difference. Let $\bar{r}_{gh^*d}$ and $\bar{r}_{gh^*d}^{identity}$ represent the average reliabilities from this benchmark exam for grade g and group d .

For each non-benchmark test, we calculate the difference between content reliability for that test compared to this benchmark test and adjust overall scores based on $\hat{\beta}$ and $\hat{\beta}^{identity}$.

Formally, we calculate

$$score'_i = score_i + N_{g(i)h(i)} \left[\hat{\beta} (\tilde{r}_{g(i),d(i)} - \bar{r}_i) + \hat{\beta}^{identity} (\tilde{r}_{g(i),d(i)}^{identity} - \bar{r}_i^{identity}) \right] \quad (G1)$$

where i indexes student-exams, $g(i)$ indicates the grade level, $d(i)$ is the student's race, and $h(i)$ indicates the calendar year of the exam. $score_i$ is the score for the student on the exam, $N_{g(i)h(i)}$ is the number of questions on the exam, and $\hat{\beta}$ is the estimated coefficient from equation 4. $\bar{r}_i$ is the average reliability for the student's exam and $\tilde{r}_{g(i),d(i)}$ is the adjusted average reliability for $d(i)$ on the benchmark exam for grade $g(i)$. We obtain $\tilde{r}_{g(i),d(i)}$ from $\bar{r}_{gh*d}$ by applying an adjustment such that the average cross-demographic group reliability is the same between the benchmark test and the non-benchmark test. This is essential so that the number of students receiving each score remains relatively stable with the score adjustment.

We assign students to performance standards categories based on this new, reliability-adjusted score, $score'_i$. Since raw scores can only be a whole number, students with non-integer $score'_i$ are partially attributed to the two nearest integers. Specifically, we assign $(\lceil score'_i \rceil - score'_i)\%$ of a student to score $\lfloor score'_i \rfloor$ and the rest to $\lceil score'_i \rceil$. For example, if 100 students have a computed score of 30.4, our method essentially counts 40 students having an adjusted score of 30 and 60 students having an adjusted score of 31. After these adjustments, we re-assign student i to one of four performance standards categories for exam $g(i)m(i)$.